KOMOREBI MATHEMATICS

CORE SERIES I FINAL SERIES EDITION

Projector-resolved seed calculus I

Typed operator systems, seed rank, and exact packet coordinates

Komorebi P.

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Abstract

This paper develops the complete linear foundation of the projector-resolved seed calculus and presents it in a form intended to be read, audited, and reused. The theory begins with a typed operator grammar, passes to a concrete realized operator algebra, studies generation from one or more seeds, and then resolves both vectors and operators into exact packet coordinates. The resulting transfer-block language supports graph constraints, quotient descent, completion, and functorial transport into later realizations.

Every theorem, proposition, lemma, and corollary is proved under explicit hypotheses. The central generator results include finite free presentations, detection of the minimal generator number modulo the Jacobson radical, the semisimple multiplicity formula over arbitrary division rings, and the single-operator formula over arbitrary fields. The central coordinate results include exact finite packet reconstruction, a countable unconditional extension, tensor-product and dual packet systems, complete transfer-block multiplication, support-preorder algebras, packet-pure source-component generation, blockwise intertwining equations, invariant quotient descent, and semantic theorem transport through realization functors.

The presentation combines the complete symbolic foundation with worked quiver, coordinate-ring, matrix, and graph models, an analytic shift example, exact computational specifications, and a visual atlas. Stable statement identifiers connect the traditional argument to the machine-native companion. The separate Lean guide records precisely which statements and certificate scopes have kernel-checked counterparts.

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Reader's orientation

Status class	Meaning
Definition	Introduced structure or notation.
Theorem, proposition, corollary, lemma	Proved in this manuscript under the printed hypotheses.
Classical input	Imported standard theorem, stated explicitly because later proofs depend on it.
Boundary	A limitation, counterexample, or non-implication that is part of the formal scope.

Companion volumes. The Lean guide explains the formal development, its build procedure, theorem mapping, and the hypotheses of each verified contract. The machine-native paper exposes the stable records, dependencies, and executable evidence interfaces. The mathematics below retains the full scope of its stated classical inputs and written proofs.

Reader roadmap

The paper has four layers. They should be read in order at least once.

1. **Grammar and realization.** Specify which compositions of operators are legal, then distinguish that abstract syntax from the operators that actually act on a chosen space.
2. **Generation.** Choose one or more seeds and ask which part of the realization is generated by their operator orbits. The minimal number of required seeds is the seed rank.
3. **Packet coordinates.** Resolve the space by a complete family of idempotents. Every vector and every operator then has exact packet coordinates.
4. **Transport.** Study how those coordinates behave under products, graphs, morphisms, quotients, completion, and realization functors.

The first two layers describe what can be generated. The last two describe how the generated object is resolved and transported.

The basic conceptual chain

$$\begin{aligned}\text{typed grammar} &\longrightarrow \text{realized algebra} \longrightarrow \text{seed hull}\\ &\longrightarrow \text{seed rank} \longrightarrow \text{packet coordinates}\\ &\longrightarrow \text{transfer blocks} \longrightarrow \text{functorial realization.}\end{aligned}$$

The words in this chain have deliberately different meanings. A grammar is a presentation of allowed operator words. A realized algebra is the image of that grammar in an endomorphism algebra. A seed hull is an orbit span inside one particular module. A packet system is an exact coordinate decomposition and need not be orthogonal. A realization functor is required to preserve a declared mathematical profile, not merely the vocabulary.

Four questions answered by Paper I

Question	Formal object	Principal answer
What operations are legal?	Quiver with relations or a unital operator algebra	Typed realizations are modules over the quotient path algebra.
What can a finite seed family generate?	Evaluation map $A^{\oplus r} \to M$	The generated hull is a quotient of a finite free module.
How many seeds are required?	$\operatorname{srank}_A(M)$	Pass to $M/\operatorname{Jac}(A)M$, then compute multiplicities in the semisimple top.
How are vectors and operators resolved?	Complete idempotents (P_α)	$X \cong \bigoplus_\alpha P_\alpha X$ and $T \leftrightarrow (P_\beta T P_\alpha)$.

Notation map

Symbol	Name	Meaning
$\mathfrak{G} = (Q, I)$	Typed grammar	A finite quiver and a two-sided ideal of source-target compatible relations.
$\Lambda_{\mathfrak{G}}$	Grammar algebra	The quotient path algebra $\Bbbk Q/I$.
$\mathcal{A}_X$	Realized algebra	The image of the grammar algebra in $\operatorname{End}_{\Bbbk}(\lvert X\rvert)$.
$S \subseteq M$	Seed family	The vectors from which the operator orbit is generated.
$\operatorname{Hull}_A(S)$	Generated hull	The submodule AS, or its declared topological closure.
$\operatorname{srank}_A(M)$	Seed rank	The minimal cardinality of a finite generating family.
$\mathcal{P} = (P_\alpha)$	Packet system	Pairwise annihilating idempotents summing to the identity.
X_α	Packet space	The range $P_\alpha X$.
$\mathcal{A}_{\mathcal{P}}$	Analysis	The map $x \mapsto (P_\alpha x)_\alpha$.
$\mathcal{S}_{\mathcal{P}}$	Synthesis	The map $(x_\alpha) \mapsto \sum_\alpha x_\alpha$.
$T_{\beta\alpha}$	Transfer block	The source-target block $P_\beta T P_\alpha : X_\alpha \to X_\beta$.
$\operatorname{Supp}_{\mathcal{P}}(x)$	Packet support	The packet indices on which x has a nonzero coordinate.
psrank	Packet-pure seed rank	The minimal number of generating seeds each supported in a single packet.

Running micro-examples

The following small systems recur throughout the explanatory boxes. The worked models and visual atlas later develop these ideas in complete calculations.

R1: one operator with several primary blocks

Let T act on a finite-dimensional vector space. Regard X as a $\Bbbk[t]$-module by $t \cdot x = Tx$. If T has three Jordan blocks for the same eigenvalue, then three seeds are necessary and sufficient for that primary component. The largest number of blocks over any irreducible polynomial is the seed rank.

R2: a matrix algebra with multiplicity

Let $A = M_2(\Bbbk)$ act diagonally on $M = (\Bbbk^2)^{\oplus 3}$. One seed can contribute at most two independent multiplicity directions, because the regular module contains two copies of the simple module. Hence

$$\operatorname{srank}_A(M) = \left\lceil \frac{3}{2} \right\rceil = 2.$$

This is the simplest nontrivial instance of the semisimple multiplicity formula.

R3: an exact but oblique packet split

On $X = \Bbbk^2$, set

$$P_1 = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}, \qquad P_2 = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}.$$

Then $P_i^2 = P_i$, $P_1P_2 = P_2P_1 = 0$, and $P_1 + P_2 = I$. The ranges form a direct sum, but the projectors are not orthogonal in the Euclidean inner product. Exact reconstruction and numerical conditioning are therefore separate questions.

R4: a directed packet system

Let $X = X_1 \oplus X_2 \oplus X_3$. Suppose the generating operators permit transfers $1 \to 2$ and $2 \to 3$, with no path entering packet 1. Packet 1 is the unique source component. Under the packet-propagation hypotheses of theorem 9.7, one packet-pure seed in X_1 can generate all three packets.

Suggested reading paths

- For the minimal algebraic foundation, read Sections 1–5, then Sections 7–8.
- For finite-dimensional representation theory, concentrate on Sections 4–6.
- For signal, wavelet, or quantum applications, read Sections 7–8 and the tensor-product additions carefully.
- For graph and residual-transport applications, read Sections 8–10.
- For later categorical claims, read Sections 10–11 together with the final semantic-realization criterion.

Do not conflate

The paper does not claim that every useful decomposition is automatically discovered, that every packet split is numerically stable, or that exact coordinates solve a nonlinear problem. Paper I establishes the linear language and its exact transport laws. Refinement, obstruction theory, nonlinear trees, arithmetic valuations, and continuum limits belong to later papers.

1 Coefficient conventions, logical status, and scope

Reader roadmap

This short section fixes the logical contract for the entire paper. It says when a statement is algebraic, when topology is present, which results are imported, and what the stable record IDs mean. Many later ambiguities disappear once these regimes are kept separate.

1.1 Ambient regimes and theorem records

Definition 1.1 (Coefficient and module regimes)**.** All modules are left modules unless stated otherwise. The algebraic regime consists of a unital ring A and a unital left A-module M. When linear dimensions, tensor products, quivers, or operator blocks are used, A is a unital $\Bbbk$-algebra over a field $\Bbbk$ and M is a $\Bbbk$-vector space. The topological regime consists of a Hausdorff topological vector space over a complete valued field, together with a unital algebra of continuous linear operators. Every infinite sum, completion, quotient topology, and closure below is interpreted only in the regime explicitly declared in its statement.

Empty-family conventions. For finite families of nonnegative integers we use $\max \varnothing = 0$. The height of an empty finite partially ordered set is 0, as is the height of a nonempty poset with no strict comparable pair. Thus the height in the support-preorder theorem counts strict steps.

Definition 1.2 (Canonical theorem record)**.** A canonical theorem record is the tuple

$$(\mathrm{ID}, \mathrm{status}, \mathrm{hypotheses}, \mathrm{statement}, \mathrm{dependencies}, \mathrm{proof}, \mathrm{exports}, \mathrm{boundary}).$$

A downstream paper cites the stable record ID. A domain specialization may add domain hypotheses, but it does not repeat a generic proof from this paper.

Boundary 1.3 (Symbolic scope). This edition proves the algebraic and topological statements printed below. It does not assert historical novelty, independent external certification, automatic discovery of useful packet systems, nonlinear obstruction theory, multilevel refinement, arithmetic congruences, quantum gravity, automorphy, or local or global Langlands reciprocity.

2 Typed operator grammars, realizations, and seed systems

Reader roadmap

The goal is to separate syntax from action. A grammar lists legal typed compositions. A realization assigns vector spaces and operators. The realized algebra is the image of that syntax in the endomorphism algebra. Seeds live only after a realization has been chosen.

2.1 Typed syntax and module realizations

Definition 2.1 (Typed operator grammar)**.** Let $Q = (Q_0, Q_1, s, t)$ be a quiver with finite vertex set. A typed operator grammar over $\Bbbk$ is a pair

$$\mathfrak{G} = (Q, I),$$

where $I \triangleleft \Bbbk Q$ is a two-sided ideal generated by $\Bbbk$-linear combinations of paths having common source and common target. Write

$$\Lambda_{\mathfrak{G}} := \Bbbk Q / I.$$

A realization of $\mathfrak{G}$ is a family $(X_v)_{v \in Q_0}$ and linear maps

$$X_a : X_{s(a)} \longrightarrow X_{t(a)} \qquad (a \in Q_1)$$

that annihilate every relation in I. A morphism $f : X \to Y$ of realizations is a family of linear maps $f_v : X_v \to Y_v$ satisfying

$$f_{t(a)} X_a = Y_a f_{s(a)} \qquad (a \in Q_1).$$

Definition 2.2 (Collapsed one-space realization)**.** For a typed realization X, define its collapsed one-space object by

$$|X| := \bigoplus_{v \in Q_0} X_v.$$

Every vertex idempotent acts as a coordinate projector on $|X|$, and every path acts as a source-target block operator.

Theorem 2.3 (Typed realization and module equivalence)**.** *The category of realizations of $\mathfrak{G}$ is equivalent to the category of unital left $\Lambda_{\mathfrak{G}}$-modules. Explicitly, a realization X determines the collapsed space $|X|$ of definition 2.2 with the vertex idempotent e_v acting as the coordinate projector onto X_v and each arrow acting in its declared source-target block. Conversely, a unital $\Lambda_{\mathfrak{G}}$-module M determines the realization*

$$M_v = e_v M, \qquad M_a(x) = ax.$$

The two constructions are inverse up to canonical natural isomorphism.

Proof. Because Q_0 is finite, $1 = \sum_{v \in Q_0} e_v$ in $\Bbbk Q$, the e_v are pairwise orthogonal, and the same identities hold in $\Lambda_{\mathfrak{G}}$. Given a realization, define the action of a path by composition of its arrow

maps and extend linearly. Every element of I acts as zero, so the action factors through $\Lambda_{\mathfrak{G}}$. The vertex idempotents give

$$|X| = \bigoplus_v e_v|X| = \bigoplus_v X_v.$$

Conversely, if M is unital, then

$$M = 1M = \sum_v e_v M.$$

Orthogonality makes the sum direct. For an arrow $a : v \to w$, the identities $a = e_w a e_v$ imply

$$a(e_v M) \subseteq e_w M.$$

The relations in I act as zero because M is a $\Lambda_{\mathfrak{G}}$-module. Applying both constructions returns the original vertex spaces and arrow maps, or the original module decomposition, through the evident identity maps. These identifications commute with morphisms, giving an equivalence of categories. □

Interpretation

The equivalence does not claim that a quiver presentation is unique. It says that once a grammar is fixed, working with typed arrows and working with a module over the quotient path algebra carry the same information. The typed description is often easier for constructing an operator system. The module description is often easier for proving generation and quotient theorems.

A two-vertex grammar

Take two vertices and one arrow $a : 1 \to 2$. A realization is simply a pair of spaces and a linear map

$$X_1 \xrightarrow{X_a} X_2.$$

After collapse, $|X| = X_1 \oplus X_2$, the vertex idempotents are the two coordinate projectors, and a is a single off-diagonal block. This elementary case already contains the packet and transfer-block viewpoint used later.

2.2 The realized action and faithful reduction

Definition 2.4 (Realized operator algebra)**.** For a realization X, the realized operator algebra is

$$\mathcal{A}_X := \mathrm{im}(\Lambda_{\mathfrak{G}} \longrightarrow \mathrm{End}_{\Bbbk}(|X|)).$$

It records the actual operators in the realization and may be a proper quotient of the abstract grammar algebra.

Proposition 2.5 (Action factorization and hull invariance)**.** *Let $S \subseteq |X|$. The generated submodule under the grammar algebra equals the generated submodule under the realized operator algebra:*

$$\Lambda_{\mathfrak{G}} S = \mathcal{A}_X S.$$

Consequently, every seed-rank statement depends only on the realized action, not on redundant relations in a chosen presentation of the grammar.

Proof. The action map $\rho : \Lambda_{\mathfrak{G}} \twoheadrightarrow \mathcal{A}_X$ is surjective by definition and satisfies $\lambda s = \rho(\lambda)s$. Hence every grammar orbit vector lies in $\mathcal{A}_X S$. The reverse inclusion holds because every element of $\mathcal{A}_X$ is $\rho(\lambda)$ for some λ. □

Key idea

Replacing the abstract grammar algebra by its image $\mathcal{A}_X$ removes syntax that acts as zero. This is a faithful reduction of the action. It does not change any orbit, hull, or seed-rank computation.

2.3 Seeds, addressed hulls, and natural presentations

Definition 2.6 (Linear seed system)**.** A linear seed system is a tuple

$$\mathfrak{S} = (X, \mathcal{A}, S, \mathcal{P}, \tau),$$

where X is a unital left $\mathcal{A}$-module, $\mathcal{A}$ is fixed independently of the target element, $S = (s_1, \ldots, s_r)$ is a finite distinguished seed family, $\mathcal{P}$ is either empty or a declared packet system, and τ is either the algebraic convention or a specified Hausdorff topology in which the declared operator action is continuous.

Definition 2.7 (Generated hull)**.** For $S \subseteq X$, define

$$\mathrm{Hull}_{\mathcal{A}}(S) := \sum_{s \in S} \mathcal{A}s.$$

In the topological regime define

$$\mathrm{Hull}_{\mathcal{A},\tau}(S) := \overline{\mathrm{Hull}_{\mathcal{A}}(S)}^{\tau}.$$

For a singleton $S = \{s\}$, write $\mathrm{Hull}_{\mathcal{A}}(s)$.

Definition 2.8 (Natural presentation)**.** A seed presentation is natural relative to a declared grammar when the primitive operators and relations are fixed independently of the target, the relation system is finite or effectively enumerable, the topology is intrinsic to the realization, no primitive operator directly stores the complete target, and the presentation exposes a reusable invariant, theorem, or computation not encoded by a target-dependent primitive.

Proposition 2.9 (Algebra generation and module generation are distinct)**.** *Every nonzero unital algebra A, regarded as a left module over itself, has*

$$\mathrm{srank}_A(A) = 1,$$

with cyclic seed 1_A. This conclusion is independent of the number of algebra generators required to present A as an algebra.

Proof. For every $a \in A$, one has $a = a1_A$. Hence $A1_A = A$. The number of operations needed to generate A as an algebra is a different invariant and does not enter the module calculation. □

Boundary 2.10 (Target-dependent operators). For a fixed seed s and target x, the existence of a specially defined operator A_x satisfying $A_x s = x$ does not establish a natural seed calculus unless A_x belongs to a target-independent grammar satisfying definition 2.8.

Do not conflate

An operator may be chosen because it sends the current seed to a desired target. Such a choice proves only an existence statement for that one target unless the operator already belongs to the fixed algebra. The algebra must be declared before the seed-rank question is asked. Otherwise the acting system changes with the answer one wants to obtain.

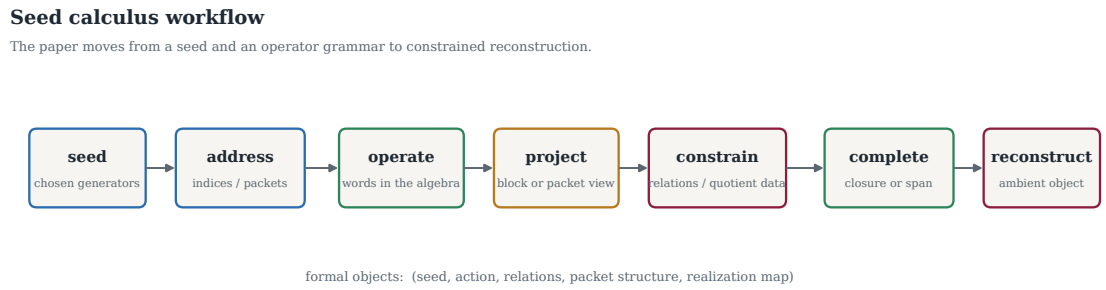


Figure 1. Schematic organization of the seed, operator, relation, and reconstruction data. This is a guide to the presentation, not an additional formal theorem.

3 Free presentations by seeds and relations

Reader roadmap

A seed family gives a canonical map from a finite free module. The generated hull is exactly its image, and the relations among addressed seeds form the kernel. This converts orbit generation into a presentation problem.

3.1 Finite free covers

Definition 3.1 (Seed evaluation map)**.** Let A be a unital ring, let M be a left A-module, and let $S = (s_1, \dots, s_r) \in M^r$. Define the A-linear map

$$\varepsilon_S : A^{\oplus r} \longrightarrow M, \qquad (a_1, \dots, a_r) \longmapsto \sum_{j=1}^{r} a_j s_j.$$

Theorem 3.2 (Finite free presentation)**.** *For the map in definition 3.1,*

$$\operatorname{im} \varepsilon_S = \operatorname{Hull}_A(S), \qquad N_S := \operatorname{Ker} \varepsilon_S \leq A^{\oplus r},$$

and the induced map is an isomorphism of left A-modules

$$\boxed{A^{\oplus r}/N_S \cong \operatorname{Hull}_A(S).}$$

Consequently, S generates M if and only if ε_S is surjective.

Proof. The image consists exactly of finite sums $\sum_j a_j s_j$, which is $\mathrm{Hull}_A(S)$. Because ε_S is A-linear, its kernel is a left submodule. The first isomorphism theorem gives

$$A^{\oplus r}/N_S \cong \operatorname{im}\varepsilon_S = \mathrm{Hull}_A(S).$$

The final equivalence is the equality $\operatorname{im}\varepsilon_S = M$. □

Interpretation

The theorem converts the geometric language of addressed seed orbits into the algebraic language of generators and relations. A word in the operators is represented by an element of the free module. Two addressed expressions represent the same vector exactly when their difference lies in the kernel. The generated hull is therefore not merely spanned by the free module. It is its canonical quotient.

A cyclic nilpotent chain

Let $A = \Bbbk[t]$ and $M = \Bbbk[t]/(t^3)$. The residue class $s = [1]$ is one seed and

$$\varepsilon_s : A \to M, \qquad f(t) \mapsto f(t)[1]$$

has kernel (t^3). Hence $M \cong A/(t^3)$, and the addressed vectors $s, ts, t^2 s$ are the visible orbit chain.

Corollary 3.3 (Minimal free-cover characterization)**.** *For every finitely generated left A-module M,*

$$\boxed{\mathrm{srank}_A(M) = \min\{r \geq 0 : \textit{there exists a surjection } A^{\oplus r} \twoheadrightarrow M\}.}$$

Proof. A seed family of size r gives a surjection by theorem 3.2. Conversely, the images of the standard basis vectors under any surjection $A^{\oplus r} \twoheadrightarrow M$ form a generating family of size at most r. □

Corollary 3.4 (One-seed presentation)**.** *For $s \in M$, the annihilator*

$$\mathrm{Ann}_A(s) := \{a \in A : as = 0\}$$

is a left ideal and

$$\boxed{A/\operatorname{Ann}_A(s) \cong \mathrm{Hull}_A(s).}$$

Proof. The one-seed evaluation map is $a \mapsto as$. Its kernel is $\mathrm{Ann}_A(s)$, which is therefore a left ideal, and theorem 3.2 gives the quotient isomorphism. □

3.2 One seed and the distinction between left and two-sided relations

Proposition 3.5 (Two-sided annihilator criterion)**.** *For $s \in M$, the left ideal $L = \mathrm{Ann}_A(s)$ is two-sided if and only if*

$$L\,\mathrm{Hull}_A(s) = 0.$$

If s is cyclic, these conditions are equivalent to

$$\mathrm{Ann}_A(s) = \mathrm{Ann}_A(M).$$

Proof. The left ideal L is two-sided exactly when $ab \in L$ for every $a \in L$ and $b \in A$. This is equivalent to

$$0 = (ab)s = a(bs) \qquad (a \in L,\ b \in A).$$

By additivity, the last identity is equivalent to $L\,\mathrm{Hull}_A(s) = 0$.

If s is cyclic, then $\mathrm{Hull}_A(s) = M$. Hence the preceding condition is $L \subseteq \mathrm{Ann}_A(M)$. The reverse inclusion always holds because an operator annihilating all of M annihilates s. Thus equality follows. Conversely, $\mathrm{Ann}_A(M)$ is a two-sided ideal, so equality implies that L is two-sided. □

Why the annihilator need not be two-sided

Let $A = M_2(\Bbbk)$ act on $M = \Bbbk^2$, and let $s = e_1$. Then $\mathrm{Ann}_A(s)$ consists of matrices whose first column is zero. It is a left ideal because left multiplication preserves a zero first column. It is not generally a right ideal. Thus the natural one-seed quotient is a quotient of the left regular module, not automatically a quotient algebra.

3.3 Topological completion of a presentation

Theorem 3.6 (Completion of a dense free presentation)**.** *Let X be a complete Hausdorff locally convex space, let $\mathcal{A} \subseteq \mathrm{End}(X)$ act continuously, and let $S = (s_1, \dots, s_r)$. Put*

$$E := \mathcal{A}^{\oplus r}/N_S$$

algebraically and let $j : E \to X$ be the injective map induced by ε_S. Give E the locally convex topology transported from $j(E)$, equivalently the topology generated by

$$\widetilde{p}(e) := p(j(e))$$

for the continuous seminorms p on X. Every $a \in \mathcal{A}$ acts continuously on E in the transported topology and therefore extends to $\widehat{E}$. If $\mathrm{Hull}_{\mathcal{A}}(S)$ is dense in X, then j extends uniquely to a topological $\mathcal{A}$-module isomorphism

$$\boxed{\widehat{E} \cong X.}$$

Proof. By theorem 3.2, j is an algebraic $\mathcal{A}$-module isomorphism from E onto $\mathrm{Hull}_{\mathcal{A}}(S)$. The transported seminorms make it a topological isomorphism onto that subspace. For $a \in \mathcal{A}$ and a continuous seminorm p on X, continuity of $a : X \to X$ supplies continuous seminorms controlling $p(aj(e))$, so the induced action on E is continuous and extends uniquely to $\widehat{E}$. The subspace is

dense by hypothesis. A Cauchy net in E is mapped to a Cauchy net in X, which converges because X is complete. This defines a continuous extension $\widehat{j} : \widehat{E} \to X$. For any $x \in X$, choose a net (e_λ) in E with $j(e_\lambda) \to x$. The transported topology makes (e_λ) Cauchy, so it converges to some $\xi \in \widehat{E}$, and continuity gives $\widehat{j}(\xi) = x$. Thus $\widehat{j}$ is surjective. If $\widehat{j}(\xi) = 0$, represent ξ by a Cauchy net (e_λ). Then $j(e_\lambda) \to 0$, so every defining seminorm $\widetilde{p}(e_\lambda)$ tends to zero. Thus $e_\lambda \to 0$ in the completion and $\xi = 0$. Hence $\widehat{j}$ is bijective. Its inverse is continuous by the universal property of completion, or directly because the topology on E was transported from X. □

Section recap

A seed family always gives a free presentation of its hull. The kernel stores all linear relations among addressed seed words. One seed gives a cyclic module, but only special annihilators permit an algebra quotient. Dense generation is handled by completing the presented image rather than by silently replacing algebraic span with closure.

4 Seed rank and functorial calculus

Reader roadmap

Seed rank is the minimal size of a finite generating family. This section establishes the operations that can only decrease it, the constructions that add or bound it, and the quotient witnesses that prove lower bounds.

4.1 Definition, cyclicity, and monotonicity

Definition 4.1 (Seed rank)**.** For a unital left A-module M, define

$$\operatorname{srank}_A(M) := \min\{|S| : AS = M\}.$$

For a topological module (X, τ), define

$$\operatorname{srank}_{\mathcal{A},\tau}(X) := \min\{|S| : \overline{\mathcal{A}S}^{\tau} = X\}.$$

Set $\operatorname{srank}_A(0) = 0$. When the value is finite, it is the minimal number of module generators.

Proposition 4.2 (Cyclic characterization)**.** *For $M \neq 0$,*

$$\operatorname{srank}_A(M) = 1 \iff M \text{ contains a cyclic vector.}$$

Proof. Both conditions say that there is $m \in M$ with $Am = M$. □

Proposition 4.3 (Monotonicity under enlargement of the acting algebra)**.** *Let $B \subseteq A \subseteq \operatorname{End}_{\Bbbk}(X)$ be unital operator algebras with the same identity. Then*

$$\boxed{\operatorname{srank}_A(X) \leq \operatorname{srank}_B(X).}$$

The same statement holds for topological seed rank when both algebras act continuously.

Proof. For every $S \subseteq X$, one has $BS \subseteq AS$. Hence every B-generating family is also an A-generating family. Taking closures gives the topological statement. □

Interpretation

Enlarging the acting algebra can only create more orbit directions, so it can only lower the number of seeds required. This is why every seed-rank statement must name the acting algebra. The same vector space may be cyclic for a large algebra and require many seeds for a smaller one.

4.2 Quotients, isomorphisms, and exact sequences

Theorem 4.4 (Quotient monotonicity)**.** *Let $\varphi : A \to B$ be a unital ring homomorphism and let*

$$\Phi : M \twoheadrightarrow N$$

be a surjective additive map satisfying

$$\Phi(am) = \varphi(a)\Phi(m) \qquad (a \in A,\ m \in M).$$

Then

$$\boxed{\operatorname{srank}_B(N) \leq \operatorname{srank}_A(M).}$$

If M and N are topological modules, Φ is continuous and surjective, and the actions are continuous, then

$$\operatorname{srank}_{B,\tau_N}(N) \leq \operatorname{srank}_{A,\tau_M}(M).$$

Proof. Let S generate M. For every $n \in N$, choose $m \in M$ with $\Phi(m) = n$ and write $m = \sum_i a_i s_i$. Then

$$n = \sum_i \varphi(a_i)\Phi(s_i),$$

so $\Phi(S)$ generates N.

For the topological statement, let S generate M densely. Given $n = \Phi(m)$, choose a net $m_\lambda \in AS$ converging to m. Continuity gives

$$\Phi(m_\lambda) \longrightarrow n,$$

and every $\Phi(m_\lambda)$ lies in $B\Phi(S)$. Thus $B\Phi(S)$ is dense in N. □

A quotient can strictly lower seed rank

Let $A = \mathbb{k}$ act by scalars on $M = \mathbb{k}^3$. Then $\operatorname{srank}_A(M) = 3$. Quotienting by a one-dimensional subspace gives a two-dimensional scalar module, whose seed rank is 2. Quotient monotonicity is therefore often strict.

Corollary 4.5 (Isomorphism invariance)**.** *An intertwining module isomorphism preserves algebraic seed rank. A topological intertwining isomorphism preserves topological seed rank.*

Proof. Apply theorem 4.4 to the isomorphism and its inverse. □

Theorem 4.6 (Short exact sequence bounds)**.** *Let*

$$0 \longrightarrow L \xrightarrow{\iota} M \xrightarrow{\pi} N \longrightarrow 0$$

be a short exact sequence of left A-modules. Then

$$\boxed{\operatorname{srank}_A(N) \le \operatorname{srank}_A(M) \le \operatorname{srank}_A(L) + \operatorname{srank}_A(N).}$$

Proof. The lower inequality is quotient monotonicity. For the upper inequality, choose generators $\ell_1, \dots, \ell_r$ of L and generators $n_1, \dots, n_s$ of N. Choose lifts $m_j \in M$ with $\pi(m_j) = n_j$. For $m \in M$, express

$$\pi(m) = \sum_{j=1}^{s} a_j n_j.$$

Then $m - \sum_j a_j m_j \in \iota(L)$ and is an A-linear combination of the $\iota(\ell_i)$. Hence the $r + s$ displayed vectors generate M. □

Why the hypotheses matter

For a short exact sequence, a generating family for M always maps to a generating family for the quotient. It does not automatically generate the submodule K. The lower bound involving K is therefore tied to the restriction or lifting hypotheses printed in the theorem. This is one place where a compact slogan can hide a real module-theoretic distinction.

4.3 Direct sums under shared and independent actions

Proposition 4.7 (Shared direct-sum bounds)**.** *If one algebra A acts diagonally on $M \oplus N$, then*

$$\boxed{\max\{\operatorname{srank}_A(M), \operatorname{srank}_A(N)\} \le \operatorname{srank}_A(M \oplus N) \le \operatorname{srank}_A(M) + \operatorname{srank}_A(N).}$$

Proof. The coordinate projections are intertwining quotients and give the lower bound. For the upper bound, embed generators of M as $(m_i, 0)$ and generators of N as $(0, n_j)$. □

Theorem 4.8 (Independent direct-sum formula)**.** *Let M be a left A-module and N a left B-module. Under the product-algebra action*

$$(a, b)(m, n) = (am, bn),$$

one has

$$\boxed{\operatorname{srank}_{A\times B}(M \oplus N) = \max\{\operatorname{srank}_A(M), \operatorname{srank}_B(N)\}.}$$

Proof. The coordinate projections give the lower bound. Let r be the maximum on the right. Pad shorter generating lists by zeros and write them as $(m_1, \dots, m_r)$ and $(n_1, \dots, n_r)$. For arbitrary $m \in M$ and $n \in N$, choose $a_i \in A$ and $b_i \in B$ with

$$m = \sum_i a_i m_i, \qquad n = \sum_i b_i n_i.$$

Then

$$(m, n) = \sum_{i=1}^{r} (a_i, b_i)(m_i, n_i),$$

so the r paired vectors generate. □

4.4 Tensor generation and lower-bound witnesses

Proposition 4.9 (Tensor-product bound)**.** *Let A and B be unital $\Bbbk$-algebras, let M be a left A-module, and let N be a left B-module. Under the natural $A \otimes_{\Bbbk} B$ action,*

$$\boxed{\operatorname{srank}_{A\otimes B}(M \otimes_{\Bbbk} N) \leq \operatorname{srank}_A(M) \operatorname{srank}_B(N).}$$

Proof. If $m_1, \ldots, m_r$ generate M and $n_1, \ldots, n_s$ generate N, then every simple tensor is a sum of vectors of the form

$$(am_i) \otimes (bn_j) = (a \otimes b)(m_i \otimes n_j).$$

Since simple tensors span the tensor product, the rs tensors $m_i \otimes n_j$ generate. □

Interpretation

The tensor-product bound is constructive. Tensor the chosen seed families and act independently in the two factors. Equality can fail because a single tensor seed may contain several useful components, or because the tensor-product algebra acquires additional relations. Later quantum applications should therefore cite the bound, not assume multiplicativity.

Boundary 4.10 (Tensor rank is not generally multiplicative). The inequality in proposition 4.9 can be strict. Equality requires additional hypotheses on the acting algebras and their module multiplicities and is not exported as a universal law.

Proposition 4.11 (Scalar-extension monotonicity)**.** *Let $K/\Bbbk$ be a field extension, let A be a unital $\Bbbk$-algebra, and let M be a finitely generated left A-module. Set*

$$A_K := K \otimes_{\Bbbk} A, \qquad M_K := K \otimes_{\Bbbk} M.$$

Then

$$\boxed{\operatorname{srank}_{A_K}(M_K) \leq \operatorname{srank}_A(M).}$$

Every generating family $S \subseteq M$ extends to the generating family $1 \otimes S \subseteq M_K$.

Proof. Let $S = \{s_1, \ldots, s_r\}$ generate M. Every pure tensor $\lambda \otimes m \in M_K$ has an expression

$$m = \sum_{i=1}^{r} a_i s_i,$$

so

$$\lambda \otimes m = \sum_{i=1}^{r} (\lambda \otimes a_i)(1 \otimes s_i).$$

Pure tensors span M_K, hence $1 \otimes S$ generates M_K over A_K. Taking the minimum over S proves the inequality. □

Why the hypotheses matter

The proposition is one-way. A smaller generating family found after extending scalars need not descend to the original field. Descent requires additional structure. The paper therefore exports monotonicity, not an unconditional equality.

4.5 Quotient witnesses for lower bounds

Definition 4.12 (Seed-rank witness)**.** A surjective intertwiner $\Phi : M \twoheadrightarrow N$ is a seed-rank witness of order r when $\operatorname{srank}(N) = r$. It certifies $\operatorname{srank}(M) \geq r$.

Proposition 4.13 (Scalar quotient witness)**.** *Let M be an A-module over a field $\Bbbk$. Suppose M has an r-dimensional quotient N on which A acts through a character*

$$\chi : A \longrightarrow \Bbbk.$$

Then

$$\boxed{\operatorname{srank}_A(M) \geq r.}$$

Proof. For every $n \in N$, the cyclic submodule An is contained in the one-dimensional space $\Bbbk n$. Therefore q seed images span a subspace of dimension at most q. Generating the r-dimensional quotient requires at least r seeds. Quotient monotonicity transfers the lower bound to M. □

Section recap

Seed rank is functorial under quotients and isomorphisms, subadditive across extensions, exactly additive for independent product actions, and bounded above under tensor products. Lower bounds come from quotients on which the action simplifies. Scalar extension cannot increase seed rank, but descent of a reduced seed family is a separate problem.

5 Generator number modulo the Jacobson radical

Reader roadmap

The radical contains the part of the algebra that cannot create new simple top-level directions. Nakayama's lemma therefore reduces the minimal-generator problem to the semisimple top. This is the main computational engine for finite-dimensional algebras.

5.1 Reduction to the module top

Classical Input 5.1 (Nakayama lemma)**.** Let A be a unital ring, let $J \subseteq \operatorname{Jac}(A)$, and let M be a finitely generated left A-module. If $M = N + JM$ for a submodule $N \leq M$, then $M = N$.

Theorem 5.2 (Generator number is detected modulo the Jacobson radical)**.** *Let A be a unital ring, let $J = \mathrm{Jac}(A)$, and let M be a finitely generated left A-module. A family $S \subseteq M$ generates M if and only if its image generates M/JM as an A/J-module. When A/J is semisimple Artinian, this quotient is the usual semisimple top; no semisimplicity of A/J is required for the general equality. Consequently,*

$$\boxed{\mathrm{srank}_A(M) = \mathrm{srank}_{A/J}(M/JM).}$$

Proof. If S generates M, its image generates every quotient. Conversely, suppose the image of S generates M/JM and put $N = AS$. Then

$$M = N + JM.$$

Nakayama's lemma gives $M = N$. Thus the same finite families generate before and after passage to the top, and the minimal cardinalities agree. □

Interpretation

The theorem says that nilpotent or radical directions can lengthen orbit chains but cannot reduce the number of independent top-level directions that must be seeded. Once the images of the seeds span M/JM, Nakayama's lemma forces the entire module to be generated.

A local nilpotent algebra

Let $A = \Bbbk[\varepsilon]/(\varepsilon^3)$ and $M = A^{\oplus 2}$. The radical is $J = (\varepsilon)$, and

$$M/JM \cong \Bbbk^2.$$

Therefore two seeds are necessary and sufficient. The extra nilpotent layers εM and $\varepsilon^2 M$ are generated automatically once the two top directions are seeded.

Corollary 5.3 (Local algebra formula)**.** *Let A be a local ring with Jacobson radical J and division-ring residue algebra $D = A/J$. For every finitely generated left A-module M,*

$$\boxed{\mathrm{srank}_A(M) = \dim_D(M/JM),}$$

where the dimension is the left D-dimension.

Proof. The A/J-module M/JM is a left vector space over D. Its minimal number of generators is its D-dimension. □

5.2 Semisimple multiplicity calculus

Classical Input 5.4 (Wedderburn–Artin decomposition)**.** Every semisimple Artinian ring is isomorphic to a finite product

$$A \cong \prod_{i=1}^{q} M_{d_i}(D_i)$$

of full matrix rings over division rings.

Lemma 5.5 (Regular-module multiplicities)**.** *Let*

$$A = \prod_{i=1}^{q} M_{d_i}(D_i)$$

and let $S_i = D_i^{d_i}$ *denote the natural simple left module for the* i*th factor, extended by zero across the other factors. Then, as a left* A*-module,*

$$\boxed{A \cong \bigoplus_{i=1}^{q} S_i^{\oplus d_i}.}$$

Proof. For one factor $M_d(D)$, let e_{jj} be the diagonal matrix units. The left regular module decomposes as

$$M_d(D) = \bigoplus_{j=1}^{d} M_d(D)e_{jj}.$$

Each minimal left ideal $M_d(D)e_{jj}$ is isomorphic to the natural column module D^d. Taking the direct product of factors gives the stated decomposition. □

Theorem 5.6 (Semisimple seed-rank formula over arbitrary division rings)**.** *Let*

$$A \cong \prod_{i=1}^{q} M_{d_i}(D_i)$$

be semisimple Artinian and let

$$M \cong \bigoplus_{i=1}^{q} S_i^{\oplus m_i}$$

be a finitely generated left A*-module. Then*

$$\boxed{\operatorname{srank}_A(M) = \max_{1 \le i \le q} \left\lceil \frac{m_i}{d_i} \right\rceil.}$$

If A *is any finite-dimensional algebra over a field, the same formula applied to the decomposition of* $M/J(A)M$ *over* $A/J(A)$ *computes* $\operatorname{srank}_A(M)$*.*

Proof. By corollary 3.3, M is generated by r elements exactly when it is a quotient of $A^{\oplus r}$. By lemma 5.5,

$$A^{\oplus r} \cong \bigoplus_{i} S_i^{\oplus r d_i}.$$

Every short exact sequence of modules over a semisimple ring splits. Therefore a quotient isomorphic to M exists if and only if

$$m_i \le r d_i \qquad \text{for every } i.$$

The smallest such r is the displayed maximum. For a finite-dimensional nonsemisimple algebra, theorem 5.2 reduces the calculation to the semisimple module $M/J(A)M$ over $A/J(A)$. □

Interpretation

For the factor $M_{d_i}(D_i)$, one free generator contributes d_i copies of the corresponding simple module. If the top of M contains m_i copies, at least $\lceil m_i/d_i \rceil$ free generators are required. The worst simple type controls the global answer.

Multiplicity consumes seed capacity

For $A = M_2(\mathbb{k})$ acting on $(\mathbb{k}^2)^{\oplus 3}$, the regular module contributes two copies of the natural simple module per generator. One generator cannot cover all three multiplicity directions. Two can, giving seed rank 2.

Corollary 5.7 (Algebraically closed multiplicity formula)**.** *If $\mathbb{k}$ is algebraically closed,*

$$A \cong \prod_i M_{d_i}(\mathbb{k}), \qquad M/JM \cong \bigoplus_i (\mathbb{k}^{d_i})^{\oplus m_i},$$

then

$$\operatorname{srank}_A(M) = \max_i \left\lceil \frac{m_i}{d_i} \right\rceil.$$

Proof. Take every $D_i = \mathbb{k}$ in theorem 5.6. □

Corollary 5.8 (Full matrix algebra)**.** *For the natural module $S = D^d$ over $M_d(D)$,*

$$\operatorname{srank}_{M_d(D)}(S) = 1.$$

More strongly, every nonzero vector of S is cyclic.

Proof. The formula gives $\lceil 1/d \rceil = 1$. Directly, let $v \neq 0$ and choose a coordinate $v_j \neq 0$. For any target $w = (w_i)$, the matrix whose jth column has entries $w_i v_j^{-1}$ and whose other columns vanish sends v to w. □

Corollary 5.9 (Multiplicity obstruction)**.** *For the diagonal action of $M_d(D)$ on $S^{\oplus m}$,*

$$\boxed{\operatorname{srank}_{M_d(D)}(S^{\oplus m}) = \left\lceil \frac{m}{d} \right\rceil.}$$

Proof. Apply theorem 5.6 to the single simple type with multiplicity m. □

Section recap

For finitely generated modules, the minimal number of seeds is detected on the Jacobson-radical quotient. In the semisimple top, each generator has a fixed multiplicity capacity d_i in the i-th simple type, and the largest capacity ratio determines the exact seed rank.

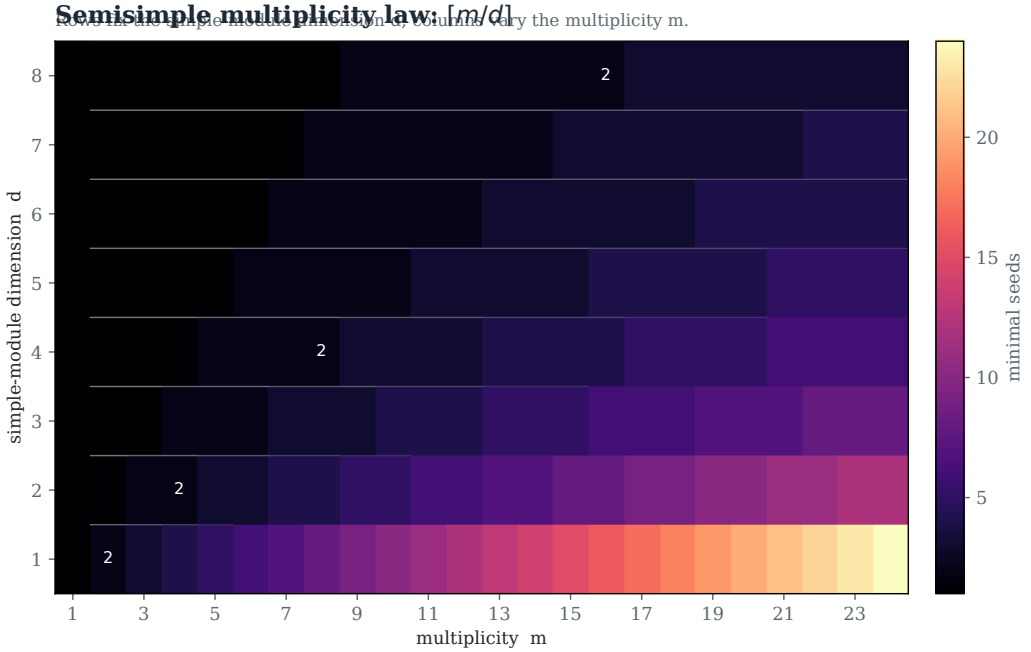


Figure 2. The semisimple multiplicity formula for one simple block. A seed can expose at most d independent multiplicity directions, so m copies require $\lceil m/d \rceil$ seeds.

6 Single-operator modules over arbitrary fields

Reader roadmap

One operator is equivalent to a module over $\Bbbk[t]$. Primary decomposition isolates the irreducible polynomial sectors, and the quotient by $p(T)$ counts how many cyclic chains begin in each sector.

6.1 Primary components and generator counts

Classical Input 6.1 (Primary decomposition for polynomial modules)**.** Every finite-dimensional $\Bbbk[t]$-module X decomposes as

$$X = \bigoplus_{p \in \mathrm{Irr}(\Bbbk[t])} X_p,$$

where only finitely many components are nonzero and every nonzero X_p is a finite direct sum of modules $\Bbbk[t]/(p^e)$.

Theorem 6.2 (Single-operator seed rank)**.** *Let X be finite-dimensional over $\Bbbk$ and let $T \in \mathrm{End}_{\Bbbk}(X)$. Regard X as a $\Bbbk[t]$-module by $t \cdot x = Tx$. For every monic irreducible p, let X_p be the p-primary component and set*

$$\mu_p(T) := \dim_{\Bbbk[t]/(p)} \big(X_p / p(T) X_p\big).$$

Then

$$\boxed{\mathrm{srank}_{\Bbbk[T]}(X) = \max_p \mu_p(T).}$$

Moreover, $\mu_p(T)$ is the number of cyclic p-primary summands in the elementary-divisor decomposition of X_p.

Proof. Let the minimal polynomial be

$$m_T(t) = \prod_p p(t)^{e_p}.$$

The acting algebra is

$$\Bbbk[T] \cong \Bbbk[t]/(m_T) \cong \prod_p R_p, \qquad R_p := \Bbbk[t]/(p^{e_p}),$$

by the Chinese remainder theorem. The factor R_p acts on X_p and annihilates every other primary component. Hence the independent direct-sum formula gives

$$\operatorname{srank}_{\Bbbk[T]}(X) = \max_p \operatorname{srank}_{R_p}(X_p).$$

Each R_p is local with maximal ideal generated by the class of p and residue division field $K_p = \Bbbk[t]/(p)$. By corollary 5.3,

$$\operatorname{srank}_{R_p}(X_p) = \dim_{K_p} X_p/p(T)X_p = \mu_p(T).$$

Finally, every summand $\Bbbk[t]/(p^e)$ contributes one copy of K_p after quotienting by p, so the dimension equals the number of elementary-divisor summands. □

Interpretation

For one operator, the quotient $X_p/p(T)X_p$ counts the number of independent primary chains for the irreducible polynomial p. Over an algebraic closure these become Jordan chains. The seed rank is the largest chain count among the primary sectors, not the largest chain length.

Block count versus block size

A nilpotent operator with Jordan block sizes $(5, 1, 1)$ has seed rank 3, even though its largest block has length 5. Three seeds are needed because there are three independent chains. Once their initial vectors are chosen, repeated application of T generates every position in those chains.

6.2 Invariant factors, Jordan blocks, and cyclicity

Corollary 6.3 (Invariant-factor count)**.** *If $X \neq 0$ and the rational canonical decomposition is written with nonconstant invariant factors*

$$X \cong \bigoplus_{j=1}^{r} \Bbbk[t]/(f_j), \qquad f_1 \mid f_2 \mid \cdots \mid f_r,$$

then

$$\boxed{\operatorname{srank}_{\Bbbk[T]}(X) = r.}$$

Proof. For each irreducible p, the number of invariant factors divisible by p equals the number of p-primary elementary divisors. Because $f_1 \mid \cdots \mid f_r$ and $f_1 \neq 1$ after zero summands are removed, some irreducible divisor of f_1 divides all r invariant factors. Hence $\max_p \mu_p(T) = r$. □

Corollary 6.4 (Geometric-multiplicity formula)**.** *If $\mathbb{k}$ is algebraically closed, then*

$$\boxed{\operatorname{srank}_{\mathbb{k}[T]}(X) = \max_{\lambda \in \operatorname{Spec}(T)} \dim_{\mathbb{k}} \operatorname{Ker}(T - \lambda I).}$$

Proof. The irreducibles are $p = t - \lambda$. On the λ-primary component, the quotient by $(T - \lambda I)X_\lambda$ has dimension equal to the number of Jordan blocks, which is also $\dim \operatorname{Ker}(T - \lambda I)$. □

Corollary 6.5 (Cyclic-matrix criterion)**.** *For finite-dimensional X, the following are equivalent:*

(i) *T has a cyclic vector.*

(ii) $\operatorname{srank}_{\mathbb{k}[T]}(X) = 1$.

(iii) *Every p-primary component is cyclic.*

(iv) *The rational canonical form has one invariant factor.*

(v) *The minimal polynomial equals the characteristic polynomial.*

(vi) *After scalar extension to an algebraic closure, every eigenvalue has geometric multiplicity one.*

Proof. The equivalence of (i) and (ii) is proposition 4.2. By theorem 6.2, (ii) holds exactly when every $\mu_p(T)$ equals one, which is (iii). Relatively prime primary cyclic modules combine into one cyclic module by the Chinese remainder theorem, so (iii) is equivalent to (iv). In rational canonical form, one invariant factor means that the unique invariant factor is both the minimal and characteristic polynomial, proving (iv)⇔(v). Scalar extension preserves the equality of minimal and characteristic polynomials. Over an algebraic closure, that equality is equivalent to one Jordan block for each eigenvalue, which is (vi). □

Section recap

A single operator is cyclic exactly when each primary sector has one cyclic chain. Over an algebraic closure this is equivalent to one Jordan block per eigenvalue and to equality of the minimal and characteristic polynomials.

7 Exact finite and countable packet coordinates

Reader roadmap

Packets are exact idempotent coordinates. The finite theory is purely algebraic. The countable theory requires an explicit convergence mode and a uniform bound. This section also adds packet support, tensor products, duality, and the universal mapping property of the packet decomposition.

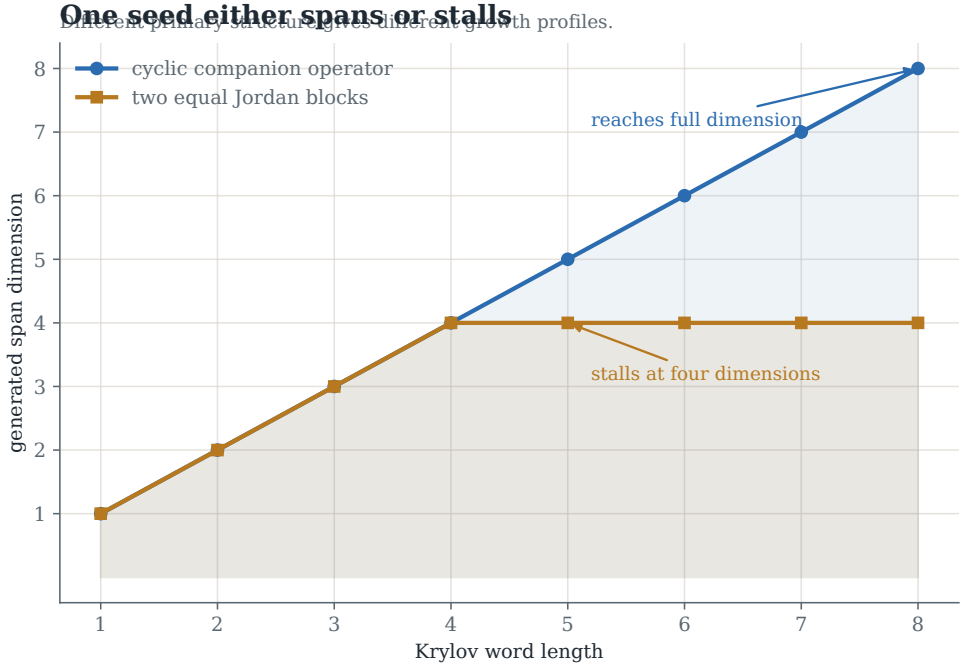


Figure 3. Krylov rank profiles for a cyclic companion operator and for two equal primary blocks. The second orbit stalls before reaching the ambient dimension because one seed cannot separate the repeated blocks.

7.1 Finite exact packets

Definition 7.1 (Finite packet system)**.** A finite packet system on a $\Bbbk$-vector space X is a family

$$\mathcal{P} = (P_\alpha)_{\alpha \in I} \subseteq \mathrm{End}_\Bbbk(X)$$

satisfying

$$P_\alpha^2 = P_\alpha, \qquad P_\alpha P_\beta = 0 \quad (\alpha \neq \beta), \qquad \sum_{\alpha \in I} P_\alpha = I_X.$$

Set $X_\alpha = P_\alpha X$. Zero packet spaces may be removed without changing any identity. No orthogonality is assumed unless stated.

Definition 7.2 (Packet support and packet-pure vectors)**.** For a finite packet system $\mathcal{P} = (P_\alpha)_{\alpha \in I}$ and $x \in X$, define

$$\mathrm{Supp}_{\mathcal{P}}(x) := \{\alpha \in I : P_\alpha x \neq 0\}.$$

The vector x is *packet-pure* if $|\mathrm{Supp}_{\mathcal{P}}(x)| \leq 1$. A finite family is packet-pure if each of its members is packet-pure, although different members may lie in different packets.

Interpretation

Packet support is coordinate support, not geometric support in the original domain. A seed may be localized in one packet, spread across many packets, or zero. The distinction becomes decisive in the source-component theorem because one mixed-support seed can meet several graph sources at once.

Lemma 7.3 (Finite packet aggregation)**.** *Let $\sigma : I \twoheadrightarrow J$ be a surjection and define*

$$Q_j := \sum_{\sigma(\alpha)=j} P_\alpha.$$

Then $(Q_j)_{j\in J}$ is a finite packet system and

$$Q_j X = \bigoplus_{\sigma(\alpha)=j} X_\alpha.$$

Proof. Pairwise annihilation of the P_α gives

$$Q_j^2 = Q_j, \qquad Q_j Q_k = 0 \quad (j \neq k).$$

Summing over j recovers $\sum_\alpha P_\alpha = I$. The range statement follows from the directness forced by pairwise annihilation. □

Definition 7.4 (Packet analysis and synthesis)**.** For a finite packet system, define

$$\mathcal{A}_\mathcal{P} : X \longrightarrow \bigoplus_{\alpha\in I} X_\alpha, \qquad x \longmapsto (P_\alpha x)_\alpha,$$

and

$$\mathcal{S}_\mathcal{P} : \bigoplus_{\alpha\in I} X_\alpha \longrightarrow X, \qquad (x_\alpha)_\alpha \longmapsto \sum_\alpha x_\alpha.$$

Theorem 7.5 (Exact packet-coordinate isomorphism)**.** *For every finite packet system,*

$$\boxed{\mathcal{S}_\mathcal{P}\mathcal{A}_\mathcal{P} = I_X, \qquad \mathcal{A}_\mathcal{P}\mathcal{S}_\mathcal{P} = I_{\oplus X_\alpha}.}$$

Hence

$$\boxed{X \cong \bigoplus_{\alpha\in I} X_\alpha.}$$

Proof. For $x \in X$,

$$\mathcal{S}_\mathcal{P}\mathcal{A}_\mathcal{P} x = \sum_\alpha P_\alpha x = x.$$

If $x_\alpha \in X_\alpha$, then $P_\beta x_\alpha = \delta_{\alpha\beta} x_\alpha$, so

$$\mathcal{A}_\mathcal{P}\mathcal{S}_\mathcal{P}(x_\alpha)_\alpha = \left(P_\beta \sum_\alpha x_\alpha\right)_\beta = (x_\beta)_\beta.$$

Thus analysis and synthesis are inverse linear maps. □

Key idea

Analysis records every packet coordinate and synthesis adds them back. The theorem is therefore an exact change of coordinates. Orthogonality is unnecessary for algebraic reconstruction. It matters only for norm, energy, and conditioning statements.

Corollary 7.6 (Packetwise equality)**.** *For $x, y \in X$,*

$$x = y \quad \Longleftrightarrow \quad P_\alpha x = P_\alpha y \text{ for every } \alpha.$$

Proof. The forward implication is immediate. For the reverse implication, sum the packet equalities and use $\sum_\alpha P_\alpha = I$. □

Corollary 7.7 (Universal mapping property of finite packet coordinates)**.** *Let $\mathcal{P} = (P_\alpha)_{\alpha \in I}$ be a finite packet system on X, and let Y be a $\Bbbk$-vector space.*

1. *Every family $f_\alpha : X_\alpha \to Y$ determines a unique linear map $f : X \to Y$ with $f|_{X_\alpha} = f_\alpha$, namely*

$$f = \sum_\alpha f_\alpha P_\alpha.$$

2. *Every family $g_\alpha : Y \to X_\alpha$ determines a unique linear map $g : Y \to X$ with $P_\alpha g = g_\alpha$, namely*

$$g = \sum_\alpha g_\alpha.$$

Equivalently,

$$\mathrm{Hom}(X, Y) \cong \prod_\alpha \mathrm{Hom}(X_\alpha, Y), \qquad \mathrm{Hom}(Y, X) \cong \prod_\alpha \mathrm{Hom}(Y, X_\alpha),$$

where the products are finite.

Proof. For $x = \sum_\alpha P_\alpha x$, any extension of the f_α must satisfy

$$f(x) = \sum_\alpha f_\alpha(P_\alpha x),$$

which proves both existence and uniqueness of the first formula. For the second, the sum $g = \sum_\alpha g_\alpha$ satisfies

$$P_\beta g = \sum_\alpha P_\beta g_\alpha = g_\beta$$

because $g_\alpha(Y) \subseteq X_\alpha$. Uniqueness follows from $\sum_\alpha P_\alpha = I$. □

7.2 Tensor products and dual packets

Theorem 7.8 (Tensor-product packet systems)**.** *Let $\mathcal{P} = (P_\alpha)_{\alpha \in I}$ be a finite packet system on X, and let $\mathcal{Q} = (Q_\gamma)_{\gamma \in J}$ be a finite packet system on Y. Then*

$$\mathcal{P} \boxtimes \mathcal{Q} := (P_\alpha \otimes Q_\gamma)_{(\alpha,\gamma) \in I \times J}$$

is a finite packet system on the algebraic tensor product $X \otimes_\Bbbk Y$, with packet spaces

$$(P_\alpha \otimes Q_\gamma)(X \otimes Y) = X_\alpha \otimes Y_\gamma.$$

Hence

$$\boxed{X \otimes Y \cong \bigoplus_{(\alpha,\gamma)\in I\times J} X_\alpha \otimes Y_\gamma.}$$

Proof. For every pair of packet indices,

$$(P_\alpha \otimes Q_\gamma)^2 = P_\alpha^2 \otimes Q_\gamma^2 = P_\alpha \otimes Q_\gamma.$$

If $(\alpha,\gamma) \neq (\beta,\delta)$, then either $P_\alpha P_\beta = 0$ or $Q_\gamma Q_\delta = 0$, so the corresponding tensor idempotents annihilate each other. Finally,

$$\sum_{\alpha,\gamma} P_\alpha \otimes Q_\gamma = \left(\sum_\alpha P_\alpha\right) \otimes \left(\sum_\gamma Q_\gamma\right) = I_X \otimes I_Y.$$

The range is generated by pure tensors $P_\alpha x \otimes Q_\gamma y$, hence equals $X_\alpha \otimes Y_\gamma$. Exact reconstruction follows from theorem 7.5. □

Proposition 7.9 (Dual packet systems)**.** *Let X be finite dimensional and let $\mathcal{P} = (P_\alpha)$ be a finite packet system. The algebraic dual maps*

$$P_\alpha^* : X^* \to X^*, \qquad P_\alpha^*(\varphi) = \varphi \circ P_\alpha,$$

form a finite packet system on X^*. Its α-packet is canonically isomorphic to X_α^*, and

$$X^* \cong \bigoplus_\alpha X_\alpha^*.$$

Proof. Because dualization reverses products,

$$P_\alpha^* P_\beta^* = (P_\beta P_\alpha)^* = \delta_{\alpha\beta} P_\alpha^*.$$

Also $\sum_\alpha P_\alpha^* = I_{X^*}$. Thus the dual maps form a packet system. Restriction gives a map from $P_\alpha^* X^*$ to X_α^*. It is injective because a covector in the range of P_α^* vanishes on every other packet, and it is surjective because any covector on X_α extends to X through the direct-sum decomposition. □

Tensoring the oblique packet split

Tensor the two oblique packets in Running Example R3 with the coordinate packets of $\mathbb{k}^3$. The result has six exact packets. No orthogonality is created by tensoring, but the analysis and synthesis identities remain exact because they are idempotent identities.

7.3 Banach stability and coordinate conditioning

Theorem 7.10 (Finite Banach packet isomorphism)**.** *Let X be Banach and let $(P_\alpha)_{\alpha\in I}$ be a finite family of bounded pairwise annihilating idempotents with sum I_X. Then every range $X_\alpha = P_\alpha X$ is closed. For any finite product norm equivalent to the maximum norm on $\bigoplus_\alpha X_\alpha$, packet analysis and synthesis are bounded inverse operators.*

Proof. The range of a bounded idempotent is its fixed-point space,

$$P_\alpha X = \operatorname{Ker}(I - P_\alpha),$$

which is closed. Because I is finite, analysis and synthesis are bounded for the maximum and sum norms, respectively, and therefore for every equivalent finite product norm. Their inverse identities are algebraic and were proved in theorem 7.5. □

Definition 7.11 (Packet-coordinate condition number)**.** Fix a Banach direct-sum norm on $\bigoplus_\alpha X_\alpha$. Define

$$\kappa(\mathcal{P}) := \|\mathcal{A}_\mathcal{P}\|\,\|\mathcal{S}_\mathcal{P}\|.$$

Proposition 7.12 (Conditioned coordinate transport)**.** *For a finite Banach packet system,*

$$\kappa(\mathcal{P}) \geq 1.$$

If $T \in \mathcal{B}(X)$ and

$$[T]_\mathcal{P} := \mathcal{A}_\mathcal{P} T \mathcal{S}_\mathcal{P},$$

then

$$\boxed{\|[T]_\mathcal{P}\| \leq \kappa(\mathcal{P})\|T\|, \qquad \|T\| \leq \kappa(\mathcal{P})\|[T]_\mathcal{P}\|.}$$

Proof. Since $I = \mathcal{S}_\mathcal{P}\mathcal{A}_\mathcal{P}$,

$$1 = \|I\| \leq \|\mathcal{S}_\mathcal{P}\|\|\mathcal{A}_\mathcal{P}\|.$$

The first displayed estimate follows from submultiplicativity. The inverse identity

$$T = \mathcal{S}_\mathcal{P}[T]_\mathcal{P}\mathcal{A}_\mathcal{P}$$

gives the second estimate. □

Interpretation

The quantity $\kappa(\mathcal{P})$ measures how strongly the chosen packet coordinates can amplify perturbations. An orthogonal Hilbert decomposition has the best possible behavior in the natural ℓ^2 sum norm. An oblique packet system may still be exact while having a large condition number.

Corollary 7.13 (Completion preserves a finite packet split)**.** *Let M be a dense normed subspace of a Banach space X. Suppose every bounded P_α leaves M invariant and the restricted projectors form a packet system on M. Then*

$$P_\alpha X = \overline{P_\alpha M}$$

and the packet decomposition of M extends uniquely to the completion X.

Proof. For $x \in X$, choose $m_n \in M$ with $m_n \to x$. Continuity gives $P_\alpha m_n \to P_\alpha x$, so $P_\alpha X \subseteq \overline{P_\alpha M}$. The reverse inclusion follows because $P_\alpha X$ is closed and contains $P_\alpha M$. The bounded extensions remain pairwise annihilating idempotents whose finite sum is the identity. □

7.4 Countable packets and unconditional convergence

Definition 7.14 (Unconditional packet sum space)**.** Let $(X_n)_{n\ge 1}$ be closed subspaces of a real or complex Banach space X. Define

$$\bigoplus_{n\ge 1}^{u} X_n$$

to be the space of sequences $x = (x_n)$ with $x_n \in X_n$ for which $\sum_n x_n$ converges unconditionally in X. Equip it with

$$\|x\|_u := \sup_{F\in\mathbb{N}} \sup_{|a_n|\le 1} \left\| \sum_{n\in F} a_n x_n \right\|.$$

Lemma 7.15 (Completeness of the unconditional packet sum)**.** *The normed space $\bigoplus_{n\ge 1}^{u} X_n$ in definition 7.14 is Banach.*

Proof. Let $x^{(k)} = (x_n^{(k)})$ be Cauchy in $\|\cdot\|_u$. For every fixed n,

$$\|x_n^{(k)} - x_n^{(\ell)}\| \le \|x^{(k)} - x^{(\ell)}\|_u,$$

so $x_n^{(k)}$ converges to some $x_n \in X_n$ because X_n is closed. Given $\varepsilon > 0$, choose k_0 such that

$$\|x^{(k)} - x^{(\ell)}\|_u < \varepsilon \qquad (k, \ell \ge k_0).$$

For finite F and $|a_n| \le 1$, pass $\ell \to \infty$ in the finite sum to obtain

$$\left\| \sum_{n\in F} a_n (x_n^{(k)} - x_n) \right\| \le \varepsilon \qquad (k \ge k_0).$$

Hence $\|x^{(k)} - x\|_u \le \varepsilon$ once x is known to be unconditionally summable.

Choose such a $k \ge k_0$. Since $x^{(k)}$ is unconditionally summable, there is a finite F_0 such that every finite $F \subseteq \mathbb{N} \setminus F_0$ and every $|a_n| \le 1$ satisfy

$$\left\| \sum_{n\in F} a_n x_n^{(k)} \right\| < \varepsilon.$$

The preceding uniform estimate then gives

$$\left\| \sum_{n\in F} a_n x_n \right\| < 2\varepsilon.$$

Thus the tails of $\sum_n x_n$ are uniformly small under bounded scalar changes, which is the unconditional Cauchy criterion. Therefore $x \in \bigoplus^u X_n$, and the earlier estimate proves $x^{(k)} \to x$ in $\|\cdot\|_u$. □

Theorem 7.16 (Countable unconditional packet isomorphism)**.** *Let X be a real or complex Banach space and let $(P_n)_{n\ge 1} \subseteq \mathcal{B}(X)$ satisfy*

$$P_n P_m = \delta_{nm} P_n$$

and

$$x = \sum_{n\geq 1} P_n x$$

unconditionally for every $x \in X$*. Assume*

$$K := \sup_{F\in\mathbb{N}} \sup_{|a_n|\leq 1} \left\| \sum_{n\in F} a_n P_n \right\| < \infty.$$

Then analysis and synthesis define inverse Banach-space isomorphisms

$$\boxed{X \cong \bigoplus_{n\geq 1}^{u} P_n X.}$$

More precisely,

$$\|\mathcal{A}_{\mathcal{P}}\| \leq K, \qquad \|\mathcal{S}_{\mathcal{P}}\| \leq 1.$$

Proof. Each $P_n X = \operatorname{Ker}(I - P_n)$ is closed. For $x \in X$,

$$\|\mathcal{A}_{\mathcal{P}} x\|_u = \sup_{F,a} \left\| \sum_{n\in F} a_n P_n x \right\| \leq K\|x\|.$$

For $y = (y_n) \in \bigoplus^u P_n X$, define

$$\mathcal{S}_{\mathcal{P}} y = \sum_n y_n.$$

Since the all-one finite partial sums are among those used in $\|y\|_u$,

$$\|\mathcal{S}_{\mathcal{P}} y\| \leq \|y\|_u.$$

The assumed expansion gives $\mathcal{S}_{\mathcal{P}}\mathcal{A}_{\mathcal{P}} x = x$. Conversely, boundedness of P_m permits termwise application to the unconditionally convergent series:

$$P_m \mathcal{S}_{\mathcal{P}} y = \sum_n P_m y_n = y_m.$$

Thus $\mathcal{A}_{\mathcal{P}}\mathcal{S}_{\mathcal{P}} = I$. Completeness of the coordinate space is lemma 7.15. □

Remark 7.17 (Uniform boundedness). If every expansion $\sum_n P_n x$ is unconditional, the family of signed finite partial-sum operators is pointwise bounded. The uniform boundedness principle therefore yields a finite constant K. It is retained explicitly in theorem 7.16 because it is the quantitative coordinate bound exported downstream.

Boundary 7.18 (Infinite packet sums). No identity $\sum_\alpha P_\alpha = I$ is used for an infinite family unless the convergence mode and the corresponding uniform boundedness statement are declared. Pointwise, strong, norm, weak, and unconditional convergence are not interchangeable.

Section recap

Finite packet systems are exact direct-sum coordinates with no orthogonality requirement. Their universal property permits arbitrary packetwise maps to be assembled uniquely. Tensor products

and duals preserve packet structure. In Banach spaces, conditioning is controlled by the analysis and synthesis norms. Countable packet systems require unconditional convergence and a uniform signed-partial-sum bound.

8 Transfer-block algebra and support patterns

Reader roadmap

Once vectors are resolved into packets, every operator becomes a matrix of maps between packet spaces. Operator multiplication becomes block convolution. Zero patterns become relations on the packet index set, and preorder patterns produce triangular algebras with nilpotent strict-transfer ideals.

8.1 Exact operator coordinates

Definition 8.1 (Transfer block)**.** For $T \in \mathrm{End}_{\Bbbk}(X)$ and a finite packet system $\mathcal{P}$, define

$$T_{\beta\alpha} := P_\beta T P_\alpha \in \mathrm{Hom}_{\Bbbk}(X_\alpha, X_\beta).$$

Theorem 8.2 (Exact operator reconstruction)**.** *Every $T \in \mathrm{End}_{\Bbbk}(X)$ satisfies*

$$\boxed{T = \sum_{\alpha,\beta \in I} T_{\beta\alpha}.}$$

Proof. Insert the finite packet identity on both sides:

$$T = \left(\sum_\beta P_\beta\right) T \left(\sum_\alpha P_\alpha\right) = \sum_{\alpha,\beta} P_\beta T P_\alpha.$$

□

Interpretation

Operator reconstruction inserts the identity on both sides of T. The source index α records where a vector begins, and the target index β records where the image lands. This is an exact operator matrix whose entries are themselves linear maps.

Theorem 8.3 (Packet matrix multiplication)**.** *For $S, T \in \mathrm{End}_{\Bbbk}(X)$,*

$$\boxed{(ST)_{\gamma\alpha} = \sum_{\beta \in I} S_{\gamma\beta} T_{\beta\alpha}.}$$

Proof. Insert $I = \sum_\beta P_\beta$ between S and T:

$$P_\gamma S T P_\alpha = \sum_\beta P_\gamma S P_\beta T P_\alpha.$$

□

Key idea

The intermediate packet index in block multiplication plays the role of a summed matrix index. Repeated multiplication therefore becomes a sum over packet paths. Paper II adds weights and filtrations to this path expansion. Paper III adds nonlinear trees.

Corollary 8.4 (Packet powers and paths)**.** *For $m \geq 1$,*

$$(T^m)_{\beta_m\beta_0} = \sum_{\beta_1,\dots,\beta_{m-1}} T_{\beta_m\beta_{m-1}} \cdots T_{\beta_1\beta_0}.$$

The same formula holds for a product of nonidentical operators with the corresponding blocks in temporal order.

Proof. Induct on m using theorem 8.3. The mixed-operator statement is the same induction with a different operator at each step. □

Corollary 8.5 (Adjoint reversal)**.** *If X is Hilbert, every P_α is orthogonal, and $T \in \mathcal{B}(X)$, then*

$$\boxed{(T_{\beta\alpha})^* = (T^*)_{\alpha\beta}.}$$

Proof. Because $P_\alpha^* = P_\alpha$,

$$(P_\beta T P_\alpha)^* = P_\alpha T^* P_\beta.$$

□

8.2 The full block algebra and intertwining equations

Corollary 8.6 (Tensor-product transfer blocks)**.** *Let $S \in \mathrm{End}(X)$ and $T \in \mathrm{End}(Y)$, with packet systems $\mathcal{P}$ and $\mathcal{Q}$. Relative to the tensor-product packet system of theorem 7.8,*

$$\boxed{(S \otimes T)_{(\beta,\delta),(\alpha,\gamma)} = S_{\beta\alpha} \otimes T_{\delta\gamma}.}$$

Proof. By definition,

$$\begin{aligned}(S \otimes T)_{(\beta,\delta),(\alpha,\gamma)} &= (P_\beta \otimes Q_\delta)(S \otimes T)(P_\alpha \otimes Q_\gamma) \\ &= (P_\beta S P_\alpha) \otimes (Q_\delta T Q_\gamma) \\ &= S_{\beta\alpha} \otimes T_{\delta\gamma}.\end{aligned}$$

□

Theorem 8.7 (Blockwise intertwining equations)**.** *Let $T \in \mathrm{End}(X)$, $S \in \mathrm{End}(Y)$, let $\mathcal{P} = (P_\alpha)_{\alpha\in I}$ and $\mathcal{Q} = (Q_\gamma)_{\gamma\in J}$ be finite packet systems, and let $U : X \to Y$. Define*

$$T_{\beta\alpha} = P_\beta T P_\alpha, \qquad S_{\gamma\delta} = Q_\gamma S Q_\delta, \qquad U_{\gamma\alpha} = Q_\gamma U P_\alpha.$$

Then

$$\boxed{UT = SU}$$

if and only if, for every $\gamma \in J$ *and* $\alpha \in I$,

$$\boxed{\sum_{\beta \in I} U_{\gamma\beta} T_{\beta\alpha} = \sum_{\delta \in J} S_{\gamma\delta} U_{\delta\alpha}.}$$

Proof. Apply Q_γ on the left and P_α on the right. Inserting $I_X = \sum_\beta P_\beta$ gives

$$Q_\gamma U T P_\alpha = \sum_\beta Q_\gamma U P_\beta T P_\alpha = \sum_\beta U_{\gamma\beta} T_{\beta\alpha}.$$

Inserting $I_Y = \sum_\delta Q_\delta$ gives

$$Q_\gamma S U P_\alpha = \sum_\delta Q_\gamma S Q_\delta U P_\alpha = \sum_\delta S_{\gamma\delta} U_{\delta\alpha}.$$

Thus $UT = SU$ implies every block equation. Conversely, if every block equation holds, then all blocks of $UT - SU$ vanish. Exact operator reconstruction yields $UT - SU = 0$. □

Interpretation

A global intertwiner is equivalent to a coupled finite system of packet equations. Later holonomy and representation-theoretic arguments use this theorem after local blocks have been constructed. The theorem does not construct those local blocks. It states the exact compatibility they must satisfy.

Theorem 8.8 (Full block-algebra representation)**.** *The map*

$$\Theta_{\mathcal{P}} : \mathrm{End}_{\Bbbk}(X) \longrightarrow \prod_{\beta,\alpha \in I} \mathrm{Hom}_{\Bbbk}(X_\alpha, X_\beta), \qquad T \longmapsto (T_{\beta\alpha})_{\beta,\alpha},$$

is an algebra isomorphism when the codomain is equipped with operator-valued block-matrix multiplication.

Proof. Injectivity follows from exact reconstruction. Given arbitrary maps $B_{\beta\alpha} : X_\alpha \to X_\beta$, define

$$T\left(\sum_\alpha x_\alpha\right) := \sum_{\alpha,\beta} B_{\beta\alpha} x_\alpha.$$

The packet decomposition is unique and the index set is finite, so this defines a linear operator whose blocks are the prescribed maps. Thus $\Theta_{\mathcal{P}}$ is surjective. Compatibility with multiplication is theorem 8.3. □

Corollary 8.9 (Propagation of packet support)**.** *Let* $T \in \mathrm{End}(X)$ *and* $x \in X$*. Then*

$$\boxed{\mathrm{Supp}_{\mathcal{P}}(Tx) \subseteq \{\beta : \exists \alpha \in \mathrm{Supp}_{\mathcal{P}}(x) \text{ with } T_{\beta\alpha} \neq 0\}.}$$

Equality holds whenever, for every target β in the displayed right-hand set, the active contributions

$$\{T_{\beta\alpha}P_\alpha x : \alpha \in \mathrm{Supp}_{\mathcal{P}}(x),\ T_{\beta\alpha} \neq 0\}$$

have nonzero sum. In particular, equality holds if at most one active source packet feeds each such β and that source coordinate does not lie in the kernel of the corresponding block.

Proof. For every target packet β,

$$P_\beta T x = \sum_{\alpha \in \mathrm{Supp}_{\mathcal{P}}(x)} T_{\beta\alpha}P_\alpha x.$$

If every block in this sum is zero, then the β-coordinate vanishes. This proves the inclusion. Equality is exactly the assertion that the displayed active sum is nonzero for every target in the right-hand set. The final sufficient condition leaves at most one nonzero summand at each target, and kernel avoidance makes that summand nonzero. □

Corollary 8.10 (Dimension count)**.** *If X is finite-dimensional and $d_\alpha = \dim X_\alpha$, then*

$$\dim \mathrm{End}_{\Bbbk}(X) = \sum_{\alpha,\beta} d_\alpha d_\beta = \left(\sum_\alpha d_\alpha\right)^2.$$

Proof. Take dimensions in the block-algebra isomorphism and use $\dim \mathrm{Hom}(X_\alpha, X_\beta) = d_\alpha d_\beta$. □

Proposition 8.11 (Block-diagonal projection and packet commutant)**.** *Define*

$$\Delta_{\mathcal{P}}(T) := \sum_\alpha P_\alpha T P_\alpha.$$

Then $\Delta_{\mathcal{P}}$ is a linear idempotent on $\mathrm{End}(X)$ whose range is the block-diagonal subalgebra. For $T \in \mathrm{End}(X)$, the following are equivalent:

(i) $TP_\alpha = P_\alpha T$ *for every* α.

(ii) $T_{\beta\alpha} = 0$ *whenever* $\beta \neq \alpha$.

(iii) $\Delta_{\mathcal{P}}(T) = T$.

Proof. Pairwise annihilation gives $\Delta_{\mathcal{P}}^2 = \Delta_{\mathcal{P}}$ and identifies its range with diagonal block families. If T commutes with all P_α, then for $\beta \neq \alpha$,

$$P_\beta T P_\alpha = P_\beta P_\alpha T = 0.$$

Thus (i) implies (ii). Exact reconstruction shows (ii) implies (iii). If $T = \sum_\alpha P_\alpha T P_\alpha$, then

$$TP_\gamma = P_\gamma T P_\gamma = P_\gamma T,$$

so (iii) implies (i). □

Upper-triangular support

Take three one-dimensional packets and permit blocks only from a lower-index source to a higher-index target, together with the diagonal. The support relation is a total preorder. The strict off-diagonal blocks form a nilpotent ideal, and every product of three strict transfers vanishes. This is the finite algebraic prototype for the weighted triangular inverses developed in Paper II.

8.3 Support patterns, preorders, and triangular structure

Definition 8.12 (Support relation and support space)**.** For a relation $R \subseteq I \times I$, with (β, α) interpreted as an allowed transfer $\alpha \to \beta$, define

$$\mathfrak{M}_R := \{T \in \mathrm{End}(X) : T_{\beta\alpha} = 0 \text{ whenever } (\beta, \alpha) \notin R\}.$$

Theorem 8.13 (Support-pattern algebra criterion)**.** *Assume $X_\alpha \neq 0$ for every $\alpha \in I$. Then:*

(i) $I_X \in \mathfrak{M}_R$ *if and only if* R *is reflexive.*

(ii) $\mathfrak{M}_R\mathfrak{M}_R \subseteq \mathfrak{M}_R$ *if and only if* R *is transitive.*

(iii) $\mathfrak{M}_R$ *is a unital subalgebra if and only if* R *is a preorder.*

Proof. The identity has diagonal blocks I_{X_α} and no off-diagonal blocks, so it belongs to $\mathfrak{M}_R$ exactly when $(\alpha, \alpha) \in R$ for every α.

Assume R is transitive. If a summand

$$S_{\gamma\beta}T_{\beta\alpha}$$

in the (γ, α) block of ST is nonzero, then (γ, β) and (β, α) lie in R. Transitivity gives $(\gamma, \alpha) \in R$, so every forbidden block of ST vanishes.

Conversely, suppose transitivity fails. Choose α, β, γ with

$$(\beta, \alpha) \in R, \qquad (\gamma, \beta) \in R, \qquad (\gamma, \alpha) \notin R.$$

Choose nonzero vectors $u \in X_\alpha$, $v \in X_\beta$, and $w \in X_\gamma$. Choose linear functionals $f \in X_\alpha^*$ and $g \in X_\beta^*$ with $f(u) = g(v) = 1$. Let T have the sole block $x \mapsto f(x)v$ from X_α to X_β, and let S have the sole block $y \mapsto g(y)w$ from X_β to X_γ. Then $S, T \in \mathfrak{M}_R$, but ST has a nonzero forbidden (γ, α) block. This proves necessity. Combining reflexivity and transitivity gives (iii). □

Corollary 8.14 ($*$-closed bounded support patterns)**.** *Let X be Hilbert, let the packet projectors be orthogonal, assume $X_\alpha \neq 0$ for every $\alpha \in I$, and put*

$$\mathfrak{M}_R^{\mathrm{b}} := \mathfrak{M}_R \cap \mathcal{B}(X).$$

Then $\mathfrak{M}_R^{\mathrm{b}}$ is closed under adjoints if and only if R is symmetric. Hence $\mathfrak{M}_R^{\mathrm{b}}$ is a unital $*$-subalgebra of $\mathcal{B}(X)$ if and only if R is an equivalence relation.

Proof. Adjoint reversal sends an allowed (β, α) block to an (α, β) block. Symmetry is therefore sufficient. If symmetry fails, choose a nonzero rank-one map in an allowed nonsymmetric block. Its adjoint occupies the forbidden reverse block, proving necessity. A reflexive, transitive, symmetric relation is an equivalence relation. □

Theorem 8.15 (Preorder support algebra decomposition)**.** *Let R be a preorder on the finite set I. Define*

$$\alpha \sim \beta \quad \Longleftrightarrow \quad (\beta, \alpha) \in R \text{ and } (\alpha, \beta) \in R.$$

Let $\overline{I} = I/\!\sim$, which carries the induced partial order, and put

$$X_C := \bigoplus_{\alpha \in C} X_\alpha.$$

Let $\mathfrak{N}_R \subseteq \mathfrak{M}_R$ consist of operators whose blocks vanish whenever source and target belong to the same equivalence class. Then:

(i) $\mathfrak{N}_R$ is a two-sided nilpotent ideal.

(ii) If h is the maximal number of strict steps in a chain of $\overline{I}$, then

$$\mathfrak{N}_R^{h+1} = 0.$$

(iii) The diagonal-class map induces an algebra isomorphism

$$\boxed{\mathfrak{M}_R/\mathfrak{N}_R \cong \prod_{C \in \overline{I}} \mathrm{End}_{\Bbbk}(X_C).}$$

Proof. A nonzero block in $\mathfrak{N}_R$ moves from one equivalence class to a strictly larger class in the quotient partial order. The block multiplication law shows that products concatenate such moves. Therefore left or right multiplication by an element of $\mathfrak{M}_R$ preserves the property of having no same-class blocks, so $\mathfrak{N}_R$ is a two-sided ideal. A product of $h+1$ elements of $\mathfrak{N}_R$ would require a strict chain of $h+1$ steps in $\overline{I}$, which does not exist. This proves nilpotence.

Let $P_C = \sum_{\alpha \in C} P_\alpha$. Define

$$\rho : \mathfrak{M}_R \longrightarrow \prod_C \mathrm{End}(X_C), \qquad \rho(T) = (P_C T P_C)_C.$$

If an intermediate class D contributes to $P_C S T P_C$, then the preorder contains transfers $C \to D$ and $D \to C$, forcing $C = D$ in the quotient poset. Hence

$$P_C S T P_C = (P_C S P_C)(P_C T P_C),$$

so ρ is an algebra homomorphism. Every collection of endomorphisms of the X_C defines an allowed block-diagonal-class operator, so ρ is surjective. Its kernel is exactly $\mathfrak{N}_R$. The first isomorphism theorem gives the quotient formula. □

Section recap

Every operator is exactly a packet matrix. Products sum over intermediate packet indices, tensor products factor blockwise, and global intertwiners are equivalent to coupled block equations. A support pattern is multiplicatively closed precisely when its relation is transitive. Preorder patterns have a block-diagonal quotient and a nilpotent strict-transfer ideal.

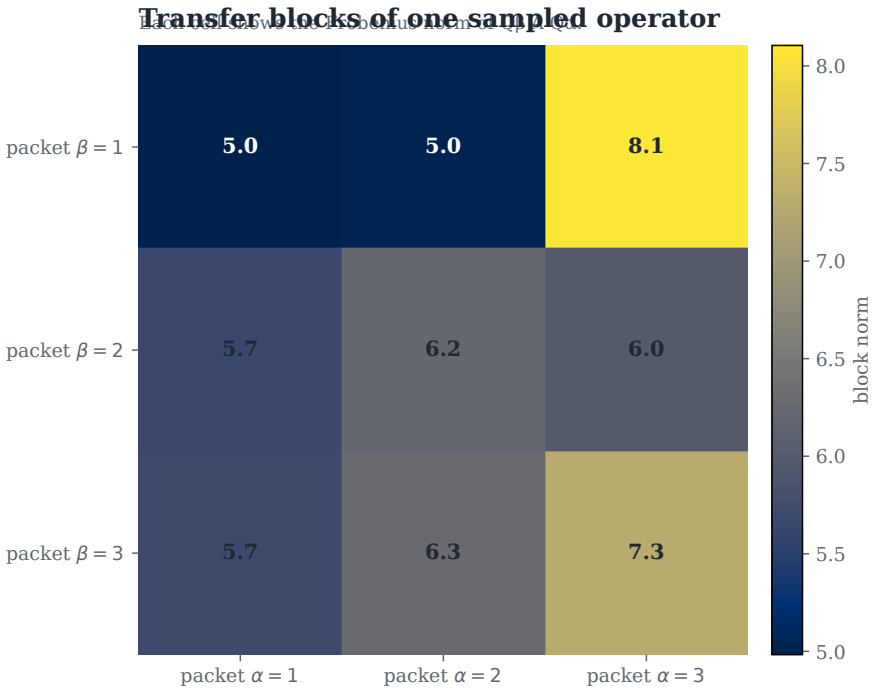


Figure 4. An operator represented by its packet-to-packet block norms. The diagonal entries preserve packets; off-diagonal entries transfer information between packets.

9 Transfer graphs and directed packet generation

Reader roadmap

The transfer graph records which packet-to-packet blocks occur in a declared generating family. Reachability constrains every word in that family. Under stronger propagation hypotheses, source strongly connected components exactly determine packet-pure seed rank.

9.1 Graphs generated by declared operators

Definition 9.1 (Transfer graph of an algebra-generating family)**.** Let $\mathcal{P} = (P_\alpha)_{\alpha \in I}$ be a finite packet system on X with every $P_\alpha \in \mathcal{A}$. Let $\mathcal{G} \subseteq \mathcal{A}$ satisfy

$$\mathcal{A} = \mathrm{alg}_{\mathbb{k}}(\{P_\alpha : \alpha \in I\} \cup \mathcal{G}),$$

where $\mathrm{alg}_{\Bbbk}(E)$ denotes the unital $\Bbbk$-subalgebra generated by E. The transfer graph $\Gamma(\mathcal{G}, \mathcal{P})$ has vertex set I and an arrow

$$\alpha \longrightarrow \beta$$

when

$$P_\beta G P_\alpha \neq 0$$

for at least one $G \in \mathcal{G}$. Reachability includes paths of length zero.

Theorem 9.2 (Reachability support of the generated algebra)**.** *For every $A \in \mathcal{A}$,*

$$P_\beta A P_\alpha = 0 \qquad \text{whenever } \beta \notin \mathrm{Reach}(\alpha).$$

Equivalently,

$$\mathcal{A} \subseteq \mathfrak{M}_{R_\Gamma}, \qquad R_\Gamma := \{(\beta, \alpha) : \beta \in \mathrm{Reach}(\alpha)\}.$$

Proof. It is enough to treat a word in the generators and packet projectors. Insert

$$I = \sum_{\gamma \in I} P_\gamma$$

between consecutive nonprojector generators. Every resulting (β, α) block is a finite sum of products

$$P_\beta G_m P_{\gamma_{m-1}} G_{m-1} \cdots P_{\gamma_1} G_1 P_\alpha.$$

A nonzero product requires every consecutive block to be nonzero, hence gives a directed path

$$\alpha \to \gamma_1 \to \cdots \to \gamma_{m-1} \to \beta.$$

Packet projectors only insert stationary steps. Thus a word has no block from α to an unreachable β. Linear combinations preserve the same zero pattern. □

Interpretation

The graph is attached to a declared generating family, not to every operator in $\mathrm{End}(X)$. Multiplying generators concatenates paths. Therefore a word in the generators cannot create a block outside graph reachability. The converse requires more than reachability because a nonzero edge block need not be surjective or generate the entire target packet.

9.2 Reachability obstructions for seeds

Proposition 9.3 (Reachability obstruction for generated hulls)**.** *Let $I_0 \subseteq I$ and let*

$$S \subseteq \bigoplus_{\alpha \in I_0} X_\alpha.$$

Then

$$\boxed{\mathrm{Hull}_{\mathcal{A}}(S) \subseteq \bigoplus_{\beta \in \mathrm{Reach}(I_0)} X_\beta.}$$

In particular, no seed supported outside a source strongly connected component can generate a nonzero vector in that source component.

Proof. For $s \in S$, write $s = \sum_{\alpha \in I_0} P_\alpha s$. If β is unreachable from I_0, then for every $A \in \mathcal{A}$,

$$P_\beta A s = \sum_{\alpha \in I_0} P_\beta A P_\alpha s = 0$$

by theorem 9.2. The same holds for every finite linear combination of orbit vectors. □

9.3 Packet-pure generation and source components

Definition 9.4 (Packet-pure seed rank)**.** A vector is packet-pure when it lies in one packet space X_α. Define

$$\operatorname{psrank}_{\mathcal{A},\mathcal{P}}(X) := \min\{|S| : S \text{ is packet-pure and } \mathcal{A}S = X\}.$$

For $C \subseteq I$, put

$$P_C := \sum_{\alpha \in C} P_\alpha, \qquad X_C := P_C X, \qquad \mathcal{A}_C := P_C \mathcal{A} P_C,$$

and define the local packet-pure rank

$$r_C := \operatorname{psrank}_{\mathcal{A}_C, \mathcal{P}|_C}(X_C).$$

Proposition 9.5 (Basic packet-pure rank bounds)**.** *For every packet-resolved A-module with $P_\alpha \in A$,*

$$\boxed{\operatorname{srank}_A(X) \le \operatorname{psrank}_{A,\mathcal{P}}(X) \le \sum_{\alpha \in I} \operatorname{srank}_{P_\alpha A P_\alpha}(X_\alpha).}$$

The right-hand corner algebra is unital on X_α with identity P_α.

Proof. Every packet-pure generating family is an unrestricted generating family, which gives the first inequality. For each α, choose a minimal generating family $S_\alpha \subseteq X_\alpha$ for the corner action $P_\alpha A P_\alpha$. The union $S = \bigcup_\alpha S_\alpha$ is packet-pure. Since every $x \in X$ has the unique decomposition $x = \sum_\alpha x_\alpha$ and every x_α lies in $(P_\alpha A P_\alpha) S_\alpha \subseteq AS$, the union generates X. Its cardinality is the displayed sum. □

Definition 9.6 (Edge-surjective packet system)**.** The packet-resolved module is edge-surjective relative to $\Gamma(\mathcal{G}, \mathcal{P})$ when, for every graph edge $\alpha \to \beta$, there exists $B \in \mathcal{A}$ such that the pure block

$$P_\beta B P_\alpha : X_\alpha \twoheadrightarrow X_\beta$$

is surjective.

Theorem 9.7 (Source-component formula for packet-pure seed rank)**.** *Assume I is finite, every packet space is nonzero, and the packet-resolved module is edge-surjective. Let $\mathfrak{C}_0$ be the set of*

source strongly connected components of the condensation graph of $\Gamma(\mathcal{G}, \mathcal{P})$*. Then*

$$\boxed{\operatorname{psrank}_{\mathcal{A},\mathcal{P}}(X) = \sum_{C \in \mathfrak{C}_0} r_C.}$$

Proof. Let S be a packet-pure generating family for X and fix a source component C. Seeds outside C cannot reach C by proposition 9.3. Moreover, for $s \in X_C$ and $A \in \mathcal{A}$,

$$P_C A s = (P_C A P_C) s \in \mathcal{A}_C s.$$

Thus the seeds of S lying in X_C must generate X_C under $\mathcal{A}_C$, and there are at least r_C of them. Summing over disjoint source components gives

$$\operatorname{psrank}_{\mathcal{A},\mathcal{P}}(X) \geq \sum_{C \in \mathfrak{C}_0} r_C.$$

For the reverse inequality, choose for each source component C a packet-pure family S_C of size r_C generating X_C under $\mathcal{A}_C$, and put $S = \bigcup_C S_C$. Then every source component lies in $\mathcal{A}S$. Traverse the finite condensation graph in a topological order. Suppose all components preceding a non-source component D have already been shown to lie in $\mathcal{A}S$. Choose an incoming condensation edge from such a component to D, represented by a graph edge $\alpha \to \beta$ with α already generated and $\beta \in D$. Edge-surjectivity gives a pure block

$$B_{\beta\alpha} : X_\alpha \twoheadrightarrow X_\beta.$$

Because $X_\alpha \subseteq \mathcal{A}S$ and $\mathcal{A}S$ is $\mathcal{A}$-invariant, surjectivity implies $X_\beta \subseteq \mathcal{A}S$. For any other vertex $\gamma \in D$, choose a directed path inside the strongly connected component from β to γ. Successive surjective edge blocks propagate the full packet space along the path, so $X_\gamma \subseteq \mathcal{A}S$. Hence $X_D \subseteq \mathcal{A}S$. Induction over the condensation graph gives $X = \mathcal{A}S$, proving the upper bound. □

Why the hypotheses matter

The source-component formula is deliberately stronger than a graph-theoretic lower bound. It requires local generation inside source components and enough edge transfer to propagate generation through the condensation graph. Without those hypotheses, a path can exist while its operator blocks reach only a proper subspace.

Definition 9.8 (Packet-propagating system)**.** An edge-surjective packet system is packet-propagating when every packet space X_α is cyclic under its diagonal corner algebra

$$P_\alpha \mathcal{A} P_\alpha.$$

Corollary 9.9 (Unweighted source-component count)**.** *Assume the packet-resolved module is packet-propagating and* $X_\alpha \neq 0$ *for every* $\alpha \in I$*. If* c_0 *denotes the number of source strongly connected components, then*

$$\boxed{\operatorname{psrank}_{\mathcal{A},\mathcal{P}}(X) = c_0.}$$

Proof. Fix a source component C and a vertex $\alpha \in C$. Choose a cyclic vector $s \in X_\alpha$ for $P_\alpha \mathcal{A} P_\alpha$. It generates all of X_α. Strong connectivity and edge-surjectivity then propagate the whole packet space to every vertex of C. Thus $r_C = 1$ for every source component. Apply theorem 9.7. □

Boundary 9.10 (Mixed-support seeds can merge source requirements). The source-component formulas concern packet-pure seeds. They do not give the unrestricted seed rank. For example, let

$$\mathcal{A} = \Bbbk \times \Bbbk, \qquad X = \Bbbk \oplus \Bbbk$$

with the coordinate action and the two coordinate packet projectors. The transfer graph has two source components and

$$\operatorname{psrank}_{\mathcal{A},\mathcal{P}}(X) = 2,$$

but the mixed vector $(1, 1)$ is cyclic, so

$$\operatorname{srank}_{\mathcal{A}}(X) = 1.$$

Why mixed seeds change the answer

Let the acting algebra be diagonal on $X = X_1 \oplus X_2$, with no transfer between the packets. Packet-pure generation requires one seed in each source component, so psrank = 2. A single mixed vector $x_1 + x_2$ may generate both one-dimensional packets under the diagonal action, so unrestricted seed rank can be 1.

Section recap

Graph reachability gives a universal zero-pattern restriction for the algebra generated by declared operators. Packet-pure generation is controlled by source components only after local cyclicity and edge propagation have been proved. Mixed-support seeds belong to a different optimization problem and may merge several source requirements.

10 Coordinate transport, morphisms, invariant exact sequences, and completion

Reader roadmap

This section answers how packet data move between spaces. Coordinate changes transport idempotents, morphisms transport blocks, invariant subspaces inherit and pass packet systems to quotients, and appropriate topological hypotheses preserve the same identities after completion.

10.1 Coordinate changes and packet-compatible morphisms

Theorem 10.1 (Coordinate transport). *Let $U : X \to Y$ be a vector-space isomorphism and let $\mathcal{P} = (P_\alpha)$ be a packet system on X. Define*

$$Q_\alpha := U P_\alpha U^{-1}.$$

Then $\mathcal{Q} = (Q_\alpha)$ is a packet system on Y. For $\widetilde{T} = UTU^{-1}$,

$$\boxed{Q_\beta \widetilde{T} Q_\alpha = U(P_\beta T P_\alpha)U^{-1}.}$$

Hence packet dimensions, zero patterns, and support preorders are invariant under coordinate transport. If a declared algebra-generating family $\mathcal{G}$ is transported to $U\mathcal{G}U^{-1}$, then its transfer graph and reachability relation are also preserved.

Proof. Conjugation preserves products and sums, so

$$Q_\alpha Q_\beta = U P_\alpha P_\beta U^{-1} = \delta_{\alpha\beta} Q_\alpha, \qquad \sum_\alpha Q_\alpha = I_Y.$$

The block identity is direct multiplication:

$$Q_\beta \widetilde{T} Q_\alpha = U P_\beta T P_\alpha U^{-1}.$$

All listed invariants are determined by these transported blocks. □

Interpretation

Coordinate transport by an isomorphism changes the packet realization without changing the abstract direct-sum structure. The transported projectors are conjugates, and the transported operator blocks are conjugates of the original blocks. This is the linear prototype for later gauge changes.

Definition 10.2 (Packet-compatible operator morphism)**.** Let $(X, \mathcal{A}, \mathcal{P})$ and $(Y, \mathcal{B}, \mathcal{Q})$ be packet-resolved operator modules with packet index sets I and J. A packet-compatible operator morphism is a triple

$$(\Phi, \varphi, \sigma)$$

consisting of a unital algebra homomorphism $\varphi : \mathcal{A} \to \mathcal{B}$, a linear map $\Phi : X \to Y$ satisfying

$$\Phi(Ax) = \varphi(A)\Phi(x),$$

and a map $\sigma : I \to J$ satisfying

$$Q_j \Phi = \Phi P_{\sigma^{-1}(j)}, \qquad P_{\sigma^{-1}(j)} := \sum_{\sigma(\alpha)=j} P_\alpha.$$

An empty fiber is interpreted as the zero projector.

Definition 10.3 (Morphism of seeded systems)**.** A morphism

$$(X, \mathcal{A}, S, \mathcal{P}) \longrightarrow (Y, \mathcal{B}, S', \mathcal{Q})$$

is a packet-compatible operator morphism satisfying the seed condition

$$\Phi(S) \subseteq \mathrm{Hull}_{\mathcal{B}}(S').$$

In the topological category all maps and actions are required to be continuous and the seed condition uses the topological hull when declared.

Theorem 10.4 (Naturality of generated hulls and transfer blocks)**.** *Let (Φ, φ, σ) be a packet-compatible operator morphism. For every subset $S \subseteq X$,*

$$\boxed{\Phi(\mathrm{Hull}_{\mathcal{A}}(S)) = \mathrm{Hull}_{\varphi(\mathcal{A})}(\Phi(S)) \subseteq \mathrm{Hull}_{\mathcal{B}}(\Phi(S)).}$$

For $T \in \mathcal{A}$ and $i, j \in J$,

$$\boxed{Q_j \varphi(T) Q_i \Phi = \sum_{\substack{\sigma(\beta)=j \\ \sigma(\alpha)=i}} \Phi P_\beta T P_\alpha.}$$

If Φ is surjective and S generates X, then $\Phi(S)$ generates Y under $\varphi(\mathcal{A})$ and hence under $\mathcal{B}$. If, in addition, the morphism is seeded, then S' generates Y.

Proof. Every element of $\Phi(\mathrm{Hull}_{\mathcal{A}}(S))$ is a finite sum of vectors

$$\Phi(As) = \varphi(A)\Phi(s),$$

so it belongs to $\mathrm{Hull}_{\varphi(\mathcal{A})}(\Phi(S))$. Conversely, every generator $\varphi(A)\Phi(s)$ equals $\Phi(As)$, giving equality.

For the block formula, use intertwining and packet compatibility twice:

$$\begin{aligned} Q_j \varphi(T) Q_i \Phi &= Q_j \varphi(T) \Phi P_{\sigma^{-1}(i)} \\ &= Q_j \Phi T P_{\sigma^{-1}(i)} \\ &= \Phi P_{\sigma^{-1}(j)} T P_{\sigma^{-1}(i)} \\ &= \sum_{\substack{\sigma(\beta)=j \\ \sigma(\alpha)=i}} \Phi P_\beta T P_\alpha. \end{aligned}$$

If Φ is surjective and $\mathrm{Hull}_{\mathcal{A}}(S) = X$, the hull equality gives $Y = \mathrm{Hull}_{\varphi(\mathcal{A})}(\Phi(S))$. If $\Phi(S) \subseteq \mathrm{Hull}_{\mathcal{B}}(S')$, then $\mathrm{Hull}_{\mathcal{B}}(\Phi(S)) \subseteq \mathrm{Hull}_{\mathcal{B}}(S')$, so S' generates Y. □

Key idea

Naturality is the statement that "analyze, apply the morphism, then compare packets" agrees with "transport the object first, then analyze." This is stronger than having two similar formulas in different domains.

Proposition 10.5 (Category of seeded packet systems)**.** *Packet-compatible operator morphisms compose, seeded morphisms compose, and identity morphisms satisfy both definitions. Thus packet-resolved operator modules form a category* **PktMod** *and seeded packet systems form a category* **SeedSys**.

Proof. Let $(\Phi, \varphi, \sigma) : (X, \mathcal{A}, \mathcal{P}) \to (Y, \mathcal{B}, \mathcal{Q})$ and $(\Psi, \psi, \tau) : (Y, \mathcal{B}, \mathcal{Q}) \to (Z, \mathcal{C}, \mathcal{R})$ be morphisms. Intertwining is preserved because

$$\Psi\Phi(Ax) = \Psi(\varphi(A)\Phi x) = (\psi \circ \varphi)(A)\Psi\Phi x.$$

For k in the target packet set,

$$\begin{aligned} R_k \Psi\Phi &= \Psi Q_{\tau^{-1}(k)} \Phi \\ &= \Psi\Phi \sum_{\tau(\sigma(\alpha))=k} P_\alpha. \end{aligned}$$

Thus the composite packet map is $\tau \circ \sigma$. If the morphisms are seeded, then

$$\Phi(S) \subseteq \mathrm{Hull}_{\mathcal{B}}(S'), \qquad \Psi(S') \subseteq \mathrm{Hull}_{\mathcal{C}}(S'').$$

By theorem 10.4,

$$\Psi\Phi(S) \subseteq \Psi(\mathrm{Hull}_{\mathcal{B}}(S')) \subseteq \mathrm{Hull}_{\mathcal{C}}(\Psi(S')) \subseteq \mathrm{Hull}_{\mathcal{C}}(S'').$$

Identity maps satisfy the definitions with identity algebra and packet maps. □

10.2 Invariant subspaces and quotient packets

Theorem 10.6 (Packet exact sequence for an invariant subspace)**.** *Let $K \leq X$ satisfy $P_\alpha K \subseteq K$ for every α. Define*

$$K_\alpha := P_\alpha K = K \cap X_\alpha$$

and

$$\overline{P}_\alpha(x + K) := P_\alpha x + K.$$

Then:

(i) The nonzero restrictions $P_\alpha|_K$ form a packet system on K.

(ii) The nonzero $\overline{P}_\alpha$ form a packet system on X/K.

(iii) For every α there is a short exact sequence

$$0 \longrightarrow K_\alpha \longrightarrow X_\alpha \longrightarrow \overline{P}_\alpha(X/K) \longrightarrow 0,$$

and

$$\overline{P}_\alpha(X/K) \cong X_\alpha/(X_\alpha \cap K) \cong (X_\alpha + K)/K.$$

(iv) The full exact sequence decomposes packetwise as the direct sum of these short exact sequences.

Proof. Restriction preserves idempotency, pairwise annihilation, and the identity sum on K. If $x \in K \cap X_\alpha$, then $x = P_\alpha x \in P_\alpha K$, proving $K_\alpha = K \cap X_\alpha$.

The quotient operator is well-defined because replacing x by $x + k$ changes $P_\alpha x$ by $P_\alpha k \in K$. The idempotent and annihilation identities descend, and the sum is the identity on X/K. The map $X_\alpha \to X/K$ has image $(X_\alpha + K)/K$ and kernel $X_\alpha \cap K$, giving the short exact sequence and the quotient isomorphisms. Exact packet coordinates identify the original sequence with the direct sum over α. □

Packet descent to a quotient

Let $X = X_1 \oplus X_2$ and let $K \subseteq X_1$ be a subspace. The quotient has packets X_1/K and X_2. Any operator preserving K descends with exactly the same block pattern, except that blocks entering or leaving the first packet are replaced by their induced quotient maps.

Theorem 10.7 (Packet descent through an invariant quotient)**.** *Under the hypotheses of theorem 10.6, the operators*

$$\overline{P}_\alpha(x + K) = P_\alpha x + K$$

are well-defined pairwise annihilating idempotents with sum $I_{X/K}$, and the quotient map satisfies

$$\boxed{\pi P_\alpha = \overline{P}_\alpha \pi.}$$

After zero packets are removed, they form the canonical quotient packet system.

Proof. All assertions are the quotient part of theorem 10.6. The intertwining identity follows directly from the definition:

$$\overline{P}_\alpha \pi(x) = P_\alpha x + K = \pi P_\alpha(x).$$

□

Corollary 10.8 (Restriction and quotient of transfer blocks)**.** *Under the hypotheses of theorem 10.6, let $T \in \mathrm{End}(X)$ preserve K. Then every block $T_{\beta\alpha}$ maps K_α into K_β, and the induced quotient operator $\overline{T}$ satisfies*

$$\boxed{\overline{T}_{\beta\alpha}\, \pi = \pi T_{\beta\alpha},}$$

where $\pi : X \twoheadrightarrow X/K$.

Proof. If $x \in K_\alpha$, then $Tx \in K$, so

$$T_{\beta\alpha} x = P_\beta T x \in K \cap X_\beta = K_\beta.$$

On the quotient,

$$\overline{T}_{\beta\alpha} \pi = \overline{P}_\beta \overline{T P}_\alpha \pi = \pi P_\beta T P_\alpha.$$

□

Corollary 10.9 (Topological packet exact sequence)**.** *If X is Banach, the P_α are bounded, and K is closed and packet invariant, then every packet sequence in theorem 10.6 is a short exact sequence of Banach spaces with bounded maps, and the quotient packets are bounded.*

Proof. Closedness of K makes X/K Banach. The restricted projectors are bounded, and the quotient projectors are bounded because

$$\|\overline{P}_\alpha(x + K)\|_{X/K} \le \|P_\alpha\|\, \|x + K\|_{X/K}.$$

The packet spaces are closed ranges of bounded idempotents. □

10.3 Dense images, quotient topology, and completion

Theorem 10.10 (Dense realization theorem)**.** *Let X and Y be Hausdorff topological vector spaces, let $\Phi : X \to Y$ be continuous, surjective, and φ-intertwining, and let S topologically generate X. Then $\Phi(S)$ topologically generates Y. If finite continuous packet systems satisfy*

$$Q_j \Phi = \Phi P_{\sigma^{-1}(j)},$$

then

$$\boxed{Q_j Y = \Phi(P_{\sigma^{-1}(j)} X)}$$

for every j.

Proof. Given $y = \Phi(x)$, choose a net $x_\lambda \in \mathrm{Hull}_{\mathcal{A}}(S)$ converging to x. Then

$$\Phi(x_\lambda) \to y,$$

and naturality places every $\Phi(x_\lambda)$ in $\mathrm{Hull}_{\mathcal{B}}(\Phi(S))$. Hence the latter hull is dense. For packets, one inclusion follows from compatibility. Conversely, if $y = Q_j y$, choose x with $\Phi(x) = y$. Then

$$y = Q_j \Phi(x) = \Phi P_{\sigma^{-1}(j)} x.$$

□

Theorem 10.11 (Quotient topology and completed realization)**.** *Let $\Phi : X \twoheadrightarrow Y$ be a continuous linear quotient map between Hausdorff topological vector spaces, and put $K = \operatorname{Ker} \Phi$. Then the induced map*

$$\widetilde{\Phi} : X/K \longrightarrow Y$$

is a topological isomorphism. If $D \subseteq X$ is a linear subspace whose image $\Phi(D)$ is dense in a complete space Y, then the completion of

$$D/(D \cap K)$$

in the topology transported from Y is canonically isomorphic to Y.

Proof. The algebraic first isomorphism theorem gives a bijection $\widetilde{\Phi}$. By definition of the quotient topology, a set $U \subseteq X/K$ is open exactly when its preimage in X is open. Because Φ is a quotient map, this is equivalent to $\widetilde{\Phi}(U)$ being open in Y. Thus $\widetilde{\Phi}$ is a homeomorphism. The second statement applies the dense completion argument of theorem 3.6 to the injective map $D/(D \cap K) \to Y$. □

Boundary 10.12 (Continuous surjections and quotient topology). Continuity and surjectivity transport density as in theorem 10.10. They do not by themselves identify the given topology on Y with the quotient topology from X. The stronger conclusion in theorem 10.11 requires the quotient-map hypothesis or an equivalent open-mapping condition.

Do not conflate

A continuous surjective linear map need not be a quotient map in arbitrary topological vector spaces. The algebraic first isomorphism theorem gives a bijection on the quotient. A topological isomorphism requires the quotient topology to agree with the target topology. Banach-space open mapping theorems often provide this automatically, but the hypothesis must still be named.

Section recap

Packet systems transport by conjugation, form a category under packet-compatible seeded morphisms, restrict to invariant subspaces, descend to invariant quotients, and survive completion under explicit boundedness and quotient-topology hypotheses.

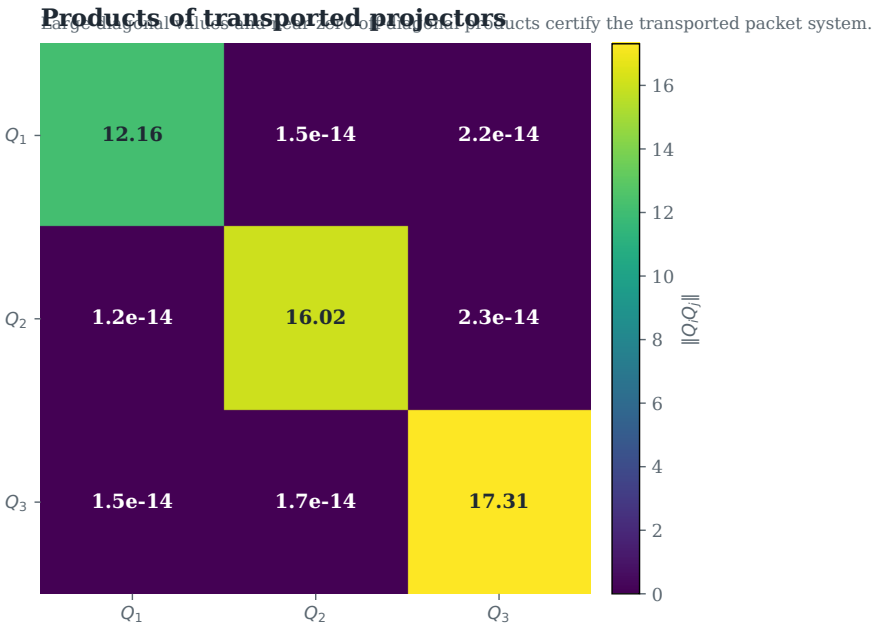


Figure 5. Numerical certificate for a nonorthogonal transported packet system. The diagonal products retain idempotent size while off-diagonal products remain at roundoff level.

11 Realization functors and semantic theorem transport

Reader roadmap

A later domain is a genuine realization only when the relevant structures and identities are preserved. The purpose of this section is to replace informal analogy by a precise preservation profile.

11.1 Preservation profiles

Definition 11.1 (Realization functor and preservation profile)**.** Let $\mathbf{SeedSys}_{\mathrm{abs}}$ be an abstract category of seed systems and let $\mathbf{C}_D$ be a domain category. A realization functor is a functor

$$\mathsf{Ev}_D : \mathbf{SeedSys}_{\mathrm{abs}} \longrightarrow \mathbf{C}_D$$

with a declared preservation profile specifying which of the following structures are preserved:

$$0,\ I,\ \text{finite sums},\ \text{composition},\ \text{biproducts},$$
$$\text{kernels},\ \text{cokernels},\ \text{quotients},\ \text{tensor products},$$
$$\text{packet idempotents},\ \text{transfer blocks},\ \text{topological completions}.$$

Preservation of a structure means preservation of its defining identity or universal property, not merely a similarity of notation.

Theorem 11.2 (Equational realization principle)**.** *Let $F : \mathbf{C} \to \mathbf{D}$ be an additive functor between preadditive categories. Any equality of morphisms constructed from declared generating morphisms by finite sums, additive inverses, identities, zero maps, and composition is preserved by F. Scalar multiplication by a coefficient ring is also preserved when the categories are enriched over that ring and F is linear for the declared enrichment. If a proof additionally uses a chosen biproduct, kernel, cokernel, quotient, tensor product, or completion and F preserves that construction in its declared profile, then the corresponding proof diagram is transported to $\mathbf{D}$.*

Proof. Functoriality gives

$$F(I) = I, \qquad F(g \circ f) = F(g) \circ F(f),$$

and additivity gives

$$F(0) = 0, \qquad F(f + g) = F(f) + F(g), \qquad F(-f) = -F(f).$$

If the declared enrichment is present, linearity additionally gives $F(af) = aF(f)$ for every coefficient a. A structural induction on the construction of a morphism expression therefore carries every permitted equational proof to its image under F. If a step invokes a universal construction, preservation means that the image diagram satisfies the same universal property. The uniquely induced morphism in the source proof is therefore carried to the uniquely induced morphism in the target proof. Repeating this argument through the finite proof diagram transports the conclusion. □

Interpretation

The equational principle covers identities built from operations the functor actually preserves. Additive identities require additivity. Scalar-weighted identities require linear enrichment. Quotients, tensor products, completions, kernels, or limits require their own preservation assumptions. The preservation profile is therefore part of the theorem, not metadata that may be omitted.

11.2 From equation transport to semantic realization

Proposition 11.3 (Realization export principle)**.** *Every canonical Paper I identity whose proof uses only the operations listed in a realization functor's verified preservation profile remains valid after applying that functor. If the proof uses a domain-specific hypothesis not encoded by the functor, that hypothesis must be verified separately before the transported statement is promoted to a theorem.*

Proof. The first assertion is theorem 11.2. The second is a logical boundary: a functor transports established identities and preserved universal constructions, but it does not create an unverified analytic, topological, positivity, finiteness, or integrality hypothesis. □

Definition 11.4 (Semantic realization criterion)**.** A domain theorem is a semantic realization of a canonical seed-calculus theorem only when:

(i) an explicit realization functor or intertwining construction is given;

(ii) the preservation profile covers every construction used in the canonical proof;

(iii) every additional analytic, finiteness, convergence, positivity, or integrality hypothesis is verified in the domain;

(iv) the resulting conclusion is stated in the domain's native objects.

Reuse of the words "seed", "packet", or "transfer" without these data is only syntactic reuse.

Do not conflate

A change of nouns is not a realization theorem. Calling a frequency band, a ramification sector, and a symmetry sector a "packet" is only terminology until one exhibits the idempotents, proves reconstruction, identifies the transfer blocks, and checks that the relevant morphisms or limits preserve those structures.

Section recap

Semantic theorem transport is controlled by a declared preservation profile. Equations transport only through operations preserved by the realization. This gives later papers a test for genuine cross-domain transfer rather than parallel notation.

12 Canonical export interface and claim boundary

Reader roadmap

The purpose of an export interface is to prevent later papers from repeating the foundational proofs. Each downstream paper should cite the stable record ID, add its domain hypotheses, and prove only the new specialization-specific step.

Definition 12.1 (Worked realization interface)**.** The foundational calculus is illustrated by four canonical algebraic realizations:

(i) a typed quiver reachability model;

(ii) an algebraic coordinate-ring presentation;

(iii) a companion and repeated-primary single-operator model;

(iv) path and cycle adjacency models.

The worked-model sections below instantiate these four interfaces. A bilateral-shift example records the analytic completion mechanism also used in Paper V. Computational specifications connect the proofs with the verification companion and are not premises of the foundational theorems.

Definition 12.2 (Paper I export object)**.** The canonical Paper I export is

$$\mathsf{PSC}_I = \left(\begin{array}{c} \text{typed grammars and realized operator algebras,} \\ \text{seed systems, free presentations, and seed rank,} \\ \text{module-top, semisimple, and single-operator formulas,} \\ \text{finite and unconditional packet coordinates,} \\ \text{transfer blocks and support algebras,} \\ \text{reachability and packet-pure source generation,} \\ \text{morphisms, invariant exact sequences, quotients, completions,} \\ \text{realization functors and equational theorem transport} \end{array} \right).$$

12.1 Cross-paper use map

Paper I export	Primary users	Later role
Typed grammar and module equivalence	II–XII	Gives each domain a legal operator syntax and a concrete module realization.
Free seed presentation	IV, IX, X	Converts generated families into quotient presentations and relation modules.
Jacobson-top seed rank	VII, X, XII	Computes minimal generators from semisimple multiplicity data.
Single-operator formula	IV, VI, IX	Resolves recurrence, companion, and spectral-chain generator counts.
Exact packet coordinates	II–XII	Supplies analysis, synthesis, and packetwise equality.
Tensor and dual packet systems	V, VII, VIII, XI	Controls product systems, observables, and paired representations.
Transfer-block multiplication	II–XII	Converts operator composition into sums over packet paths.
Block intertwining equations	III, VII, XI	Expresses global equivalence or gluing as local packet equations.
Support-preorder algebra	II, VIII, X	Produces triangular transfer structure and nilpotent strict-transfer ideals.
Source-component seed rank	VIII, X	Identifies the packet-pure source requirements under propagation hypotheses.
Packet exact sequences and quotient descent	III, VI, X, XII	Carries packets through constraints, reductions, and deformation quotients.
Semantic realization criterion	IV–XII	Specifies when a domain theorem is a true realization of the common calculus.

Boundary 12.3 (Exact limits of Paper I)*.* The existence of an exact packet decomposition does not imply that the decomposition is computationally discoverable, numerically well conditioned, statistically identifiable, dynamically invariant, or mathematically useful. The condition number

in definition 7.11 quantifies only coordinate stability once the packets are supplied. Refinement towers, filtered transfer, nonlinear obstruction objects, automatic packet recovery, continuum limits, arithmetic lifts, and automorphic occupancy require additional theorems in later papers.

13 A typed quiver model

The path-algebra language can be seen in a small model that also displays a genuine missing-packet obstruction.

Let the quiver have vertices U, V and arrows

$$R : U \to U, \qquad A : U \to V, \qquad B : V \to U,$$

with the path-algebra relation

$$BA = e_U,$$

whose realization is the operator identity $BA = I_U$. Realize

$$U = \Bbbk^2, \qquad V = \Bbbk^3,$$

by

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad A(u_1, u_2) = (u_1, u_2, 0), \qquad B(v_1, v_2, v_3) = (v_1, v_2).$$

Starting from the seed $e_1 \in U$, the R-orbit spans U, and A carries it onto the first two coordinates of V. Any path from U to V ends with A and therefore lands in $\operatorname{im} A$. The third coordinate, which is $\ker B$, is unreachable from a seed supported only at U. Therefore the generated representation is

$$U \oplus \operatorname{span}\{(1,0,0), (0,1,0)\} \subsetneq U \oplus V.$$

Adding the seed $(0,0,1) \in V$ generates the full representation.

Proposition 13.1 (Typed reachability certificate)**.** *In the preceding quiver realization, no seed supported at U can generate the full representation, whereas the two seeds*

$$e_1 \in U, \qquad e_3 \in V$$

do generate it.

Proof. The first claim follows because every path that starts at U and ends in V has final arrow A, so its image lies in $\operatorname{im} A$. For the second, e_1 and Re_1 span U; their A-images span $\operatorname{im} A$; and e_3 spans $\ker B$, the missing complementary direction in V. □

This example shows why a typed grammar is more than a notational convenience. The obstruction lies in the subspace $\ker B \subset V$, which is unreachable from seeds supported at U, and is visible before any coordinate calculation.

14 Worked algebraic model: an elliptic coordinate ring

Let $\operatorname{char}\Bbbk \neq 2$ and let

$$A = \Bbbk[x, y]/(y^2 - x^3 + x).$$

The affine equation $y^2 = x^3 - x$ has classical discriminant 64, which is nonzero under this hypothesis; standard algebraic geometry then identifies its smooth projective completion as an elliptic curve [9]. The Lean development checks the algebraic implication $2 \neq 0 \Rightarrow 64 \neq 0$, the quotient normal form, and the operator identities below. It does not formalize projective smoothness, genus, or the geometric identification of the discriminant. The packet computation itself uses only the displayed quotient-ring relation. Every class has a unique normal form

$$f(x, y) = a(x) + yb(x).$$

Indeed, $y^2 - (x^3 - x)$ is monic of degree two in y, so division in $\Bbbk[x][y]$ gives a unique remainder of degree less than two in y. Concretely, every power y^m with $m \geq 2$ can be reduced using

$$y^2 = x^3 - x.$$

14.1 One module seed and two address operators

Take the seed

$$s = 1$$

and let M_x, M_y be multiplication by x and y. Then

$$M_x^i M_y^j s = x^i y^j,$$

so the algebra, regarded as a module over its multiplication algebra, is generated by one seed.

The two packet spaces are

$$A_0 = \Bbbk[x], \qquad A_1 = y\Bbbk[x].$$

Let

$$P_0(a + yb) = a, \qquad P_1(a + yb) = yb.$$

These are $\Bbbk$-linear pairwise annihilating idempotents and satisfy $P_0 + P_1 = I$. They are not A-linear module maps: multiplication by y exchanges the two packets.

Identify the packet coefficients by

$$A_0 \oplus A_1 \longrightarrow \Bbbk[x] \oplus \Bbbk[x], \qquad (a, yb) \longmapsto (a, b).$$

In these ordered coefficient coordinates, with M_x and M_{x^3-x} denoting multiplication on $\Bbbk[x]$,

$$M_x = \begin{pmatrix} M_x & 0 \\ 0 & M_x \end{pmatrix}, \qquad M_y = \begin{pmatrix} 0 & M_{x^3-x} \\ I & 0 \end{pmatrix}.$$

The relation is reconstructed as an operator identity:

$$M_y^2 = M_x^3 - M_x.$$

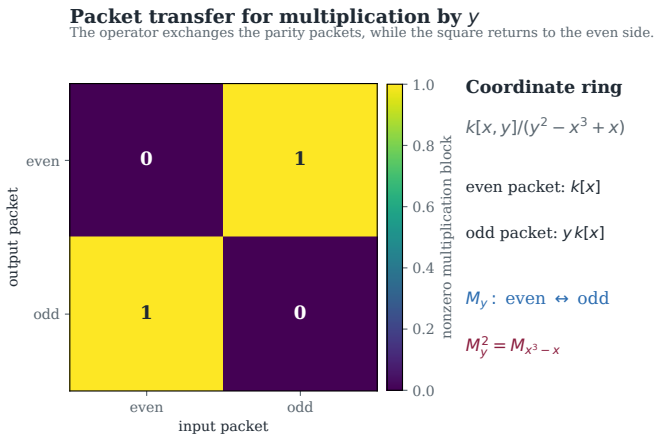


Figure 6. Multiplication by y exchanges the even and odd packets. The odd-to-even transfer carries the relation weight $x^3 - x$.

14.2 Exact product law

Write

$$f = a_0 + ya_1, \qquad g = b_0 + yb_1.$$

Then

$$fg = (a_0b_0 + (x^3 - x)a_1b_1) + y(a_0b_1 + a_1b_0).$$

This is a two-packet convolution law. It is already visible at the linear operator level because multiplication by a fixed element is a transfer operator.

Example 14.1 (Worked reduction)**.** Take

$$f = x^2 + y, \qquad g = x - y.$$

Then

$$\begin{aligned} fg &= x^3 - x^2y + xy - y^2 \\ &= x^3 - x^2y + xy - (x^3 - x) \\ &= x + (x - x^2)y. \end{aligned}$$

Thus

$$P_0(fg) = x, \qquad P_1(fg) = (x - x^2)y.$$

The symbolic certificate in the companion code performs the same quotient reduction exactly.

14.3 What this example establishes

The coordinate ring illustrates four distinct parts of the calculus:

(i) the unit is a module seed;

(ii) M_x and M_y are address operators;

(iii) the curve equation is an operator relation;

(iv) parity in y gives exact packet analysis, transfer, and reconstruction.

It does not imply that every algebra has a one-generator algebra presentation. It shows instead that unital algebras are cyclic as modules over their full multiplication algebra.

15 Worked signal model: a bilateral shift

Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and let

$$X = \ell^2(\mathbb{Z}; \mathbb{F})$$

with the explicit integer-indexed family of bounded reindexing operators

$$(S_m x)[n] = x[n - m] \qquad (m, n \in \mathbb{Z}).$$

Thus $S_0 = I$, $S_{m+r} = S_m S_r$, and S_1 is the usual bilateral shift. We use the standard Hilbert-space and bounded-operator conventions of [3]. The impulse δ_0 satisfies

$$S_m \delta_0 = \delta_m$$

for every $m \in \mathbb{Z}$. Therefore finite-support sequences lie in its algebraic orbit hull and are dense in $\ell^2(\mathbb{Z})$.

Proposition 15.1 (Shift seed)**.** *For $n \in \mathbb{N}$, let $G(n)$ mean that some n-element finite seed set has dense hull under the declared family $\{S_m : m \in \mathbb{Z}\}$. Then*

$$1 = \min\{n \in \mathbb{N} : G(n)\}.$$

Equivalently, one seed suffices and no empty seed family generates densely.

Proof. The orbit of δ_0 contains every standard basis vector. Their finite linear span is dense, so $G(1)$ holds. The hull of the empty seed set is zero, while $\ell^2(\mathbb{Z}; \mathbb{F})$ is nonzero, so $G(0)$ fails. Therefore 1 is the least admissible finite cardinality. □

For $N \geq 1$, define residue projectors

$$(P_{N,j} x)[n] = \begin{cases} x[n], & n \equiv j \pmod{N}, \\ 0, & \text{otherwise.} \end{cases}$$

They form an orthogonal packet system and satisfy

$$S_1 P_{N,j} = P_{N,j+1} S_1,$$

with indices modulo N. The shift is therefore a cyclic transfer between residue packets.

Boundary 15.2. This section uses residue packets only to demonstrate the foundational transfer algebra. Nested refinement and filter-bank structures require additional definitions and are outside Paper I.

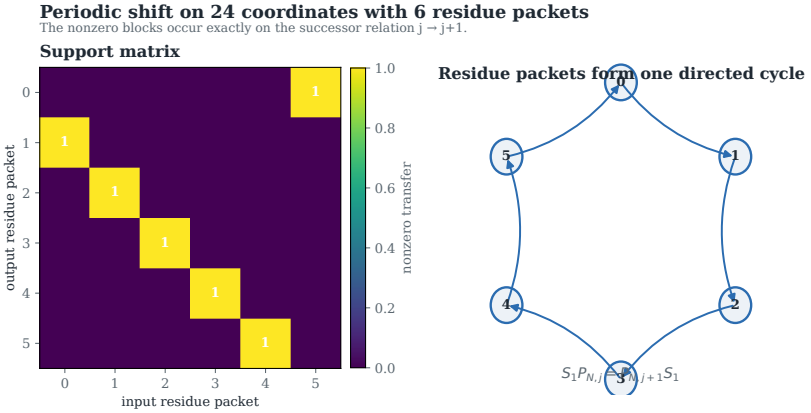


Figure 7. Residue-packet support in the 24-coordinate periodic shift model with six packets. Each input residue transfers into the next output residue. The infinite bilateral-shift theorem is established separately in the text.

16 Worked matrix models

16.1 The full matrix algebra

Let $X = \Bbbk^n$ and let $\mathcal{A} = M_n(\Bbbk) = \mathrm{End}_{\Bbbk}(X)$, with the full matrix algebra declared before a seed or target is chosen. By corollary 5.8, every nonzero seed is cyclic. This is the maximal linear grammar on X, not a compression result; under a smaller declared operator algebra, the same vector need not be cyclic.

16.2 Companion and repeated-block operators

For one operator T, theorem 6.2 turns verified primary data into an exact minimum-seed certificate. The numerical companion compares:

(a) a nilpotent companion operator with one Jordan chain, for which the Krylov rank is $1, 2, \ldots, n$;

(b) two equal Jordan blocks, for which one diagonal seed produces a chain only as long as one block and then stalls.

The observed stall is not numerical failure. The associated two-dimensional scalar quotient is the lower-bound certificate in theorem 6.2; the identification of that quotient with the geometric-multiplicity space is the classical Jordan input stated above.

17 Worked graph model: adjacency algebras

Let G be a nonempty finite graph with adjacency matrix A_G, and regard $\Bbbk^{V(G)}$ as a module over $\Bbbk[A_G]$. The general result applies directly to a spectral certificate. For a symmetric real adjacency matrix, the classical spectral theorem supplies such data.

Corollary 17.1 (Certified graph seed number)**.** *Let $\Bbbk$ be a field and let Λ be a nonempty finite set. Suppose there are pairwise distinct scalars $a_\lambda \in \Bbbk$, multiplicities $m_\lambda \in \mathbb{N}$, and an eigenbasis*

$$\{b_{\lambda,j} : \lambda \in \Lambda,\ 1 \le j \le m_\lambda\}$$

of $\Bbbk^{V(G)}$ *satisfying*

$$A_G b_{\lambda,j} = a_\lambda b_{\lambda,j}.$$

If $m_\lambda \le R$ *for all* λ *and* $m_{\lambda_*} = R$ *for some* λ_**, then* R *is the least natural number of adjacency seeds:*

$$R = \min\left\{ r \in \mathbb{N} : \begin{array}{l} \text{there exist } s_1, \dots, s_r \text{ with} \\ \Bbbk^{V(G)} = \operatorname{span}_\Bbbk\{p(A_G)s_j : p \in \Bbbk[t],\ 1 \le j \le r\} \end{array} \right\}.$$

Proof. The distinct eigenvalue labels give Lagrange polynomials that project onto the individual eigenspace packets. Padding one basis vector from every packet into each of R seeds gives the upper bound. Projection to the m_λ eigenspace coordinates is a surjective scalar quotient, so any generating family has at least m_λ members for every λ. The label λ_* gives the matching lower bound R. □

Boundary 17.2 (Spectral-classification input). For a finite undirected graph over $\mathbb{C}$, the real-symmetric spectral theorem classically supplies a certificate of the displayed kind, and the numbers m_λ are the usual eigenvalue multiplicities; see [2]. The formal declaration verifies the seed-number formula from the supplied eigenbasis and distinct eigenvalue labels. It does not derive diagonalizability from symmetry or implement an eigensolver.

17.1 Path graphs

For every field $\Bbbk$ and every $n \ge 1$, an endpoint of P_n is a cyclic vector for the adjacency operator. Consequently one is the least finite adjacency-seed number:

$$1 = \min\left\{ r \in \mathbb{N} : \exists s_1, \dots, s_r, \quad \Bbbk^n = \operatorname{span}_\Bbbk\{p(A_{P_n})s_j : p \in \Bbbk[t],\ 1 \le j \le r\} \right\}.$$

Over $\mathbb{C}$, the familiar spectral description has the distinct eigenvalues

$$\lambda_k = 2\cos\frac{k\pi}{n+1}, \qquad 1 \le k \le n,$$

and the normalized eigenvectors have endpoint coordinate proportional to

$$\sin\frac{k\pi}{n+1} \neq 0.$$

There is also an all-field combinatorial proof. Let

$$K_n = [e_1, Ae_1, \dots, A^{n-1}e_1].$$

Its (i,j) entry counts walks of length $j-1$ from the first vertex to vertex i. This entry is zero when $i > j$, while the diagonal entry is one because there is a unique shortest walk of length $i-1$ from vertex 1 to vertex i. Hence K_n is upper triangular with unit diagonal, so

$$\det K_n = 1 \qquad (n \ge 1).$$

Thus the endpoint is cyclic for every path size, not merely for the tested range.

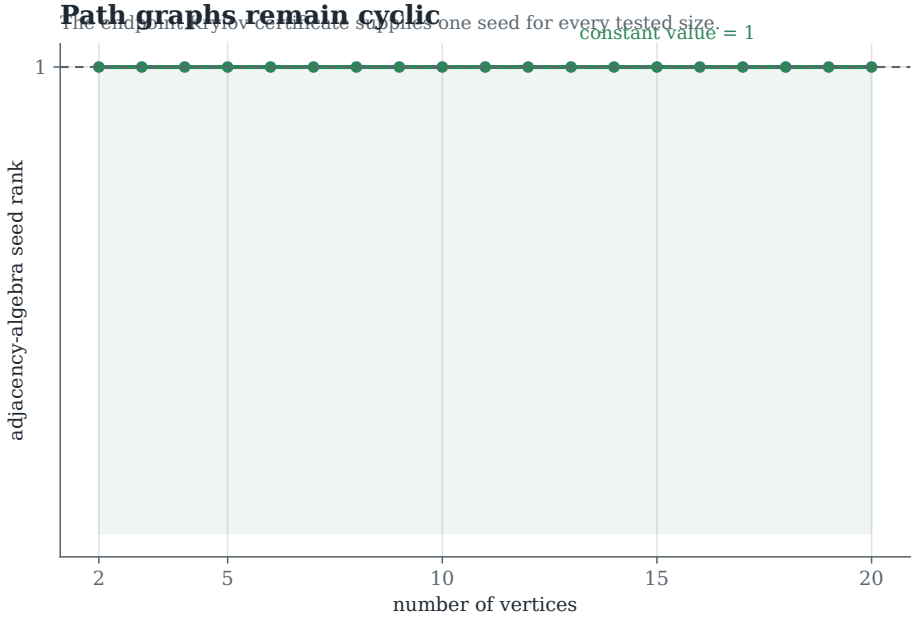


Figure 8. The endpoint Krylov certificate gives every nonempty path one adjacency seed.

17.2 Cycle graphs

Two adjacent vertex vectors generate the adjacency module of every simple cycle. Over $\mathbb{C}$, a primitive-root Fourier quotient supplies a matching two-dimensional scalar obstruction. Hence, for every $n \geq 3$, two is the least finite adjacency-seed number:

$$2 = \min\{r \in \mathbb{N} : \exists s_1, \dots, s_r, \quad \mathbb{C}^n = \operatorname{span}_{\mathbb{C}}\{p(A_{C_n})s_j : p \in \mathbb{C}[t],\ 1 \leq j \leq r\}\}.$$

To see the upper bound directly, index the vertices modulo n. The identity $Ae_j = e_{j-1} + e_{j+1}$ recovers successive vertex vectors from e_0 and e_1, over any field. For the lower bound over $\mathbb{C}$, let ζ be a primitive nth root of unity and define

$$q(x) = \left(\sum_{j=0}^{n-1} \zeta^j x_j, \sum_{j=0}^{n-1} \zeta^{-j} x_j\right).$$

Then $q(Ax) = (\zeta + \zeta^{-1})q(x)$. The images of e_0 and e_1 are $(1, 1)$ and (ζ, ζ^{-1}), which are independent because $\zeta \neq \zeta^{-1}$ for $n \geq 3$. Thus q is a two-dimensional scalar quotient, and proposition 4.13 gives the matching lower bound.

The usual spectral description writes the eigenvalues as

$$\lambda_k = 2\cos\frac{2\pi k}{n}, \qquad 0 \leq k < n.$$

The values for k and $n - k$ coincide. Apart from the simple characters at $k = 0$ and, for even n, $k = n/2$, the eigenspaces occur in pairs. Therefore the maximum eigenspace multiplicity is two, although some eigenspaces have multiplicity one, and the certified minimum above is two.

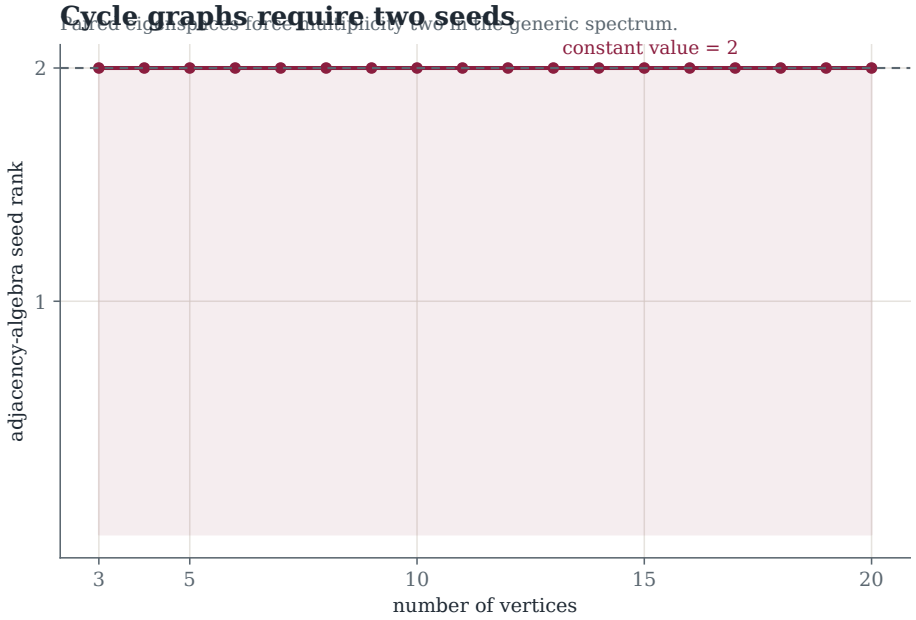


Figure 9. Cycle graphs have maximum adjacency-eigenspace multiplicity two and therefore require two adjacency seeds, although one or two exceptional eigenspaces are simple.

The path and cycle examples use nearly identical local operators. Their minimum finite seed numbers differ because the path has an endpoint Krylov basis, whereas the cycle has a two-dimensional Fourier quotient obstruction.

18 The common operator language

The preceding examples can be written without changing the foundational notation.

Table 3. Foundational realizations of the common operator language.

Realization	Seed	Operator grammar	Packets	Completion or quotient
Coordinate ring	1	M_x, M_y	even/odd y degree	quotient by $y^2 - x^3 + x$
Shift system	δ_0	$(S_m)_{m\in\mathbb{Z}}$	residue classes of n	ℓ^2 closure
Full matrix module	$s \neq 0$	$M_n(\Bbbk)$	chosen idempotents	finite-dimensional
Single operator	$s_1, \ldots, s_r$	$\Bbbk[T]$	primary or spectral	finite-dimensional
Graph adjacency	vertex seeds	$\Bbbk[A_G]$	eigenspaces	finite-dimensional
Typed grammar	vertex seeds	path algebra $\Bbbk Q/I$	vertex idempotents	relation quotient

The invariant content is:

$$\begin{aligned}
&\text{generated hull: } \mathrm{Hull}_{\mathcal{A}}(S),\\
&\text{seed complexity: } \mathrm{srank}_{\mathcal{A}}(X),\\
&\text{packet coordinates: } x \leftrightarrow (P_\alpha x)_\alpha,\\
&\text{operator coordinates: } T \leftrightarrow (P_\beta T P_\alpha)_{\beta,\alpha},\\
&\text{relations: } F^m/N_S,\\
&\text{coordinate invariance: } P_\alpha \mapsto U P_\alpha U^{-1},\\
&\text{realization: } \Phi A = \varphi(A)\Phi.
\end{aligned}$$

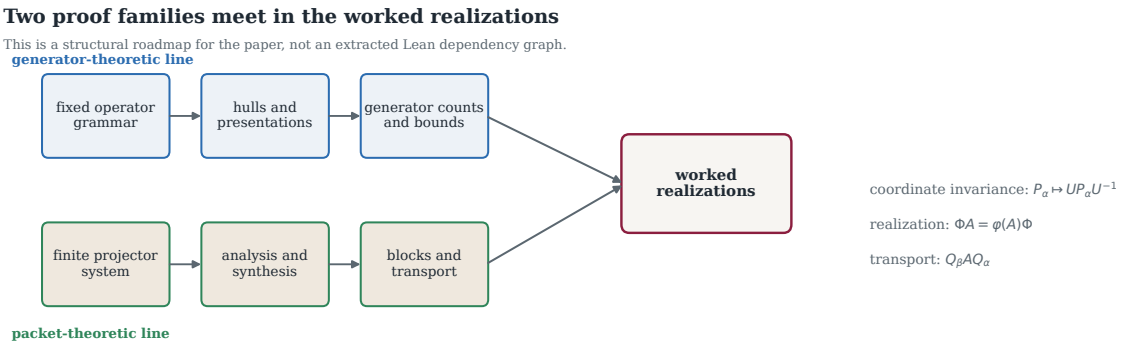


Figure 10. Structural overview of the two proof families. The generator and packet arguments meet in the worked realizations. This diagram is not an exhaustive graph of Lean declaration dependencies.

19 Computational certificates

The accompanying Python package is not used as a substitute for proof. It provides exact symbolic checks and finite numerical regression tests for formulas already proved. The computed ranges and random seeds are recorded explicitly. These checks do not extend a theorem beyond its hypotheses or certify an unrestricted numerical implementation. The journal companion also tests integer counts beyond floating-point precision and rejects the unsupported complex-symmetric spectral shortcut.

19.1 Test suite

The test suite checks:

(i) projector idempotence, annihilation, and completeness;

(ii) exact vector analysis and synthesis;

(iii) exact operator reconstruction from all transfer blocks;

(iv) coordinate transport under a dense random similarity transform;

(v) cyclicity of a companion operator;

(vi) the two-seed obstruction for repeated Jordan blocks;

(vii) the semisimple multiplicity formula in representative cases;

(viii) full orbit dimension for two representative nonzero seeds under the full matrix algebra;

(ix) exact reduction and packet multiplication in the elliptic coordinate ring;

(x) seed ranks of path and cycle adjacency modules over the reported finite ranges;

(xi) the cyclic residue transfer pattern of a finite shift.

19.2 Recorded results

Table 4. Archived representative checks from the v1.3.1 journal-edition run. Fresh companion runs are reported separately in the release verification record.

Quantity	Result
Projector idempotence residual	0
Projector annihilation residual	0
Projector completeness residual	0
Operator reconstruction residual	0
Transported-projector residual, maximum	1.85×10^{-15}
Companion seed rank	1
Two equal Jordan blocks, seed rank	2
$M_3(\mathbb{C})$ acting on seven copies, seed rank	3
Sampled path graph seed ranks, $2 \le n \le 12$	1
Sampled cycle graph seed ranks, $3 \le n \le 12$	2
Full $M_4(\mathbb{R})$ orbit dimensions for two sampled nonzero seeds	$4, 4$
Automated tests	33/33 passed

The finite shift pattern recovered by the experiment is

$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

which is the exact packet adjacency of a cyclic four-phase transfer.

19.3 Algorithms

The following finite constructions state mathematical specifications for the companion. Their Lean counterparts and exact scope are listed in the separate guide. They do not assert code extraction or a particular backend's performance.

Construction 19.1 (Orbit-hull closure specification)**.** Let V be finite-dimensional, let $(O_i)_{i\in I}$ be a finite operator family, and let $S \subseteq V$. Define

$$W_0 = \operatorname{span}_{\Bbbk} S, \qquad W_{n+1} = W_n + \sum_{i\in I} O_i(W_n).$$

Then there is an $n \in \mathbb{N}$ such that

$$W_{n+1} = W_n \qquad \text{and} \qquad W_n = \operatorname{Hull}_{(O_i)}(S),$$

where the right side is the least subspace containing S and invariant under every O_i. Thus the ascending subspace iteration terminates and its stable value is exactly the finite operator hull.

Construction 19.2 (Transfer-block table specification)**.** Given finite projector systems $(P_i)_{i\in I}$ and $(Q_j)_{j\in J}$ and a linear map T, form the table

$$B_{ji} = Q_j T P_i.$$

Its assembly reconstructs the input map exactly,

$$\sum_{j\in J}\sum_{i\in I} B_{ji} = T,$$

and its reported support relation is exactly $\{(i,j) : B_{ji} \neq 0\}$. The latter equality is the specification of the transfer-graph output, not a claim that Lean extracts a decidable edge-list routine for every coefficient field.

Construction 19.3 (Certified exact graph seed-rank specification)**.** For a finite square matrix A over a field K, define the noncomputable exact spectral quantity

$$r_{\mathrm{spec}}(A) = \max_{\mu\in \mathrm{Spec}_K(A)} \dim_K \ker(A - \mu I),$$

with value zero for an empty spectrum. Suppose an exact certificate supplies a finite eigenbasis indexed by labels and multiplicities, the eigenvalue equations, a multiplicity bound attained by one label, projector polynomials that extract the padded eigenbasis generators, coverage of every spectral value by a label, equality of the declared and actual eigenspace dimensions, and membership of the maximizing eigenvalue in the spectrum. Then $r_{\mathrm{spec}}(A)$ is the least natural number of polynomial-orbit seeds for A.

This is a verified certificate checker/specification. It does not construct the eigenbasis, prove that a symmetric matrix is diagonalizable, or implement an exact eigensolver. The companion Python code performs symbolic computations in the reported finite graph range; the all-size path and cycle results are proved separately above.

20 Reading guide: common confusions and diagnostic questions

20.1 Common confusions

Do not conflate	Correct distinction
Grammar with realized algebra	The grammar may contain relations or words that act as zero. Seed rank depends only on the image action.
Seed with basis vector	A seed is a generator under an algebra action. One seed may generate a high-dimensional module.
Cyclic vector with eigenvector	A cyclic vector must generate the whole module under operator words. It is usually not an eigenvector.
Packet with orthogonal subspace	A packet is the range of an idempotent in a complete annihilating family. Orthogonality is optional.
Exact reconstruction with good conditioning	Algebraic inverse identities may coexist with large analysis and synthesis norms.
Nonzero transfer edge with surjective transfer	A nonzero block can have small rank. Graph reachability alone does not prove generation.
Packet-pure seed rank with unrestricted seed rank	Mixed-support seeds may meet several source components simultaneously.
Continuous surjection with quotient isomorphism	The target topology must agree with the quotient topology.
Repeated notation with semantic realization	A true realization must preserve a declared collection of operations and identities.

20.2 Diagnostic questions for a new application

Before claiming that a new subject is a realization of Paper I, answer the following.

1. What is the fixed operator grammar or acting algebra?
2. What is the concrete realization space or module?
3. Which seeds are declared before the target is chosen?
4. Is generation algebraic, topological, probabilistic, or approximate?
5. What are the idempotent packets, and in what sense do they sum to the identity?
6. Are the packets finite, countable unconditional, or merely formal?
7. What are the transfer blocks, and which blocks are known to vanish?
8. Which morphisms preserve the packet system?
9. Which quotient, completion, tensor, or limit operations must be preserved?

10. What exact theorem from Paper I is being specialized, and what new domain theorem remains to be proved?

20.3 One-page theorem spine

free seed presentation	packet idempotents
$\Downarrow$	$\Downarrow$
minimal generator number	exact analysis/synthesis
$\Downarrow$	$\Downarrow$
Jacobson-radical reduction	transfer-block algebra
$\Downarrow$	$\Downarrow$
semisimple or primary computation	graphs, quotients, and realization

21 Notation and convention ledger

Address.
An operator word applied to a seed.

Analysis.
The map sending an object to all packet coordinates.

Block support.
The set of packet pairs (β, α) for which $P_\beta T P_\alpha \neq 0$.

Collapsed realization.
The direct sum of all typed vertex spaces.

Condensation graph.
The acyclic graph obtained by collapsing strongly connected components.

Generated hull.
The submodule generated by a seed family, or its declared closure.

Grammar.
A typed quiver and relation ideal describing legal operator words.

Packet.
The range of one idempotent in a complete annihilating family.

Packet-pure.
Supported in at most one packet.

Preservation profile.
The operations a realization functor is required to preserve.

Realized algebra.
The image of the grammar algebra in the endomorphism algebra of a realization.

Seed rank.
The minimal size of a finite generating family.

Source component.
A strongly connected component with no incoming edge in the condensation graph.

Synthesis.
The map adding packet coordinates into the original object.

Transfer block.
The map from one packet space to another obtained by sandwiching an operator between packet projectors.

Transfer graph.
The directed graph formed from nonzero blocks of a declared algebra-generating family.

How to use the paper after the first reading

For a concrete calculation, first reduce the action to the realized algebra. Next compute or bound seed rank using quotients, the module top, or primary decomposition. Then declare the packet system and use exact analysis and synthesis. Finally move all operator equations into transfer blocks, where support, graph, quotient, and functorial constraints become explicit.

22 Proof dependency spine

Canonical record	Immediate proof dependencies
PSC-I-T-TYPED-MODULE-EQUIVALENCE	Typed grammar and path-algebra action.
PSC-I-T-FREE-PRESENTATION	Seed evaluation map and first isomorphism theorem.
PSC-I-T-COMPLETED-PRESENTATION	Free presentation and uniqueness of completion.
PSC-I-T-QUOTIENT-MONOTONICITY	Intertwining identity and generated hull.
PSC-I-T-EXACT-SEQUENCE-BOUNDS	Quotient monotonicity and lifting generators.
PSC-I-T-MODULE-TOP	Nakayama lemma.
PSC-I-T-SEMISIMPLE-SEED-RANK	Module-top theorem and regular-module multiplicities.
PSC-I-T-SINGLE-OPERATOR	Primary decomposition, local formula, and product action.
PSC-I-T-EXACT-PACKET-COORDINATES	Complete finite idempotent family.
PSC-I-T-COUNTABLE-UNCONDITIONAL-PACKETS	Completeness of the unconditional coordinate space.
PSC-I-T-BLOCK-ALGEBRA	Reconstruction and block multiplication.
PSC-I-T-SUPPORT-PATTERN-ALGEBRA	Block multiplication and rank-one necessity tests.
PSC-I-T-PREORDER-ALGEBRA-DECOMPOSITION	Preorder support criterion and finite-chain nilpotence.
PSC-I-T-ALGEBRA-REACHABILITY-SUPPORT	Generator-word packet expansion.
PSC-I-T-SOURCE-COMPONENT-SEED-RANK	Reachability obstruction and edge-surjective propagation.
PSC-I-T-NATURALITY	Intertwining and packet aggregation.
PSC-I-T-PACKET-EXACT-SEQUENCE	Invariant packet projectors and first isomorphism theorem.
PSC-I-T-DENSE-REALIZATION	Continuity and naturality of hulls.
PSC-I-T-REALIZATION-EXPORT	Additive functor axioms and preserved universal constructions.

23 Conclusions and further use

The main structural conclusions can be collected in the following form:

I.1. The two-sided annihilator criterion is proved in proposition 3.5.

I.2. Generator detection modulo the Jacobson radical is proved in theorem 5.2.

I.3. The arbitrary-division-ring semisimple formula is proved in theorem 5.6 without a left-right row-rank ambiguity.

I.4. The arbitrary-field single-operator formula and all cyclic criteria are proved in theorem 6.2 and corollary 6.5.

I.5. The countable unconditional packet theorem is proved in theorem 7.16.

I.6. The $*$-support classification is proved in corollary 8.14.

I.7. The source-component theorem is strengthened to the weighted formula in theorem 9.7.

I.8. Seed-system composition, packet exact sequences, quotient block descent, dense realization, quotient completion, and semantic theorem transport are proved in proposition 10.5, theorems 10.6, 10.10, 10.11 and 11.2, and corollary 10.8.

I.9. Additional omitted structure is supplied by theorems 4.6 and 8.15 and propositions 4.3, 7.12 and 8.11.

A Stable statement concordance

These identifiers preserve the expanded foundation's 108-record namespace. The display numbers refer to this edition. They identify mathematical statements, not a count of Lean-verified theorems. The worked-model statements added after the foundation retain their source labels and are mapped separately by the Lean guide.

Stable identifier	No.	Statement
PSC-I-D-COEFFICIENT-REGIME	1.1	Coefficient and module regimes
PSC-I-D-THEOREM-RECORD	1.2	Canonical theorem record
PSC-I-B-SYMBOLIC-SCOPE	1.3	Symbolic scope
PSC-I-D-TYPED-GRAMMAR	2.1	Typed operator grammar
PSC-I-D-COLLAPSED-REALIZATION	2.2	Collapsed one-space realization
PSC-I-T-TYPED-MODULE-EQUIVALENCE	2.3	Typed realization and module equivalence
PSC-I-D-REALIZED-ALGEBRA	2.4	Realized operator algebra
PSC-I-P-ACTION-FACTORIZATION	2.5	Action factorization and hull invariance
PSC-I-D-SEED-SYSTEM	2.6	Linear seed system
PSC-I-D-GENERATED-HULL	2.7	Generated hull
PSC-I-D-NATURAL-PRESENTATION	2.8	Natural presentation
PSC-I-P-ALGEBRA-MODULE-GENERATION	2.9	Algebra generation and module generation are distinct
PSC-I-B-TARGET-DEPENDENT-OPERATORS	2.10	Target-dependent operators
PSC-I-D-SEED-EVALUATION	3.1	Seed evaluation map
PSC-I-T-FREE-PRESENTATION	3.2	Finite free presentation
PSC-I-C-MINIMAL-FREE-COVER	3.3	Minimal free-cover characterization
PSC-I-C-ONE-SEED-PRESENTATION	3.4	One-seed presentation
PSC-I-P-ANNIHILATOR-TWO-SIDED	3.5	Two-sided annihilator criterion
PSC-I-T-COMPLETED-PRESENTATION	3.6	Completion of a dense free presentation
PSC-I-D-SEED-RANK	4.1	Seed rank
PSC-I-P-CYCLIC-CHARACTERIZATION	4.2	Cyclic characterization
PSC-I-P-ACTING-ALGEBRA-MONOTONICITY	4.3	Monotonicity under enlargement of the acting algebra
PSC-I-T-QUOTIENT-MONOTONICITY	4.4	Quotient monotonicity
PSC-I-C-ISOMORPHISM-INVARIANCE	4.5	Isomorphism invariance
PSC-I-T-EXACT-SEQUENCE-BOUNDS	4.6	Short exact sequence bounds
PSC-I-P-SHARED-DIRECT-SUM	4.7	Shared direct-sum bounds
PSC-I-T-INDEPENDENT-DIRECT-SUM	4.8	Independent direct-sum formula
PSC-I-P-TENSOR-BOUND	4.9	Tensor-product bound
PSC-I-B-TENSOR-NONEQUALITY	4.10	Tensor rank is not generally multiplicative
PSC-I-P-SCALAR-EXTENSION-MONOTONICITY	4.11	Scalar-extension monotonicity
PSC-I-D-SEED-RANK-WITNESS	4.12	Seed-rank witness
PSC-I-P-SCALAR-QUOTIENT-WITNESS	4.13	Scalar quotient witness
PSC-I-CI-NAKAYAMA	5.1	Nakayama lemma
PSC-I-T-MODULE-TOP	5.2	Generator number is detected modulo the Jacobson radical
PSC-I-C-LOCAL-ALGEBRA-FORMULA	5.3	Local algebra formula
PSC-I-CI-WEDDERBURN-ARTIN	5.4	Wedderburn–Artin decomposition
PSC-I-L-REGULAR-MODULE-MULTIPLICITIES	5.5	Regular-module multiplicities
PSC-I-T-SEMISIMPLE-SEED-RANK	5.6	Semisimple seed-rank formula over arbitrary division rings
PSC-I-C-ALGEBRAICALLY-CLOSED-FORMULA	5.7	Algebraically closed multiplicity formula

Stable identifier	No.	Statement
PSC-I-C-FULL-MATRIX	5.8	Full matrix algebra
PSC-I-C-MULTIPLICITY-OBSTRUCTION	5.9	Multiplicity obstruction
PSC-I-CI-PID-PRIMARY-DECOMPOSITION	6.1	Primary decomposition for polynomial modules
PSC-I-T-SINGLE-OPERATOR	6.2	Single-operator seed rank
PSC-I-C-INVARIANT-FACTOR-COUNT	6.3	Invariant-factor count
PSC-I-C-ALGEBRAICALLY-CLOSED-SINGLE-OPERATOR	6.4	Geometric-multiplicity formula
PSC-I-C-CYCLIC-MATRIX	6.5	Cyclic-matrix criterion
PSC-I-D-PACKET-SYSTEM	7.1	Finite packet system
PSC-I-D-PACKET-SUPPORT	7.2	Packet support and packet-pure vectors
PSC-I-L-PACKET-AGGREGATION	7.3	Finite packet aggregation
PSC-I-D-ANALYSIS-SYNTHESIS	7.4	Packet analysis and synthesis
PSC-I-T-EXACT-PACKET-COORDINATES	7.5	Exact packet-coordinate isomorphism
PSC-I-C-PACKETWISE-EQUALITY	7.6	Packetwise equality
PSC-I-C-PACKET-UNIVERSAL-PROPERTY	7.7	Universal mapping property of finite packet coordinates
PSC-I-T-TENSOR-PACKET-SYSTEM	7.8	Tensor-product packet systems
PSC-I-P-DUAL-PACKET-SYSTEM	7.9	Dual packet systems
PSC-I-T-BANACH-PACKET-ISOMORPHISM	7.10	Finite Banach packet isomorphism
PSC-I-D-PACKET-CONDITION-NUMBER	7.11	Packet-coordinate condition number
PSC-I-P-CONDITIONED-COORDINATES	7.12	Conditioned coordinate transport
PSC-I-C-COMPLETION-PACKET-SPLIT	7.13	Completion preserves a finite packet split
PSC-I-D-UNCONDITIONAL-PACKET-SPACE	7.14	Unconditional packet sum space
PSC-I-L-UNCONDITIONAL-COMPLETENESS	7.15	Completeness of the unconditional packet sum
PSC-I-T-COUNTABLE-UNCONDITIONAL-PACKETS	7.16	Countable unconditional packet isomorphism
PSC-I-R-UNIFORM-BOUNDEDNESS	7.17	Uniform boundedness
PSC-I-B-INFINITE-PACKET-TOPOLOGY	7.18	Infinite packet sums
PSC-I-D-TRANSFER-BLOCK	8.1	Transfer block
PSC-I-T-OPERATOR-RECONSTRUCTION	8.2	Exact operator reconstruction
PSC-I-T-BLOCK-MULTIPLICATION	8.3	Packet matrix multiplication
PSC-I-C-BLOCK-POWERS	8.4	Packet powers and paths
PSC-I-C-ADJOINT-REVERSAL	8.5	Adjoint reversal
PSC-I-C-TENSOR-TRANSFER-BLOCKS	8.6	Tensor-product transfer blocks
PSC-I-T-BLOCK-INTERTWINING	8.7	Blockwise intertwining equations
PSC-I-T-BLOCK-ALGEBRA	8.8	Full block-algebra representation
PSC-I-C-PACKET-SUPPORT-PROPAGATION	8.9	Propagation of packet support
PSC-I-C-DIMENSION-COUNT	8.10	Dimension count
PSC-I-P-BLOCK-DIAGONAL-COMMUTANT	8.11	Block-diagonal projection and packet commutant
PSC-I-D-SUPPORT-RELATION	8.12	Support relation and support space
PSC-I-T-SUPPORT-PATTERN-ALGEBRA	8.13	Support-pattern algebra criterion
PSC-I-C-SUPPORT-PATTERN-STAR	8.14	$*$-closed bounded support patterns
PSC-I-T-PREORDER-ALGEBRA-DECOMPOSITION	8.15	Preorder support algebra decomposition
PSC-I-D-TRANSFER-GRAPH	9.1	Transfer graph of an algebra-generating family
PSC-I-T-ALGEBRA-REACHABILITY-SUPPORT	9.2	Reachability support of the generated algebra

Stable identifier	No.	Statement
PSC-I-P-REACHABILITY-OBSTRUCTION	9.3	Reachability obstruction for generated hulls
PSC-I-D-DIRECTED-SEED-RANK	9.4	Packet-pure seed rank
PSC-I-P-PACKET-PURE-RANK-BOUNDS	9.5	Basic packet-pure rank bounds
PSC-I-D-EDGE-SURJECTIVE	9.6	Edge-surjective packet system
PSC-I-T-SOURCE-COMPONENT-SEED-RANK	9.7	Source-component formula for packet-pure seed rank
PSC-I-D-PACKET-PROPAGATING	9.8	Packet-propagating system
PSC-I-C-SOURCE-COMPONENT-COUNT	9.9	Unweighted source-component count
PSC-I-B-PACKET-PURE-BOUNDARY	9.10	Mixed-support seeds can merge source requirements
PSC-I-T-COORDINATE-TRANSPORT	10.1	Coordinate transport
PSC-I-D-PACKET-MORPHISM	10.2	Packet-compatible operator morphism
PSC-I-D-SEED-MORPHISM	10.3	Morphism of seeded systems
PSC-I-T-NATURALITY	10.4	Naturality of generated hulls and transfer blocks
PSC-I-P-SEED-SYSTEM-CATEGORY	10.5	Category of seeded packet systems
PSC-I-T-PACKET-EXACT-SEQUENCE	10.6	Packet exact sequence for an invariant subspace
PSC-I-T-QUOTIENT-PACKET-DESCENT	10.7	Packet descent through an invariant quotient
PSC-I-C-QUOTIENT-PACKET-DESCENT	10.8	Restriction and quotient of transfer blocks
PSC-I-C-TOPOLOGICAL-PACKET-EXACT-SEQUENCE	10.9	Topological packet exact sequence
PSC-I-T-DENSE-REALIZATION	10.10	Dense realization theorem
PSC-I-T-QUOTIENT-TOPOLOGY-COMPLETION	10.11	Quotient topology and completed realization
PSC-I-B-TOPOLOGICAL-QUOTIENT	10.12	Continuous surjections and quotient topology
PSC-I-D-REALIZATION-FUNCTOR	11.1	Realization functor and preservation profile
PSC-I-T-REALIZATION-EXPORT	11.2	Equational realization principle
PSC-I-P-REALIZATION-EXPORT	11.3	Realization export principle
PSC-I-D-SEMANTIC-REALIZATION	11.4	Semantic realization criterion
PSC-I-ALG-EXAMPLE-SLOTS	12.1	Worked realization interface
PSC-I-EXPORT	12.2	Paper I export object
PSC-I-B-FINAL-CLAIM-BOUNDARY	12.3	Exact limits of Paper I

B Visual atlas

The following twenty-two plates give exact finite models, explanatory diagrams, and one analytic density example. All graphics are generated from mathematical source. Cross-references are aligned with this edition. The atlas illustrates the proofs and hypotheses in the main text.

Plate	Route through the examples
1–2	Fixed grammars, seed power, and the objects of the calculus.
3–4	Two linked chains and a quotient proving the minimum seed count.
5–6	Left-ideal relations and typed reachability with a missing direction.
7–8	Oblique exact coordinates and complete transfer-block multiplication.
9–10	Cancellation along paths and the algebra of admissible zero patterns.
11–12	Coordinate transport, shared action, and independent control.
13–14	Semisimple multiplicity budgets and primary extraction polynomials.
15–16	Path endpoints and the cycle's two-dimensional quotient obstruction.
17	A coordinate-ring relation as a two-packet product rule.
18–19	Dense shift generation, residue coordinates, and address carry.
20–21	Exact orbit-hull stabilization and all tensor cross terms.
22	The scope of mathematical, formal, and computational evidence.

The seed examples can be read consecutively, while the packet and graph plates form independent entry points. The displayed matrices are exact. Numerical illustrations in the earlier figures retain their explicitly stated finite ranges. Every diagram is accompanied by the equation or obstruction it represents.

PLATE 1 / PAPER I / A WORKED-EXAMPLE VISUAL APPENDIX

What can one seed actually do?

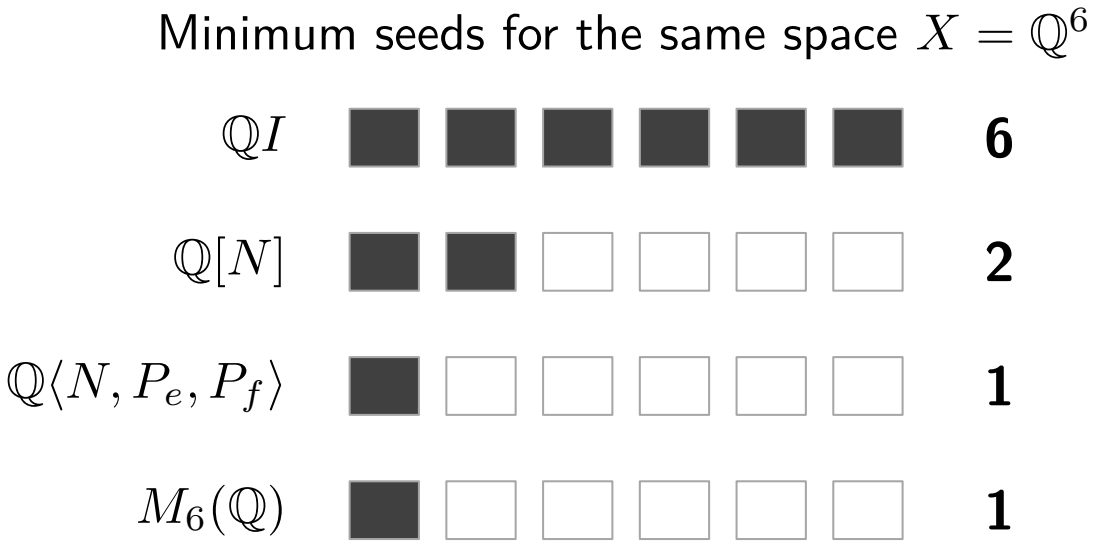
Main text: definition 4.1, proposition 4.3, and theorem 7.5

A seed does not carry a fixed amount of generative power by itself. The answer depends on the operators that are allowed to act on it. Paper I makes that dependence part of the mathematical object rather than leaving it implicit.

Fix the same six-dimensional space throughout the first comparison:

$$X = \mathbb{Q}^6, \qquad N = J_3(0) \oplus J_3(0), \qquad s = e_0 + f_0.$$

Here each $J_3(0)$ moves a basis vector forward along a three-step chain. The projectors P_e, P_f select the two chains. Their formulas and the minimum-seed proofs appear on Plate 3–Plate 4.



A count of minimum generators, not a timing comparison. The ambient space is unchanged. Enlarging the acting algebra changes what one seed can separate.

Three questions to keep apart

Generation	Can the fixed grammar reach every vector by finite linear combinations of operator words?
Coordinates	Can every vector be split into packets and then reconstructed exactly?
Completion	When finite words do not suffice, does their span become dense in the specified topology?

A packet projector can be available as a way to *inspect* a vector without belonging to the *generation* grammar. Admitting it as a new primitive can lower seed rank. That distinction explains the second and third rows of the figure.

> **Read this as:** The useful question is not "How powerful is this seed?" It is "What does this seed generate under this particular, previously fixed algebra of operators?"

PLATE 2 / ORIENTATION / SYMBOLS WITH JOBS

An atlas of the objects

Main text: definitions 2.1, 2.6, 4.1, 7.1 and 8.1

Most of the paper can be read by tracking a small collection of objects. The table distinguishes their mathematical roles and names a concrete instance used in this appendix.

Object	What it means	Instance here
X	The ambient vector space or module. It specifies the objects that might be generated.	$\mathbb{Q}^6$ with two length-three chains.
$S \subseteq X$	The selected seed family. A family can contain mixed-packet vectors.	$\{e_0 + f_0\}$ or $\{e_0, f_0\}$.
$\mathcal{A}$	The fixed unital algebra of admissible operators. It is not chosen anew for each target.	$\mathbb{Q}[N]$ versus $\mathbb{Q}\langle N, P_e, P_f\rangle$.
$\mathrm{Hull}_{\mathcal{A}}(S)$	The smallest invariant linear subspace containing the seeds.	$\mathrm{span}\{e_j + f_j : 0 \le j \le 2\}$.
P_α	A finite family of pairwise annihilating idempotents whose sum is I.	Chain projections, oblique projections, or residue masks.
X_α	The packet subspace $P_\alpha X$. It can have dimension greater than one.	One three-dimensional chain.
$T_{\beta\alpha}$	The block $P_\beta T P_\alpha$: input α, output β.	A numerical entry for one-dimensional coordinate packets.
F/I_s	A module of operator words modulo relations that kill the seed.	A left-module quotient, not necessarily a quotient algebra.
U	An invertible change of realization coordinates.	A shear that makes orthogonal packets oblique.
Φ	An intertwiner that need not be invertible.	Projection from two chains to the first chain.
τ	The topology used when taking the closure of an algebraic hull.	The ℓ^2 norm for a bilateral shift.

Read products from right to left

In $P_\beta T P_\alpha x$, first select the input packet, then apply T, then select the output packet. In a matrix, this is why the output label belongs to the row and the input label to the column.

Where to begin

For seed rank, start with Plate 3. For exact reconstruction, start with Plate 7. For graphs, use Plate 15–Plate 16. For infinite-dimensional completion, use Plate 18–Plate 19. The formal-proof boundary is collected on Plate 22.

PLATE 3 / MODEL A / A SIX-DIMENSIONAL OPERATOR

Two chains can stay linked

Main text: theorem 6.2; matrices over $\mathbb{Q}$

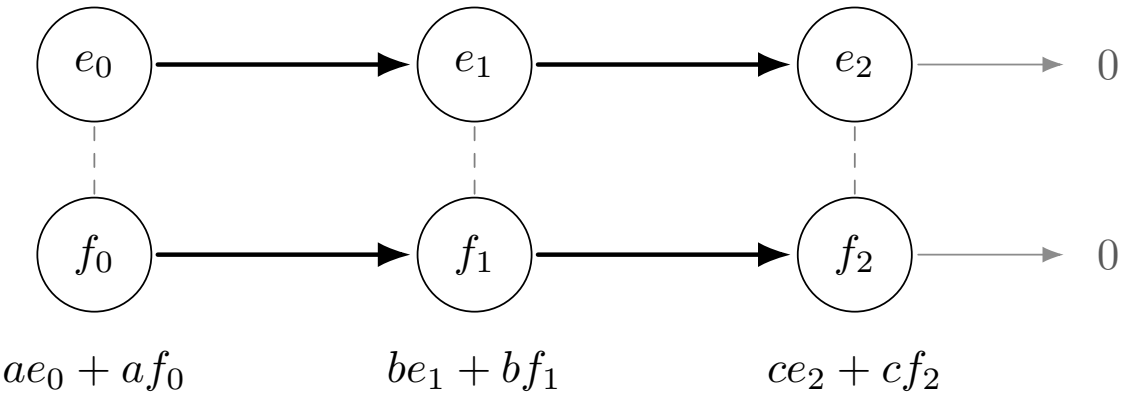
Use the ordered basis $(e_0, e_1, e_2, f_0, f_1, f_2)$. Define

$$Ne_0 = e_1, \quad Ne_1 = e_2, \quad Ne_2 = 0, \qquad Nf_0 = f_1, \quad Nf_1 = f_2, \quad Nf_2 = 0.$$

Starting with $s = e_0 + f_0$, the two chains advance together:

$$s = e_0 + f_0, \qquad Ns = e_1 + f_1, \qquad N^2 s = e_2 + f_2, \qquad N^3 s = 0.$$

$$Ne_j = e_{j+1}, \qquad Nf_j = f_{j+1}$$



The vertical dotted pairings record equal coefficients, not an additional allowed operator. Every polynomial in N uses the same coefficients on both chains.

Since $N^3 = 0$, every polynomial orbit vector has the form

$$p(N)s = a(e_0 + f_0) + b(e_1 + f_1) + c(e_2 + f_2).$$

The generated hull is therefore three-dimensional, even though the seed has a nonzero component in each three-dimensional packet. The Krylov ranks are $1, 2, 3, 3, 3, 3$.

What changes when a projector becomes admissible?

Let P_e keep the e-coordinates and P_f the f-coordinates. If these are merely measurement maps, the preceding hull is unchanged. If they become allowed generators, then

$$P_e s = e_0, \qquad P_f s = f_0,$$

and powers of N now recover all six basis vectors. The same seed becomes cyclic for the larger algebra $\mathbb{Q}\langle N, P_e, P_f\rangle$.

> **Read this as:** Nonzero components in every packet are not enough. One must also be able to separate the independent directions inside and across those packets using the allowed operations.

PLATE 4 / MODEL A / A LOWER-BOUND CERTIFICATE

A quotient proves two seeds are necessary

Main text: proposition 4.13 and theorem 6.2

The failure of the particular seed on Plate 3 does not prove that *every* single seed fails. A lower-bound certificate does.

Define a surjective linear map

$$q : X \longrightarrow \mathbb{Q}^2, \qquad q\Big(\sum a_j e_j + \sum b_j f_j\Big) = (a_0, b_0).$$

The quotient keeps only the two chain heads

e_0	e_1	e_2	f_0	f_1	f_2
1	0	0	0	0	0
0	0	0	1	0	0

$$q(a_0, a_1, a_2, b_0, b_1, b_2) = (a_0, b_0)$$

An actual 2×6 quotient matrix. It forgets the advanced positions in both chains and retains two independent head coordinates.

Because $qN = 0$, one has, for every polynomial p,

$$q(p(N)x) = p(0)q(x).$$

Thus every operator acts as a *scalar* on the quotient. The image of the orbit of one seed stays on one line in $\mathbb{Q}^2$. It cannot span this two-dimensional quotient.

Lower and upper bounds meet

If r seeds generated X, their images would generate the quotient. Since scalar multiplication cannot create a new independent direction, $r \geq 2$. Conversely, the two seeds e_0, f_0 generate their respective chains, so

$$\boxed{\operatorname{srank}_{\mathbb{Q}[N]}(X) = 2.}$$

This proof uses the displayed operator and quotient. It does not ask a computer to search every possible seed, and it does not need an existence theorem for Jordan form.

A general pattern worth recognizing

A surjective intertwiner $q : X \twoheadrightarrow Y$ turns any generating family of X into a generating family of Y. Therefore a quotient on which the grammar loses mixing power can certify a lower bound in the original, much larger space.

> **Read this as:** To rule out one seed, find a quotient in which its entire orbit is forced onto one line while the quotient still has at least two independent directions.

PLATE 5 / PRESENTATIONS / WHY A LEFT IDEAL APPEARS

Relations identify words, not just outputs

Main text: theorem 3.2, corollary 3.4, and proposition 3.5

Let $F = \langle z_1, \dots, z_r \rangle$ be the algebra of noncommuting operator words. Evaluation at a fixed seed is

$$\Phi_s : F \to X, \qquad f \mapsto f(T_1, \dots, T_r)s.$$

Two words represent the same generated vector exactly when their difference lies in $I_s = \operatorname{Ker} \Phi_s$. Consequently

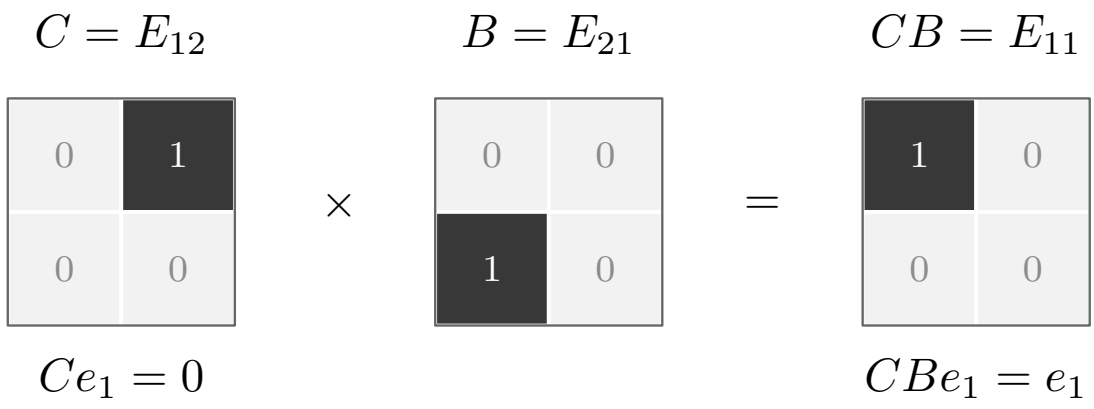
$$F/I_s \cong \operatorname{Hull}_F(s)$$

as *left modules*. The relation $f(T)s = 0$ implies $g(T)f(T)s = 0$ for every word g, which explains closure under multiplication on the left.

The distinction is visible in two dimensions

Use $X = \mathbb{Q}^2$, seed $e_1 = (1, 0)^{\mathsf{T}}$, and the full matrix algebra. Its seed annihilator is the set of matrices with zero first column:

$$I_{e_1} = \left\{ \begin{pmatrix} 0 & a \\ 0 & b \end{pmatrix} : a, b \in \mathbb{Q} \right\}.$$



The relation $Ce_1 = 0$ is destroyed by multiplying C on the right by B. The product CB sends the seed to itself.

For arbitrary D, $(DC)e_1 = D(Ce_1) = 0$. But $(CB)e_1 = e_1 \neq 0$. Thus I_{e_1} is a left ideal and is not a two-sided ideal. Calling this quotient a ring would lose the distinction that makes the presentation correct.

Several seeds give a module of relations

For $(s_1, \dots, s_m)$, evaluation is $(f_1, \dots, f_m) \mapsto \sum_j f_j(T)s_j$. Its kernel is a submodule of F^m and can include relations *between* seeds, not only relations annihilating each seed separately.

> **Read this as:** A presentation says which operator expressions are identified in a particular realization. A seed annihilator is not automatically an algebra relation ideal.

PLATE 6 / TYPED MODEL / $U = \mathbb{Q}^2$ AND $V = \mathbb{Q}^3$

A reached type can still hide a direction

Main text: proposition 13.1 and definition 9.4

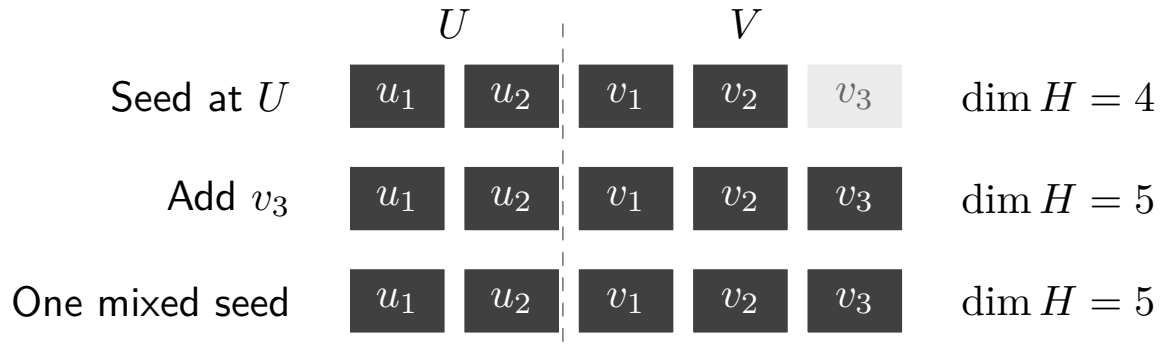
The paper's typed model has a rotation $R : U \to U$, an inclusion $A : U \to V$, and a retraction $B : V \to U$:

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

The typed relation is $BA = I_U$. In the opposite order,

$$AB = \operatorname{diag}(1, 1, 0) \neq I_V.$$

The missing direction is $\operatorname{Ker} B = \mathbb{Q}v_3$, not an entirely missing vertex.



Dark slots indicate independent directions in the generated hull. The same two vertex types are present in every row, but their generated subspaces differ.

One seed supported at U

The vectors u_1, Ru_1 span U. Their images under A span $\mathbb{Q}v_1 + \mathbb{Q}v_2$. Every path from U to V ends with A, so it cannot supply v_3. The hull has dimension four. Adding the pure seed v_3 gives all five dimensions.

Why this does not prove an unrestricted minimum of two

The bound path algebra includes the vertex idempotents P_U, P_V. A mixed seed

$$s = u_1 + v_3 \in U \oplus V$$

satisfies $P_U s = u_1$ and $P_V s = v_3$. It therefore generates the entire representation by itself. The paper proves that no U-*supported* seed is enough and that two displayed pure seeds suffice. It does not claim that two arbitrary mixed seeds are necessary.

> **Read this as:** Keep three notions separate: reaching a vertex, filling its entire vector space, and minimizing seeds under a packet-purity restriction. They answer different questions.

PLATE 7 / PACKET COORDINATES / AN OBLIQUE SPLITTING

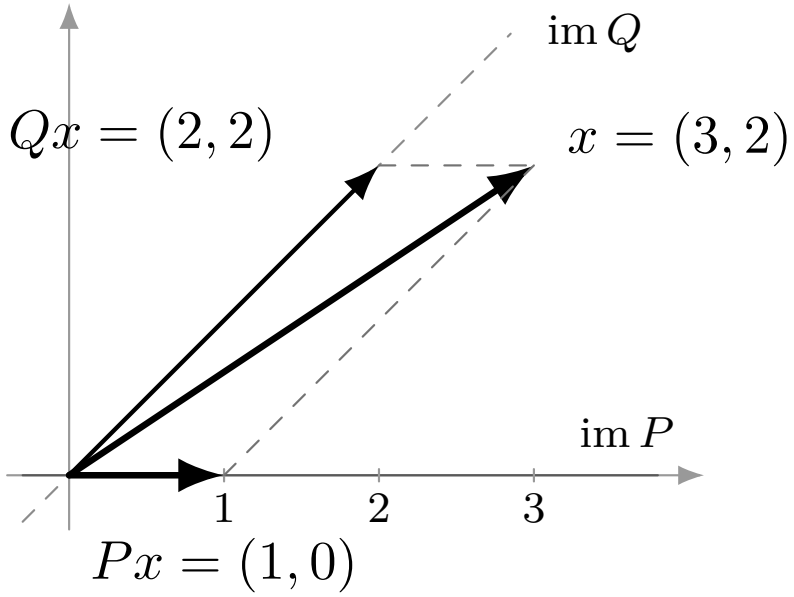
Exact does not mean orthogonal

Main text: theorem 7.5 and corollary 8.5

In $\mathbb{R}^2$, take

$$P = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}.$$

These maps satisfy $P^2 = P$, $Q^2 = Q$, $PQ = QP = 0$ and $P + Q = I$. Their images are the horizontal line and the diagonal line, respectively. They are not self-adjoint in the usual Euclidean inner product.



Actual Euclidean vector geometry for $x = (3, 2)$. The parallelogram reconstructs x from $Px = (1, 0)$ and $Qx = (2, 2)$. Neither component is an orthogonal projection of x.

Analysis records (Px, Qx) and synthesis adds the two vectors. For a packet tuple $(u, v) \in \operatorname{im} P \times \operatorname{im} Q$,

$$P(u + v) = u, \qquad Q(u + v) = v.$$

Both inverse identities follow from annihilation and completeness, without any angle condition.

What the algebra does not preserve

For the displayed x,

$$\|x\|^2 = 13, \qquad \|Px\|^2 + \|Qx\|^2 = 1 + 8 = 9.$$

There is no Parseval identity here. The cross term is nonzero because the packet directions are not perpendicular. Likewise, the simple adjoint-reversal formula for packet blocks requires self-adjoint projectors; it is not justified for this oblique pair.

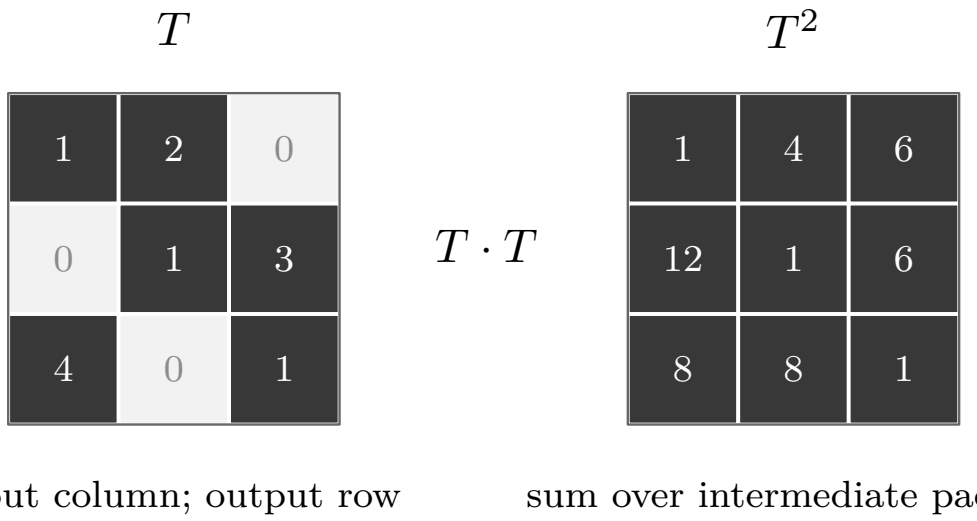
> **Read this as:** Exact reconstruction is an algebraic inverse statement. Energy preservation and adjoint identities require additional geometric hypotheses.

PLATE 8 / OPERATOR COORDINATES / MULTIPLICATION AS A SUM

A transfer block has an input and an output

Main text: theorems 8.2 and 8.3 and corollary 8.4

Use the three one-dimensional coordinate packets of $\mathbb{Q}^3$. The block $T_{ij} = P_i T P_j$ sends input packet j into output packet i. For this example, its nonzero scalar is the (i, j) matrix entry.



All displayed entries are exact integers. The square is computed from the operator, not from a thresholded graph or a matrix of block norms.

Reconstruction retains every block

The identity $P_1 + P_2 + P_3 = I$ gives

$$T = \sum_{i,j=1}^{3} P_i T P_j.$$

Selecting only the diagonal would change the operator. The entries $2, 3, 4$ carry transfers between different packets.

Composition sums over intermediate packets

The $(1, 2)$ entry of the square has two contributing routes:

$$(T^2)_{12} = T_{11}T_{12} + T_{12}T_{22} + T_{13}T_{32} = 1 \cdot 2 + 2 \cdot 1 + 0 = 4.$$

For input 3 and output 1, only the route through packet 2 survives:

$$(T^2)_{13} = T_{12}T_{23} = 2 \cdot 3 = 6.$$

For higher-dimensional packets, the same law uses actual linear maps as entries. The order of their composition still matters.

> **Read this as:** An edge says a block is nonzero. The block itself carries the coefficient, sign, rank and action. A transfer graph alone does not retain those data.

PLATE 9 / REACHABILITY / NECESSARY IS NOT SUFFICIENT

Two visible routes can cancel exactly

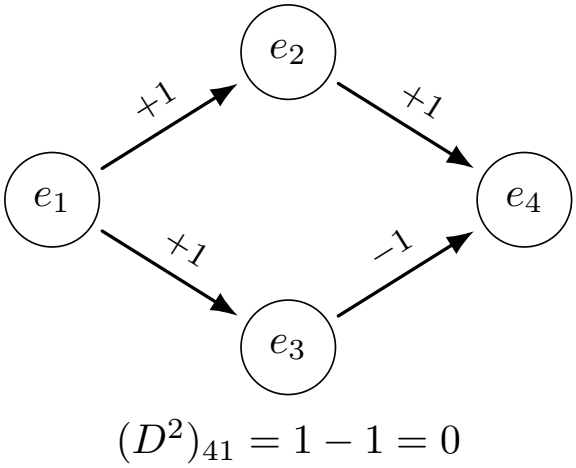
Main text: corollary 8.4 and proposition 9.3

Let a single operator D on $\mathbb{Q}^4$ act by

$$De_1 = e_2 + e_3, \qquad De_2 = e_4, \qquad De_3 = -e_4, \qquad De_4 = 0.$$

Every vertex of its coordinate-packet transfer graph is reachable from vertex 1. Nevertheless,

$$D^2 e_1 = e_4 - e_4 = 0, \qquad D^2 = 0.$$



$$[e_1, De_1, D^2 e_1]$$

1	0	0
0	1	0
0	1	0
0	0	0

$\mathsf{rank} = 2$

The diamond is the actual weighted action of D. The two length-two paths exist, but their contributions to the same transfer block sum to zero.

What the seed generates

With the fixed algebra $\mathbb{Q}[D]$ and seed e_1,

$$\mathrm{Hull}_{\mathbb{Q}[D]}(e_1) = \mathrm{span}\{e_1, e_2 + e_3\}.$$

It has dimension two, not four. Both the independent difference $e_2 - e_3$ and the last coordinate e_4 are missing.

Why projecting in the middle changes the experiment

If P_2 becomes an admissible primitive, the word DP_2D satisfies

$$DP_2De_1 = DP_2(e_2 + e_3) = De_2 = e_4.$$

But P_2 was not an allowed generator of the original $\mathbb{Q}[D]$ grammar. Using it to argue that the original seed already reached e_4 would silently enlarge the algebra.

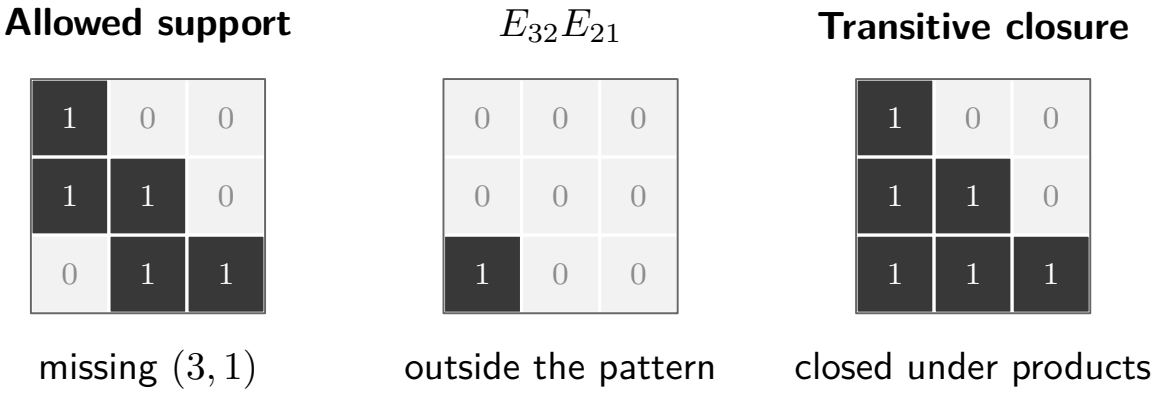
> **Read this as:** Unreachable nonzero packets obstruct generation. Reachability by itself does not prove generation: coefficients can cancel, blocks can have small images, and intermediate projectors may not be admissible.

PLATE 10 / SUPPORT ALGEBRAS / A UNIVERSAL TEST

Which zero patterns survive multiplication?

Main text: theorem 8.13; all packet spaces nonzero

Suppose every matrix supported in a specified pattern is allowed. Then multiplication can create a new entry whenever two allowed edges compose. Let E_{ij} denote the matrix that sends e_j to e_i and kills the other basis vectors.



Here 1 means "allowed," not that every admissible operator has that numerical value. The middle matrix is an actual product that violates the original pattern.

The first pattern contains the diagonal and permits $1 \to 2$ and $2 \to 3$, but forbids $1 \to 3$. Yet

$$E_{32}E_{21} = E_{31} \neq 0.$$

Both factors obey the pattern and their product does not. Thus the whole supported matrix space is not an algebra.

Reflexivity and transitivity have different jobs

The identity has a nonzero diagonal block on every nonzero packet, so including the identity requires all self-pairs. Multiplicative closure requires every composite pair. Consequently, for arbitrary algebraic block maps on nonzero packet spaces,

unital universal support algebra $\iff$ reflexive, transitive support relation.

The transitive closure in the figure is the lower-triangular pattern. It does define a unital algebra.

Why this does not contradict cancellation

On Plate 9, one particular operator has cancelling paths. The support theorem asks whether multiplication works for *every* assignment of allowed block maps. Choosing two suitable rank-one maps removes the possibility of hiding a failed composite behind special cancellation.

> **Read this as:** A zero pattern is a structural rule only when it survives all permitted block assignments. A coincidence in one matrix is not a closure theorem.

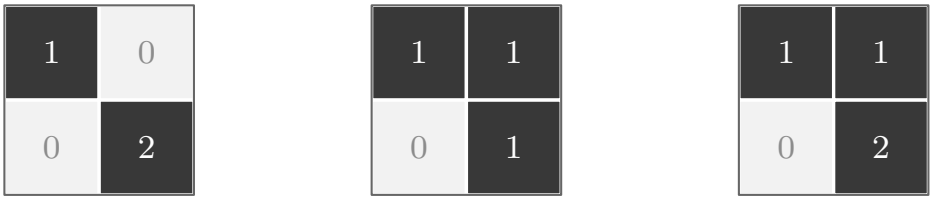
PLATE 11 / CONJUGACY / STRUCTURE VERSUS RAW SPARSITY

Move the packets with the coordinates

Main text: theorems 10.1 and 10.4 and corollary 4.5

Start with $T = \mathrm{diag}(1, 2)$ and the two coordinate projections. Apply the invertible shear U. The transformed operator is $T^U = UTU^{-1}$, and the transformed projectors are $Q_i = UP_iU^{-1}$.

T U UTU^{-1}



$Q_1 = UP_1U^{-1}$ $Q_2 = UP_2U^{-1}$



The transformed operator has a nonzero off-diagonal entry in the standard basis. Relative to its transported, oblique packets, however, both off-diagonal blocks remain zero.

Indeed,

$$Q_1T^UQ_2 = Q_2T^UQ_1 = 0, \qquad Q_iT^UQ_j = U(P_iTP_j)U^{-1}.$$

A change in raw matrix sparsity is therefore not necessarily a change in the packet transfer graph. Conjugacy preserves the exact block identities and their zero pattern; a nonunitary shear need not preserve their Euclidean norms.

When the map is not invertible

For an intertwiner $\Phi A = \varphi(A)\Phi$, the hull identity uses the *image algebra*:

$$\Phi(\mathrm{Hull}_{\mathcal{A}}(S)) = \mathrm{Hull}_{\varphi(\mathcal{A})}(\Phi S).$$

Replacing $\varphi(\mathcal{A})$ with a larger target algebra can enlarge the target hull. Seed rank need not be equal relative to that larger algebra. A surjective intertwiner gives the appropriate monotonicity inequality, not automatic equality.

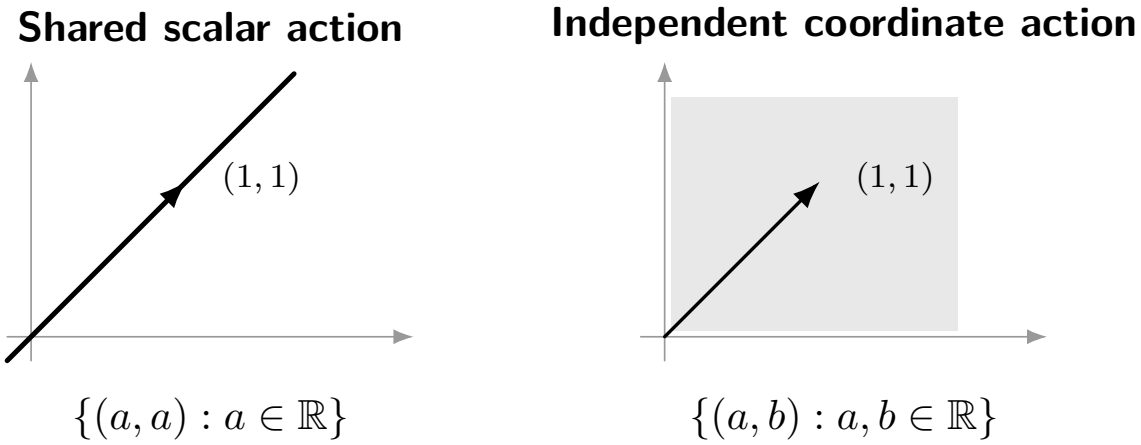
> **Read this as:** Transport the space, operators and packet maps together. An unchanged-looking matrix is not the invariant; the commuting algebraic relationships are.

PLATE 12 / DIRECT SUMS / ONE SEED CAN CARRY TWO COMPONENTS

Shared control and independent control differ

Main text: proposition 4.7 and theorem 4.8

Take $X = \urcorner \oplus \urcorner$ and $s = (1, 1)$. With one shared scalar acting on both coordinates, every orbit vector is (a, a). With independent scalars, the orbit contains every (a, b).



The geometric picture uses $\urcorner = \mathbb{R}$. On the right, shading shows a finite window of the whole plane; on the left, only the diagonal line is generated. The algebraic distinction holds over any field.

The corresponding minimum seed numbers are

$$\operatorname{srank}_{\urcorner I}(\urcorner^2) = 2, \qquad \operatorname{srank}_{\urcorner \times \urcorner}(\urcorner \oplus \urcorner) = 1.$$

In the independent case, the algebra contains $(1, 0)$ and $(0, 1)$, which recover the two coordinates of s separately.

This is why the independent formula uses a maximum

For independent actions of $\mathcal{A}$ on X and $\mathcal{B}$ on Y, a common list of seed positions can carry the seeds for both spaces. Pad the shorter list with zeros. Independent operators then select the two components, giving

$$\operatorname{srank}_{\mathcal{A}\times\mathcal{B}}(X \oplus Y) = \max\{\operatorname{srank}_{\mathcal{A}}(X), \operatorname{srank}_{\mathcal{B}}(Y)\}.$$

With a shared action, this selection need not be available; the general bound is the sum.

One operator can supply independent control

Over $\mathbb{Q}$, $T = \operatorname{diag}(1, 2)$ has polynomial projectors $2I - T$ and $T - I$. Thus $\mathbb{Q}[T]$ already separates the two eigenspaces. If the eigenvalues coincide, these formulas no longer separate them. Distinct primary labels play the same role on Plate 14.

> **Read this as:** Counting occupied components is not the same as counting seeds. One mixed seed may generate several components when the algebra can act on them independently.

PLATE 13 / SPLIT SEMISIMPLE MODEL / A CONSTRUCTIVE FORMULA

Multiplicity is a row-space budget

Main text: theorem 5.6 and corollary 5.7

Let $M_2(\mathbb{Q})$ act on $X = M_{2\times 5}(\mathbb{Q})$ by left multiplication. Think of the five columns as five copies of the same two-dimensional natural module. One seed is a 2×5 matrix V.

Every row of AV is a linear combination of the two rows of V. Hence one seed exposes at most two independent directions in the five-dimensional multiplicity space. Two seeds expose at most four. At least three are necessary.

Each row is one direction in the five-dimensional multiplicity space.

V_1

1	0	0	0	0
0	1	0	0	0

cumulative rank 4

V_2

0	0	1	0	0
0	0	0	1	0

cumulative rank 8

V_3

0	0	0	0	1
0	0	0	0	0

cumulative rank 10

The three fixed seeds cover multiplicity directions $1, 2$, then $3, 4$, then 5. The exact dimensions of their cumulative matrix orbits are $4, 8, 10$.

An explicit synthesis, not just dimension counting

For

$$W = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 6 & 7 & 8 & 9 & 10 \end{pmatrix},$$

choose A_1 from columns $1, 2$ of W, A_2 from columns $3, 4$, and A_3 with first column equal to column 5 of W and second column zero. Then

$$A_1V_1 + A_2V_2 + A_3V_3 = W.$$

The same construction works for arbitrary entries of W, so the lower bound is attained:

$$\boxed{\operatorname{srank}_{M_2(\mathbb{Q})} M_{2\times 5}(\mathbb{Q}) = \lceil 5/2 \rceil = 3.}$$

Several split simple blocks

If the acting algebra and module are supplied in compatible product coordinates, the row budget is computed separately in each block and the independent-action formula takes the maximum. For block data $(d, m) = (2, 5)$ and $(3, 4)$, the result is $\max(3, 2) = 3$, not $3 + 2$.

> **Read this as:** The formula measures multiplicity directions per seed. Producing the split algebra and module equivalences is a separate classification input in the formal contract.

PLATE 14 / PRIMARY DATA / AN ACTUAL EXTRACTION CERTIFICATE

Distinct labels let one polynomial separate chains

Main text: theorem 6.2 and corollary 6.3

Keep the six-dimensional model, but compare two different operators:

$$N = J \oplus J, \qquad T = J \oplus (2I + J), \qquad J = J_3(0).$$

For the same mixed seed $s = e_0 + f_0$, the Krylov matrices are different:

Equal primary labels

1	0	0	0	0	0
0	1	0	0	0	0
0	0	1	0	0	0
1	0	0	0	0	0
0	1	0	0	0	0
0	0	1	0	0	0

Krylov rank 3

Distinct primary labels

1	0	0	0	0	0
0	1	0	0	0	0
0	0	1	0	0	0
1	2	4	8	16	32
0	1	4	12	32	80
0	0	1	6	24	80

Krylov determinant 512

Exact rational calculations. Changing the primary label from 0 to 2 changes the operator, not merely its coordinates. The right-hand determinant is $512 \neq 0$.

The missing separating operation is now a polynomial

Define

$$p_2(t) = \frac{3t^5 - 15t^4 + 20t^3}{16}, \qquad p_0(t) = 1 - p_2(t).$$

These polynomials satisfy

$$p_2 \equiv 0 \pmod{t^3}, \qquad p_2 \equiv 1 \pmod{(t-2)^3}.$$

Consequently $p_0(T) = P_e$ and $p_2(T) = P_f$. Applying them to s extracts both chain heads, after which polynomial powers generate all six directions.

> **Read this as:** Repeated labels can lock chains together. Distinct primary labels can admit polynomial extraction. Here the block data and extractor are explicit; no Jordan-existence theorem is inferred from this example.

PLATE 15 / GRAPH MODEL / THE PATH CERTIFICATE

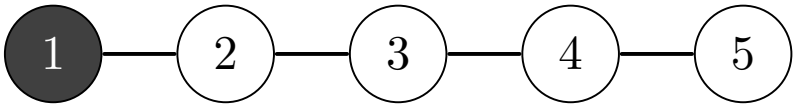
Walk counts make an endpoint cyclic

Main text: Worked graph model: all-size endpoint Krylov proof

Let A be the adjacency matrix of the five-vertex path. The (i, j) entry of

$$K = [e_1, Ae_1, A^2e_1, A^3e_1, A^4e_1]$$

counts walks of length $j - 1$ from vertex 1 to vertex i.



k	0	1	2	3	4
	1	0	1	0	2
	0	1	0	2	0
	0	0	1	0	3
	0	0	0	1	0
	0	0	0	0	1

unit diagonal; no entries below it

The entries are walk counts. At each new power, a new furthest vertex appears with coefficient one. The dark graph vertex is the seed.

A walk cannot reach vertex i in fewer than $i - 1$ steps. Therefore entries below the diagonal vanish. There is exactly one walk of length $i - 1$ from the first vertex to vertex i: the shortest path. The diagonal entries are all one, so $\det K = 1$.

The same reason works at every size

For a path with any $n \geq 1$, the endpoint Krylov matrix is upper triangular with unit diagonal. This is a combinatorial proof of cyclicity over *every field*; it does not rely on numerical eigenvalues. The appendix checks the finite cases $1 \leq n \leq 12$ as implementation examples, separately from the paper's all-size proof.

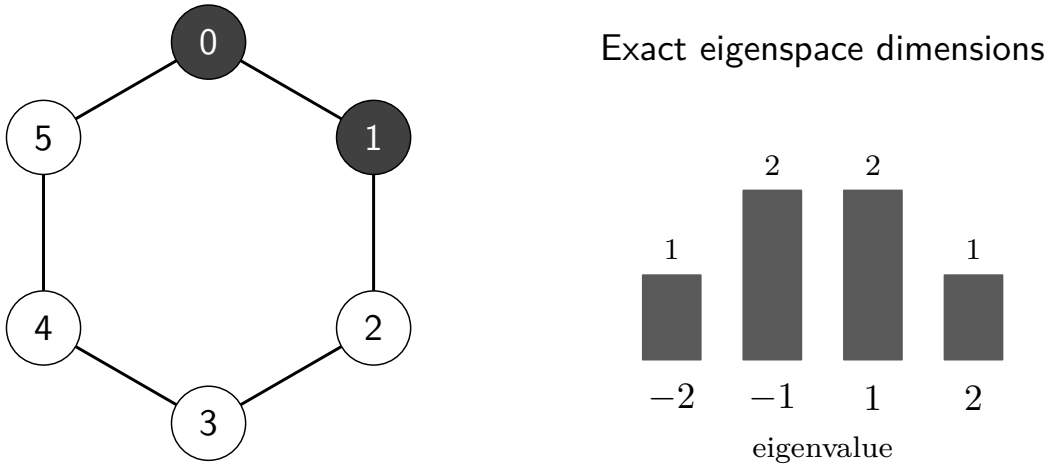
> **Read this as:** The endpoint distinguishes distances. Each extra operator power reveals one genuinely new coordinate before the finite space is exhausted.

PLATE 16 / GRAPH MODEL / A SIX-VERTEX CYCLE

Closing a path changes the seed obstruction

Main text: Worked graph model: complex minimum-seed proof for cycles

For the adjacency matrix A of C_6, the distinct eigenvalues are $2, 1, -1, -2$, with multiplicities $1, 2, 2, 1$. A single vertex seed spans only four directions. Two adjacent vertex seeds span all six.



The two dark vertices are the displayed generating pair. The bars show exact eigenspace dimensions, not estimated spectral peaks.

A lower bound without a numerical eigensolver

The two rows of the following matrix are independent and obey $qA = q$:

$$q = \begin{pmatrix} 2 & 1 & -1 & -2 & -1 & 1 \\ 0 & 1 & 1 & 0 & -1 & -1 \end{pmatrix}.$$

Thus $q : \mathbb{Q}^6 \twoheadrightarrow \mathbb{Q}^2$ is a scalar quotient for the eigenvalue 1. Its extension over $\mathbb{C}$ gives the same two-dimensional obstruction. Every one-seed orbit projects to a line, so no single seed can generate the full module.

The matching construction

From adjacent vertex vectors e_0, e_1, the identity

$$e_{j+1} = Ae_j - e_{j-1}$$

recovers successive vertices around the cycle. The lower and upper bounds therefore give two seeds for this example. The paper proves the corresponding minimum for every complex simple cycle of size at least three.

> **Read this as:** Connectedness is not cyclicity for the adjacency grammar. If a directed cyclic shift were allowed instead, one vertex would generate the whole cycle; that would be a different operator algebra.

PLATE 17 / COORDINATE ALGEBRA / NORMAL FORM RATHER THAN A CURVE SKETCH

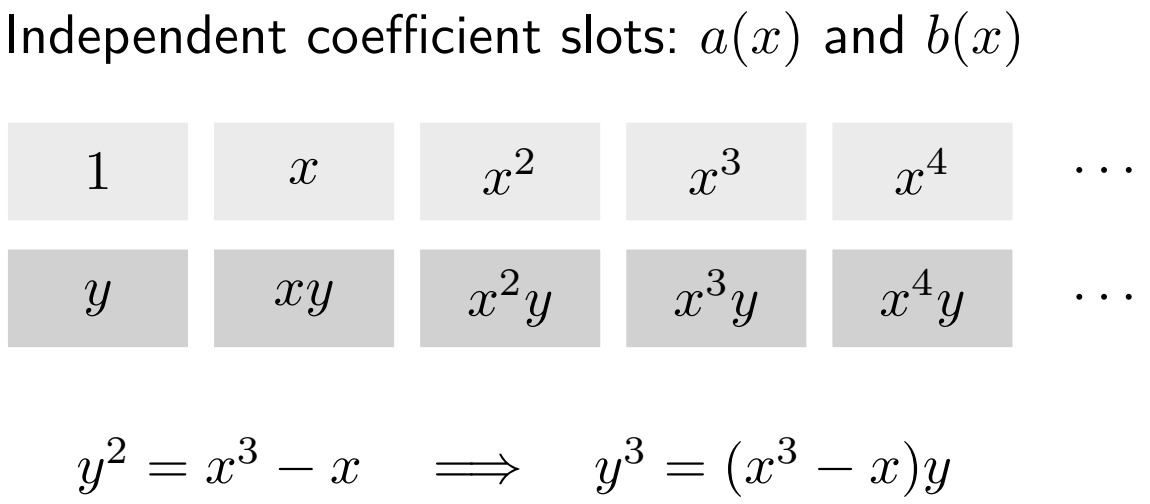
A relation becomes a packet product rule

Main text: Worked coordinate-ring model: normal form and exact multiplication

Work over $\mathbb{Q}$ in

$$A = \mathbb{Q}[x, y]/(y^2 - x^3 + x).$$

Because the relation is monic of degree two in y, each element has one unique form $a(x) + yb(x)$. The two packet coefficients are polynomials, not scalars.



A finite window of the two infinite coefficient lanes. Higher powers of y are rewritten using the relation; they are not simply discarded. No geometric genus or smoothness claim is encoded by this picture.

Multiplication by y transfers between the lanes

In the ordered coefficient coordinates (a, b),

$$M_y = \begin{pmatrix} 0 & x^3 - x \\ 1 & 0 \end{pmatrix}, \qquad M_y^2 = (x^3 - x)I.$$

The entries denote multiplication maps on $\mathbb{Q}[x]$. For $f = a_0 + ya_1$ and $g = b_0 + yb_1$,

$$fg = (a_0b_0 + (x^3 - x)a_1b_1) + y(a_0b_1 + a_1b_0).$$

In particular,

$$(x^2 + y)(x - y) = x + (x - x^2)y.$$

One module seed does not mean one algebra generator

Starting from 1, the fixed multiplication operators M_x, M_y generate every normal-form monomial. Thus the algebra is cyclic as a module over its multiplication algebra. This does not remove either address operator, and does not say that the algebra is generated by one algebra element.

> **Read this as:** The relation controls an exact weighted transfer between coefficient lanes. Packet reconstruction is $a + yb$, while packet multiplication retains the weight $x^3 - x$.

PLATE 18 / INFINITE MODEL / THE BILATERAL SHIFT

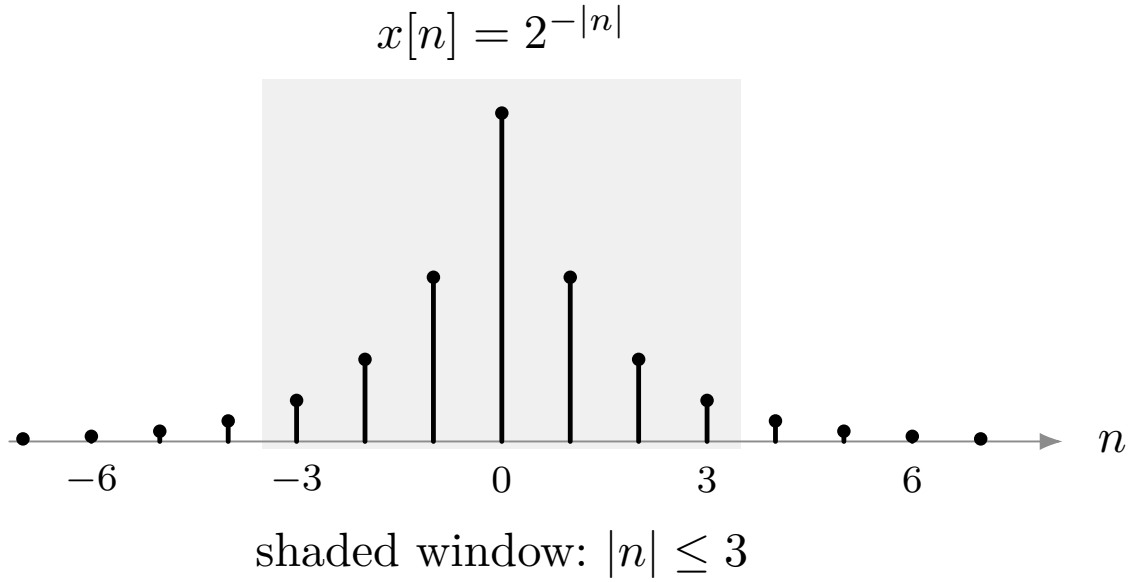
Density is not finite generation of each vector

Main text: proposition 15.1 and theorem 3.6

On $\ell^2(\mathbb{Z})$, declare every integer shift

$$(S_m x)[n] = x[n-m].$$

The seed δ_0 generates $\delta_m = S_m \delta_0$. Finite linear combinations give precisely the finite-support sequences. Their norm closure is all of $\ell^2(\mathbb{Z})$.



A finite window of the actual sequence $x[n] = 2^{-|n|}$. Outside every finite window, coefficients remain nonzero. The shaded window illustrates the truncation $x^{(3)}$, not a periodic wraparound.

This x cannot be a finite linear combination of shifted impulses: it has infinite support. But if $x^{(N)}$ agrees with x on $|n| \le N$ and is zero elsewhere, then

$$\left\| x - x^{(N)} \right\|_2^2 = 2 \sum_{n>N} 4^{-n} = \frac{2}{3} 4^{-N} \longrightarrow 0.$$

The exact squared errors for $N = 1, 3, 5$ are $1/6$, $1/96$, and $1/1536$. These are identities from a geometric series, not floating residuals.

What the topology contributes

The statement is that *one seed has dense hull*. It is not that every vector is a finite word expression. The completion theorem additionally specifies a normed presentation, a dense linear isometry into a Banach target, and the bounded intertwining needed to extend operators.

> **Read this as:** Completion adds limits in a declared topology. A finite numerical window can illustrate the approximants, but cannot be substituted for the infinite object or its convergence proof.

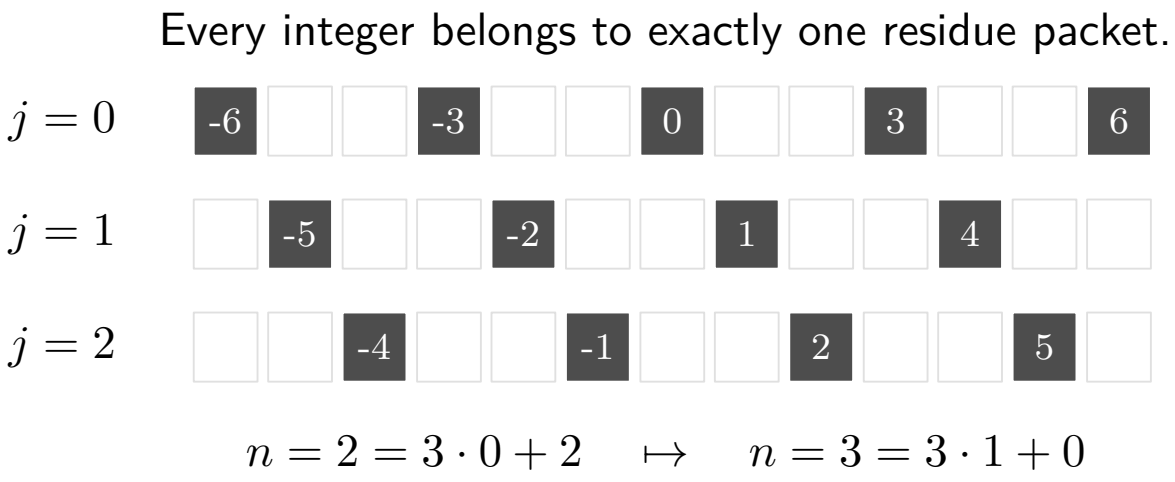
PLATE 19 / INFINITE MODEL / THREE EXACT SUBSEQUENCES

Residue packets retain an address carry

Main text: theorem 7.10; residue packets

For $j = 0, 1, 2$, let P_j keep the entries with $n \equiv j \pmod 3$. These are bounded orthogonal projectors whose sum is the identity. The packet coordinates can be written as

$$x_j[k] = x[3k + j], \qquad k \in \mathbb{Z}.$$



Each occupied tile represents one integer address in a displayed window. The three rows partition the addresses exactly. Negative indices use the same residue convention.

What one shift does in these coordinates

Let $S = S_1$ and $z = Sx$. Substituting $3k + j$ into the shift definition gives

$$z_1[k] = x_0[k], \qquad z_2[k] = x_1[k], \qquad z_0[k] = x_2[k - 1].$$

The passage from phase 2 to phase 0 carries a shift of the coarse address. If $Lh[k] = h[k - 1]$, the exact packet matrix is

$$S_{\text{packets}} = \begin{pmatrix} 0 & 0 & L \\ I & 0 & 0 \\ 0 & I & 0 \end{pmatrix}.$$

Replacing L by a scalar 1 would preserve the phase graph but would lose the operator's action on the infinite sequences.

Synthesis keeps the positions

To reconstruct x, place $x_j[k]$ at its original address $3k + j$. In the ambient-space formulation, this is the finite sum $x = P_0x + P_1x + P_2x$. Boundedness makes the same formulas valid after taking the ℓ^2 completion.

> **Read this as:** A cyclic packet graph is only the routing pattern. The block at the phase boundary still carries an actual shift operator.

PLATE 20 / EXACT COMPUTATION / AN INVARIANCE CERTIFICATE

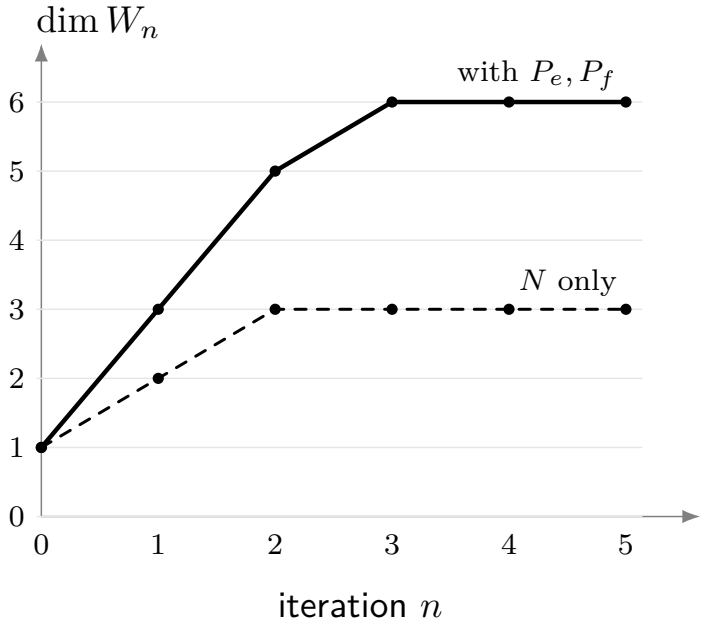
Stop when the subspace stops growing

Main text: construction 19.1

For a fixed finite operator family (O_i), start from $W_0 = \operatorname{span} S$ and iterate

$$W_{n+1} = W_n + \sum_i O_i W_n.$$

This is an ascending chain. When two consecutive subspaces agree, every generator maps the stable subspace into itself. The stable value is therefore the entire word-orbit hull, not just the span of the words tested so far.



Exact dimensions for the same mixed seed in Model A. The dashed curve allows only N. The solid curve also admits the two chain projectors. Plateau values are repeated to show the certified stop.

A finite-dimensional termination argument

If an inclusion is strict, the dimension rises by at least one. It cannot rise beyond $\dim X$. At equality, invariance and containment of the seeds prove both inclusions between the stable space and the least invariant hull.

What the companion code actually does

The new checker retains exact column bases using rational row reduction. It then verifies that adjoining the images of those bases under every generator does not increase rank. No singular-value cutoff or floating-point tolerance is involved in these examples.

The Lean statement is a mathematical termination and correctness specification. It is not a claim that this Python basis-maintenance implementation was extracted from Lean or that a particular runtime bound was proved. The original numerical companion and its tolerances remain separate.

> **Read this as:** "No new directions" is a stopping certificate only after checking the full declared operator family. Stopping after a convenient selection of words is not the same test.

PLATE 21 / SEED-RANK OPERATIONS / A RECTANGLE, NOT A DIAGONAL

Tensor products require every cross term

Main text: proposition 4.9; algebraic tensor products

If (x_i) generates X under $\mathcal{A}$ and (y_j) generates Y under $\mathcal{B}$, then the candidate tensor seeds are *all* $x_i \otimes y_j$. For finite expansions,

$$x = \sum_i A_i x_i, \qquad y = \sum_j B_j y_j,$$

bilinearity gives the required double sum:

$$x \otimes y = \sum_i \sum_j (A_i \otimes B_j)(x_i \otimes y_j).$$

The coefficient rectangle

	$5f_1$	$7f_2$
$2e_1$	10	14
$3e_2$	15	21

$$10(e_1 \otimes f_1) + 14(e_1 \otimes f_2)$$

$$+15(e_2 \otimes f_1) + 21(e_2 \otimes f_2)$$

The exact expansion of $(2e_1 + 3e_2) \otimes (5f_1 + 7f_2)$. The off-diagonal terms 14 and 15 cannot be omitted.

Keeping only the matched-index tensors $e_1 \otimes f_1$ and $e_2 \otimes f_2$ spans a two-dimensional subspace of a four-dimensional tensor product. It misses both cross directions. Under scalar actions on $\mathbb{Q}^2$ and $\mathbb{Q}^2$, all four basis tensors are necessary and sufficient.

The resulting upper bound

Since simple tensors span the algebraic tensor product, the complete rectangular family proves

$$\boxed{\operatorname{srank}_{\mathcal{A}\otimes\mathcal{B}}(X \otimes Y) \leq \operatorname{srank}_{\mathcal{A}}(X)\, \operatorname{srank}_{\mathcal{B}}(Y).}$$

This is not a universal equality claim. Changing the algebra changes the answer: the full independent matrix action can make a single nonzero pure tensor cyclic. Nor does the algebraic argument itself establish a theorem for completed topological tensor products.

> **Read this as:** The generator construction is indexed by a Cartesian product. Its proof must retain the finite double sum and the scalar-action hypotheses that make the tensor action well-defined.

PLATE 22 / PROOF BOUNDARY / RETURN TO THE MANUSCRIPT

What the pictures do—and do not—certify

Main text: Reading the theorem hypotheses and companion evidence

The mathematical statements in the main text have explicit hypotheses and written proofs. The Lean guide separately identifies the checked declarations and certificate assumptions. The machine-native paper identifies the 108 stable foundation records and the checks executed on their structured representations. These are distinct forms of evidence.

Topic	Mathematical content	What must be retained
Presentations and hulls	Evaluation, module quotients, and generation under a fixed algebra.	The grammar must be fixed independently of the target.
Finite packets	Idempotence, annihilation, completeness and exact finite synthesis.	Orthogonality requires an additional hypothesis.
Semisimple formula	Division-ring multiplicities determine the minimal free cover.	The stated classical inputs and decomposition data.
Primary and graph formulas	Primary components and scalar quotients give sharp generator counts.	The coefficient field and multiplicity conventions.
Paths and cycles	All-size path and complex-cycle formulas have separate proofs.	Finite tests illustrate rather than prove all-size claims.
Coordinate algebra	Normal form, multiplication, packet identities and a unit seed.	Smooth projective geometry needs additional classical input.
Completion	Density and bounded extraction in the stated topology.	Infinite sequences are distinct from periodic finite arrays.

Evidence travels with the examples

The release includes mathematical source, exact model definitions, deterministic graphics, Python checks, Lean source and theorem mapping, and the machine-native record archive. The visual models use integer, rational, symbolic, or explicitly analytic calculations. The separate verification record distinguishes fresh executions from archived evidence.

> **Read this as:** The pictures make the hypotheses visible. Read the traditional proof, the formal contract, and the executable record at their respective stated scopes.

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